NEW ZEALAND
PETROLEUM & MINERALS

New Zealand
Petroleum Basins



MINISTRY OF BUSINESS,
INNOVATION & EMPLOYMENT
HĪKINA WHAKATUTUKI

2014/2015 Revised Edition

Published by New Zealand Petroleum & Minerals,
Ministry of Business, Innovation and Employment
PO Box 1473, Wellington 6140, New Zealand

Website: www.nzpam.govt.nz
Email: nzpam@mbie.govt.nz
Phone: +64 3 962 6179

Scientific content supplied by GNS Science
www.gns.cri.nz

ISSN (print): 2324-397X

This publication is also available online via the NZP&M website and on USB

ISSN (online): 2324-3988

Disclaimer: Although this information has been derived from sources believed to be accurate and reliable, no liability is accepted for any opinions expressed or for any errors or omissions.

© New Zealand Petroleum & Minerals 2014

New Zealand Government

Contents

Why New Zealand?	**2**
Government Commitment to Oil and Gas Development	**4**
Legislative Environment	**5**
Annual Block Offer Permitting Rounds	**6**
History and Track Record	**7**
Accessing Data	**9**
The Continent of New Zealand	**10**
New Zealand Petroleum Systems	**16**
Taranaki Basin	**20**
Reinga-Northland Basin	**30**
East Coast Province	**38**
Great South-Canterbury Province	**50**
Discoveries and Developments	**64**
Kapuni Gas-Condensate Field	66
Mangahewa Gas-Condensate Field	68
Maui Gas-Condensate Field	70
McKee Oil and Gas Field	72
Maari-Manaia Oil Field	74
Tawn Gas-Condensate and Oil Fields	76
Kupe Gas-Condensate Field	78
Pohokura Gas-Condensate Field	80
Tui Area Oil Fields	82
Kawau Gas-Condensate Discovery	84
Galleon Gas-Condensate Discovery	86
Kora Oil Discovery	88
Titihaoa Gas Show	90
Kauhauroa Gas Discovery	92
Karewa Gas Discovery	94
Waka Nui-1	96
Future Frontiers	**98**
Recommended Reading	**103**

Why New Zealand?

New Zealand has a proven record of successful discoveries, vast potential for resources to be discovered, and an open, low risk and straightforward permitting system. New Zealand is also ranked as one of the best countries in the world to do business.

New Zealand rates internationally as a country in which to invest and is prominent in a number of reports:

- New Zealand tops Forbes' 2012 Best Countries for Business list, which labels New Zealand as a free market economy that can compete globally
- New Zealand is rated the least corrupt country in the world on Transparency International's 2011 Global Corruption Index; New Zealand is consistently at or very near the top of this Index
- New Zealand is ranked 1st in the world for protecting investors by the World Bank in its 2013 Doing Business report and ranked third for ease of doing business
- In 2012, credit-reporting company Dun & Bradstreet named New Zealand Asia Pacific's fourth safest country. New Zealand was described as "a low-risk environment for business investment"
- The 2011 Index of Economic Freedom, compiled by The Heritage Foundation and The Wall Street Journal, ranked New Zealand's economy as fourth most-free in the world
- According to AUPEC (July 2009), New Zealand has the fourth lowest effective royalty and taxation rate of approximately 42 percent

New Zealand offers a stable economy and political system, and a government that actively encourages foreign investment and the safe and responsible development of oil, gas and minerals resources.

Global benchmarking from the Fraser Institute has highlighted the attractiveness of New Zealand, as one of the top 10 countries in the world for investment in petroleum exploration.

New Zealand's oil and gas industry is built on the back of early exploration and development dating back to the 1950s. The modern industry in New Zealand has a well-developed oil and gas infrastructure, experienced support industry, and a proactive globally-competitive oil and gas regime. It is attracting industry participation from national and international investors.

New Zealand is highly prospective for further discoveries of oil and gas, with many offshore basins remaining largely, or entirely, unexplored. New Zealand provides explorers with multiple basins, excellent opportunities for portfolio development, as well as cost-competitive operating conditions.

An established oil and gas sector is supported by a wide range of expertise, including consultancy to underpin entry strategies, assistance with land access, provision of

New Zealand has a deregulated economy, an excellent business environment and high prospectivity. It has:

- Abundant natural resources
- Safe and secure investment environment
- Economic and political stability
- World-class transportation and communications infrastructure
- Highly skilled workforce
- High quality technical services
- Extensive free trade agreements
- Active Government support for international investment
- Extensive databases of geological and exploration information
- Comfortable standard of living

A full range of New Zealand economic statistics, such as GDP, migration and employment, can be sourced from Statistics New Zealand (www.stats.govt.nz).

professional services, collection and analysis of seismic acquisition, district planning, geology and geophysics, environmental advice, assistance with health & safety, engineering, design, electrical and mechanical work, and transportation.

The development of New Zealand's oil, gas and mineral resources is regulated by agencies responsible for allocation of rights to explore for and produce Crown resources, as well as for the protection of the environment and for the safety of workers. Recent changes to the legislative framework are designed to ensure New Zealand's overall regulatory regime is transparent, robust, future-proof, and reflects best international practice.

AN ISLAND NATION

New Zealand lies 1,150 nautical miles east of Australia, across the Tasman Sea. It comprises of two main islands, the North Island and the South Island.

It has a total land area of 250,000 km², but that's just the emergent parts of a mainly submerged continental landmass. Approximately 96% of the continent is under sea water. To the northwest, south and east of New Zealand are large areas of relatively shallow sea, underlain by plateaux and ridges that border the deep ocean basins of the Pacific Ocean and Tasman Sea.



SOVEREIGN RIGHTS

In August 2008, a United Nations Commission confirmed New Zealand's custodial rights to more than 5.7 million km² of seabed. The New Zealand Government exercises authority to allocate rights to petroleum over this extensive subsea area – called the Exclusive Economic Zone [EEZ]. This extends out to 200 nautical miles from the coastline and the Extended Continental Shelf beyond that [United Nations Convention on the Law of the Sea [UNCLOS]].

New Zealand has the fourth largest EEZ in the world, with a maritime jurisdiction up to 22 times the size of its land area; equivalent in size to the European Union, the North Sea, and a quarter of the Mediterranean combined.

Government Commitment to Oil and Gas Development

The New Zealand Government has placed oil and gas development as a key priority in its economic policy. Launched in 2012, the government's Business Growth Agenda is based on six key "ingredients" businesses need to grow: export markets, innovation, infrastructure, skilled and safe workplaces, natural resources and capital. The Building Natural Resources workstream of the Business Growth Agenda aims to make better use of New Zealand's abundant natural resources, to continue to grow the economy and look after the environment.

The business unit responsible for managing the Crown's oil, gas and mineral resources, including permitting, Block Offer promotion and provision of free data on New Zealand's resources is New Zealand Petroleum & Minerals.

To realise the potential of New Zealand's resources, New Zealand Petroleum & Minerals introduced annual Block Offer permitting rounds from 2012. The annual permitting rounds are the basis for exploration permit allocation that is promoted internationally.

To attract international interest in exploration, there has been significant government investment in providing an understanding of New Zealand's frontier basins through seismic survey acquisition and releases.

The government is committed to ensuring that New Zealand has a world-class and robust regulatory environment for the safe and environmentally responsible exploration, production and transportation of our petroleum resources. A number of initiatives aim to strengthen the regulatory regime, including:

- strengthening upfront provisions in the Crown Minerals Act 1991 regarding the health and safety and environmental credentials of operators
- putting in place a comprehensive regime to manage the environmental effects of all petroleum exploration activities in the EEZ and on the extended continental shelf
- strengthening the health and safety regulation of wells and well drilling activities following international best practice
- the establishment of a new High Hazards Unit with an increase in the number of inspectors and the appointment of a Chief Inspector
- strengthening guidelines for minimising acoustic disturbance to marine mammals from seismic operations.

Legislative Environment

LEGISLATION

Development of New Zealand's oil, gas and mineral resources is governed by various pieces of legislation which are managed by different regulatory agencies. The government's goal is for New Zealand's regime to be transparent, robust, future-proofed, and reflecting best international practice.

PERMITS

Awarding a permit is the first legislative step in developing a Crown-owned resource. Permitting is managed by New Zealand Petroleum & Minerals.

The requirements for awarding and managing permits and collecting royalties are set out in the Crown Minerals Act 1991 and supporting regulations and minerals programmes (known as the Crown Minerals Act regime). The Crown Minerals Act regime is currently under review, with a new regime expected to be in place in early 2013.

+ More information can be found at www.nzpam.govt.nz/cms/petroleum/legislation

LAND ACCESS

A permit does not give its holder a right to go on to land covered by the permit. If a permit holder needs access to explore or mine they must have a land access arrangement with the landowner and land occupiers. The exception is where access is required for minimum impact activities (e.g. geological, geochemical, and geophysical surveying, taking samples by hand or hand held methods, aerial surveying, and land surveying).

The Crown Minerals Act 1991 sets out the process for reaching a land access arrangement.

+ More information can be found at www.legislation.co.nz

ENVIRONMENTAL MANAGEMENT

Before a company can undertake certain activities such as exploration drilling, they must apply for environmental consents from the appropriate authority.

Onshore and offshore - up to 12 nautical miles from the coastline

Local authorities manage environmental consenting processes in their region under the Resource Management Act.

+ More information on the Resource Management Act 1991 can be found at www.mfe.govt.nz/rma

Offshore - more than 12 nautical miles from the coastline

Exploration activities more than 12 nautical miles from the coastline will need to comply with new legislation that is expected to come into force by 2014, and will be managed by the Environmental Protection Authority. For more information, visit www.epa.govt.nz

+ More information on the Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012 can be found at www.mfe.govt.nz

Other marine protection

Our marine environment is also protected under the Maritime Transport Act 1994, and the marine protection rules that come under the Act. This legislation applies to vessels, installations and ports, and is managed by Maritime New Zealand and by local councils.

+ More information can be found at www.maritimenz.govt.nz

HEALTH AND SAFETY

Under the Health and Safety in Employment Act, all New Zealand employers have a general duty to provide and maintain a safe working environment for the workforce.

Petroleum and mineral mining operators must also comply with specific health and safety regulations for the oil and gas industry and for underground mining.

The Ministry of Business, Innovation & Employment's High Hazards Unit administers the Health and Safety in Employment Act 1992 and other regulations for extractive industries.

Updated regulations for the oil and gas industry are expected to come into force mid-2013.

+ More information can be found at www.dol.govt.nz

Annual Block Offer
Permitting Rounds

The New Zealand Government allocates the right to exploration through the annual release of petroleum exploration acreage; called Block Offers.

Release Areas are selected to offer the global petroleum exploration industry a variety of investment opportunities; areas vary in size, level of existing geological knowledge and are located in a range of water depths. All release areas are supported by a pre-competitive geological and geophysical data and analysis either undertaken by the Crown or previous industry exploration.

Areas can be offered in a graticular system, or by a predefined block selection method.

+ View the current Block Offer at www.nzpam.govt.nz/cms/petroleum/block-offers

History
and Track Record

EARLY DISCOVERIES IN TARANAKI

New Zealand has a proven record of oil and gas successes that make an important contribution to the national economy. Discoveries can be traced to pre-1865 when early settlers in Taranaki noticed an oily residue on a New Plymouth beach. This led to the Alpha well, the first oil well in the Commonwealth and one of the first in the world. Further wells located both east and west of Alpha-1 were subsequently drilled to take advantage of oil seepage from the shallow Moturoa oil field.

The Taranaki Basin was the first sedimentary basin in New Zealand to be explored for hydrocarbons and is currently New Zealand's only producing province. Located predominately offshore, the Taranaki Basin is found on the west coast of the North Island and covers approximately 330,000 km^2.

KAPUNI (1959) AND MAUI (1969)

In the 1950s, the modern age of exploration emerged in New Zealand. With the advantage of seismic reflection data, Shell discovered the onshore Kapuni field in 1959. The ability to explore and drill offshore led to the discovery of New Zealand's largest hydrocarbon field, Maui, in 1969. At the time, Maui-1 was New Zealand's second offshore well drilled and provided evidence for a prolific offshore Taranaki petroleum system through the discovery of the approximately 3.4 tcf Maui oil/gas field. To date, this is New Zealand's largest hydrocarbon field.

RECENT YEARS

Today New Zealand has a series of established commercial fields.

The large gas discoveries of Maui and Kapuni fields have suggested that New Zealand is gas-condensate prone. However, geochemical research suggests similar oil potential to that of prolific oil provinces in South East Asia. This is substantiated with the discovery of the Taranaki McKee Oil Field (Australasia's largest onshore oilfield), and offshore oil fields such as the Maari and Tui fields.

Maui still contains the majority of New Zealand's proven reserves, but this field is mature. Two other fields, Maari and Pohokura, contain about 49% of oil and condensate reserves, with Pohokura also accounting for 45% of gas reserves. In total there are 16 producing fields in the Taranaki Basin.

The complex tectonic and sedimentary history of New Zealand has led to a variety of successful plays within the Taranaki Basin. Discoveries have been made at all potential reservoir levels except the Cretaceous. The economics of New Zealand's gas market has controlled the Taranaki basin's exploration, with the early discovery of the large Kapuni and Maui fields limiting offshore exploration. However, new discoveries have been made at a steady rate in recent years, and new plays are still being tested, both onshore and offshore. Of the offshore fields to have recently come onstream, Maari and Kupe have been known for over two decades; whereas Tui oilfield was brought onstream well within five years of its 2003 discovery.

There are underexplored basins, with the possibility of significant unrealised reserves. Readily available, free seismic information covers some of these basins. Exploration includes both conventional and unconventional oil and gas.

Highlights

- Fields brought into production: Pohokura (2006), Turangi (2007), Tui (2008), Maari (2009), Kowhai (2009) and Kupe (2010)
- Ahuroa started commercial gas storage (2010)
- McKee LPG plant (2011)
- 4000 km^2 multi-client seismic data acquisition in the Northern Graben, offshore Taranaki (2013)
- New Zealand exploration and production (E&P) sector has had at least five significant independently net present value (NPV) positive projects in the last five years
- New Zealand E&P market has an increasing number of high quality players, most actively reinvesting.

BASINS

New Zealand has had limited exploration outside of the Taranaki Basin with discoveries in the East Coast Basin, Canterbury Basin and Great South Basin; Kauhauroa-1, Galleon-1 and Kawau-1A respectively.

Although the Taranaki Basin is New Zealand's premier oil and gas exploration region, the Great South Basin, Canterbury, Reinga-Northland, East Coast and the deep water possibilities of New Zealand's petroleum acreage, have attracted significant interest as emerging basins of petroleum potential.

Several other nearshore and onshore basins have had a small number of wells drilled, but these, and the far offshore frontier basins, remain largely unexplored.

Limited exploration began in the East Coast and Westland basins before 1900, with shallow wells drilled near oil seeps. Until 1970, most exploration was conducted onshore, but increasing acquisition of marine seismic reflection data led to offshore drilling in several basins.

In offshore Canterbury, the Galleon-1 well (1985) produced 10 mmscf/d of gas and 2,300 bbl/d of condensate on testing. In the Great South Basin, Kawau-1A (1977) flowed 6.8 mmscf/d gas and has estimated reserves of 461 bcf.

In 1998, an onshore gas discovery was made at Kauhauroa-1, on the North Island's East Coast. At the time with cheap, readily accessible gas from Maui dominating the local marketplace, these discoveries were all deemed un-commercial. While their size still precludes economic viability, they prove the existence of effective petroleum systems outside of the Taranaki Basin.

Since the mid-2000s, New Zealand Petroleum & Minerals has undertaken concerted data acquisition. Together with advances in deepwater drilling and production technology, the focus is shifting further offshore – where the biggest new discoveries are anticipated.

Large international companies currently hold licences to parts of four frontier offshore basins: Great South, Canterbury, Pegasus, and Deepwater Taranaki.

Accessing Data

ONLINE EXPLORATION DATABASE

New Zealand Petroleum & Minerals maintains a world class collection of petroleum exploration data, collected as part of permit holders' work programme obligations, and from the academic community's voluntary donations.

Data includes geological/ geophysical/ geochemical reports, geophysical studies (2D and 3D seismic, gravity and magnetic), and well results (geophysical data and physical samples).

Access to this data is free, allowing you to easily discover, preview and access exploration information by browsing database fields and digital reports online.

+ Access the Exploration Database at data.nzpam.govt.nz

PETROLEUM BASINS EXPLORER (PBE)

PBE is an interactive web-based tool that allows you to discover and access a wide range of petroleum exploration and geoscience data from across the New Zealand sub-continent using both geographically-based and text searches.

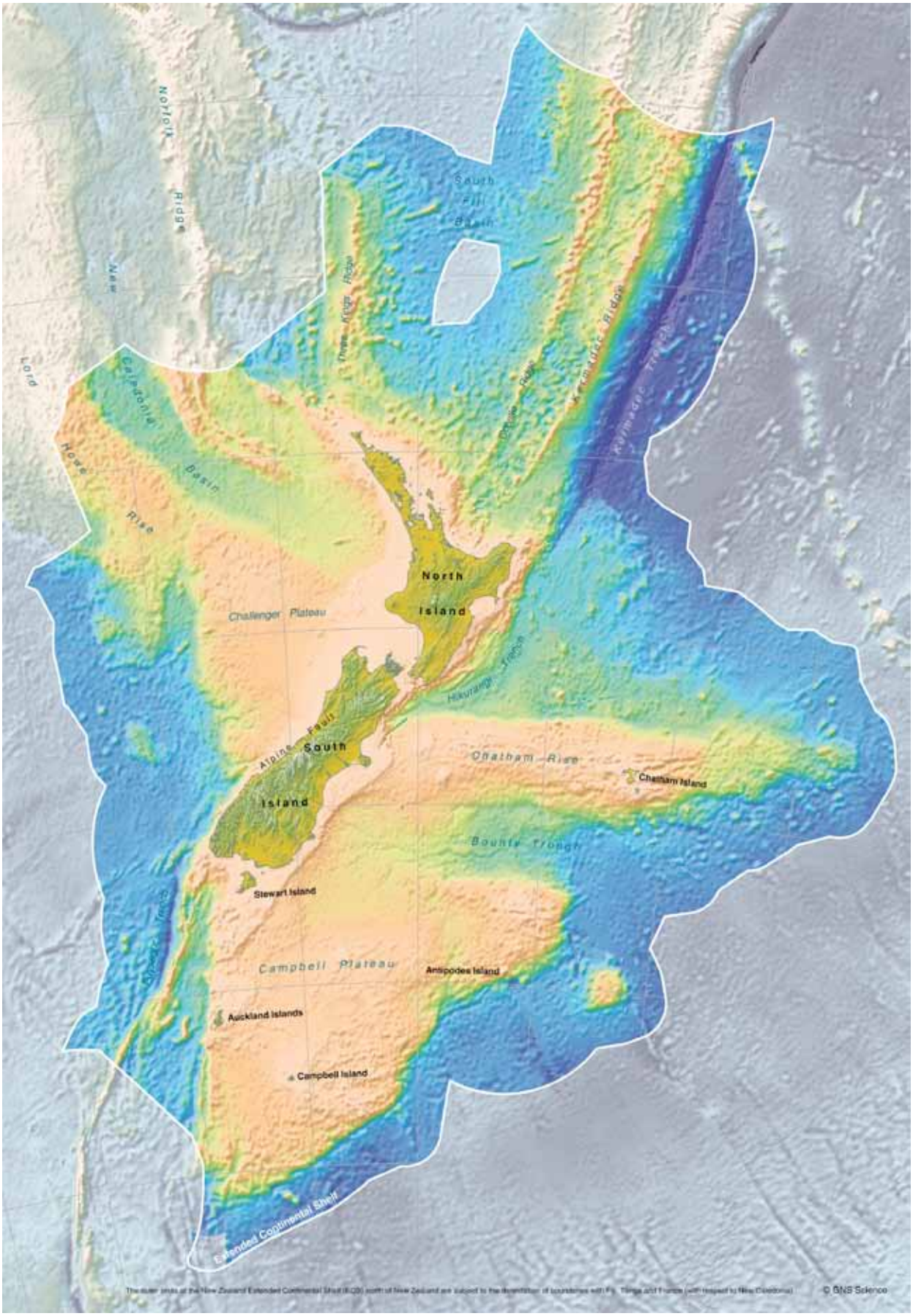
PBE contains a wide variety of data and products, ranging from simple GIS layers of the New Zealand coast to detailed oil and gas geochemical data, and includes outputs from the PEGI project and GNS Science's petroleum and basin studies research.

+ Access PBE at data.gns.cri.nz/pbe

DATA ACQUIRED

A large area of New Zealand's offshore territory is covered only by reconnaissance surveys; however the available data suggest large sedimentary basins that may host oil and gas cover about 20% of New Zealand's territory - over a million km^2.

For some of the basins, present understanding is based on modern, industry-standard seismic surveys (e.g., Deepwater Taranaki 2001–2009, Raukumara 2005–2007, Reinga 2009 and Pegasus 2010). For other basins there is a range of seismic data acquired from the 1970s onward, of variable quality. For remote areas there are limited seismic surveys completed for the 2008 New Zealand continental shelf submission to the United Nations.


Norfolk Ridge
New
Lord
Caledonia Basin
Howe Rise
Challenger Plateau
North Island
South Fiji Basin
Three Kings Ridge
Reinga Ridge
Kermadec Ridge
Kermadec Trench
Hikurangi Trench
Alpine Fault
South Island
Chatham Rise
Chatham Island
Bounty Trough
Stewart Island
Campbell Plateau
Antipodes Island
Auckland Islands
Campbell Island
Solander Trough
Extended Continental Shelf
© GNS Science

The Continent of New Zealand

A SUBMERGED CONTINENT

The islands of New Zealand have a total area of about 268,000 km^2 and are the emergent parts of an extensive, mainly submerged continental landmass with a total area of more than 6 million km^2. Northwest, south and east of New Zealand are large areas of relatively shallow sea underlain by plateaux and ridges that border deep ocean basins in the Pacific Ocean and Tasman Sea.

New Zealand established an Exclusive Economic Zone (EEZ) in 1977, defined by a line 200 nautical miles from the New Zealand coastline and, in 2004 established maritime boundaries with Australia. New Zealand's entitlement to an Extended Continental Shelf (ECS) beyond the EEZ, as defined in the United Nations Convention on the Law of the Sea (UNCLOS), was confirmed by the United Nations in August 2008. The outer limits of the ECS north of New Zealand are currently subject to delimitation with Fiji, Tonga and France in respect of New Caledonia. New Zealand now has sovereign rights over more than 5.7 million km^2 of seabed. This is an area more than 21 times greater than its land area and is equivalent in size to the European Union, the North Sea, and a quarter of the Mediterranean combined.

NEW ZEALAND'S SEDIMENTARY BASINS

A large area of New Zealand's offshore territory is covered only by reconnaissance surveys, however the available data suggest large sedimentary basins that may host oil and gas cover about 20% of New Zealand's territory – more than 1 million km^2. For some of the basins, present understanding is based on modern, industry-standard seismic surveys (e.g. Deepwater Taranaki 2001–2009, Raukumara 2005–2007, Reinga 2009 and Pegasus 2010). For other basins there is a range of seismic data acquired from the 1970s onward, of variable quality. For remote areas there are limited seismic surveys completed for the 2008 New Zealand continental shelf submission to the United Nations. Additional information from gravity and magnetic surveys and satellite data help define little known basins.

The basin boundaries shown on the map (on page 14) are mainly determined by major geological structures or seafloor physiography. In general, regions with stratigraphic continuity and a common geological history are included within a single basin. In places, the basin limits are set at a minimum sediment thickness. For some, sub-basins and provinces can be differentiated on geological or geographical criteria respectively.

MODERN SETTING

New Zealand's land area straddles the active boundary between the Australian and Pacific tectonic plates, and is above sea level mainly because of deformation and uplift in the last 20 million years. The North Island is a part of the Australian Plate and the South Island is mainly on the Pacific Plate. In the east, the Pacific Plate is moving southwest and downward beneath the North Island but southwest of the South Island the polarity is the reverse; the Australian Plate is being forced eastward and down beneath the South Island. These opposing subduction systems (represented by the Hikurangi Trough and Puysegur Trench) are connected by the Alpine Fault, a major strike-slip zone that borders the Southern Alps. Close to the active plate boundary, deformation is moderate to intense but, away from it, seismic surveys show that in many of the basins the thick successions of sedimentary rocks are little disrupted.

MESOZOIC BASIN INITIATION

In Paleozoic and early Mesozoic time, the basement rocks of New Zealand's islands and offshore plateaux were part of the Pacific margin of the Gondwana supercontinent, adjacent to the continental hinterlands of Australia and Antarctica. Triassic and Jurassic sedimentation in back-arc settings close to the Gondwana margin formed the thick marine and non-marine, structurally simple, low metamorphic grade Murihiku rocks of western and southern New Zealand, traditionally considered as economic basement.

Evidence from radiometric dating indicates that subduction persisted at the active continental margin until Early Cretaceous time (about 100 to 120 million years ago). After subduction ceased, the subsequent Gondwana continental break-up was preceded by a period of extensional tectonics, including rifting, with one margin-parallel rift basin becoming the Tasman Sea.

Basins initiated during this period contain both marine and terrestrial sediments, including coals. Although many onshore outcrops exhibit low-grade metamorphism, samples from offshore wells remain un-metamorphosed and immature for petroleum generation. Although no petroleum accumulations have yet been geochemically typed to source rocks older than Late Cretaceous age, where carbon-rich facies are present under the right conditions, they may be effective source rocks, particularly in the deep-water frontier basins.

RIFTS AND A PASSIVE MARGIN

In onshore and near-shore New Zealand, an unconformity representing Early Cretaceous uplift and erosion separates the Cretaceous and Cenozoic sedimentary rocks from underlying basement. A large part of the New Zealand region was land in Early Cretaceous time. Early basin-fill sedimentary rocks remain poorly dated. By the late Early Cretaceous, non-marine clastic sediments were accumulating in fault-controlled basins. The oldest basin-fill rocks are typically coarse-grained alluvial fan, fluviatile and lesser lacustrine facies, restricted to grabens and half-grabens. Deposition of thick passive margin marine sequences also began in the New Zealand region of the Gondwana margin in the late Early Cretaceous.

Magnetic anomalies show seafloor spreading in the Tasman Sea was well established in the Late Cretaceous (83-79 million years ago), by which time most of the major seafloor physiographic features of the New Zealand region had been formed. Large river systems developed and thick accumulations of non-marine and paralic sediments, including coal measures, accumulated in the valleys and extensive coastal plains. A thick progradational sequence that is present offshore from Taranaki represents the delta of a major river, built out into the accommodation space of a failed rift basin. It is capped by Late Cretaceous coal measures. In Taranaki, these coal measures became the source rocks for a large proportion of the oil discovered to date. Late Cretaceous marine shelf and slope sediments accumulated adjacent to the paleo-Pacific continental margin and in the more restricted seaways elsewhere.

QUIESCENCE AND TRANSGRESSION

Active seafloor spreading in the Tasman Sea and southern Pacific Ocean prevailed in Late Cretaceous, Paleocene and earliest Eocene time, when the New Zealand region was tectonically stable. With post-rift thermal subsidence and associated marine transgression, early-formed rift basins were progressively inundated. By Paleocene time, rift sedimentation was confined to small sub-basins while, at the margins of the land, coastal plain, marginal marine and shelf deposits accumulated. They include thick units of coal measures that are source rocks for oil and gas accumulations. Eocene deposits represent late-rift and post-rift transgressive sequences and, by Middle Eocene time, the reduced landmass was surrounded, in the west and south, by extensive coastal plains. Fine-grained clastic sediments and carbonates accumulated in marine settings distal from land areas.

PLATE BOUNDARY PROPAGATION AND INUNDATION

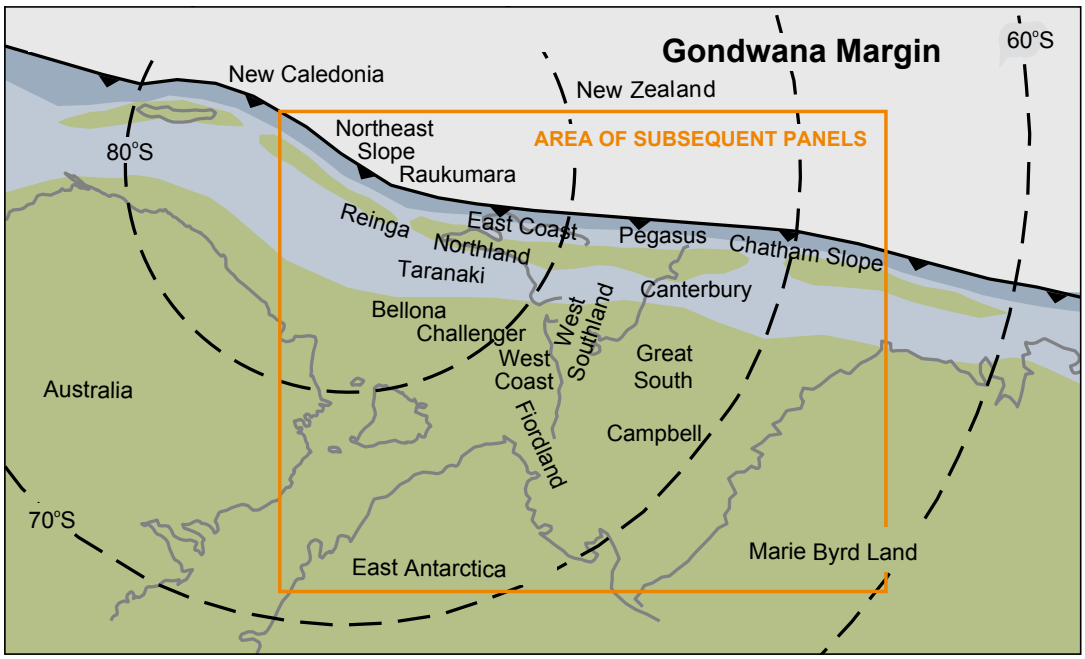
Seafloor spreading ceased in the Tasman Sea in Early Eocene time but continued in the southern Pacific Ocean. A new Australia-Pacific plate boundary formed south of New Zealand, where opening of the Emerald Basin resulted in anticlockwise rotation of eastern New Zealand relative to the west. For much of the region there was only minor deformation. In Southland there was rifting and the rotation resulted in compression in the Reinga Basin. By Late Oligocene time, the land area was greatly reduced and New Zealand may have been completely submerged. The Oligocene rocks are mainly calcareous. Differential compaction across basement highs and deformation associated with the new plate boundary formed a range of structures during this time.

NEOGENE PLATE BOUNDARY – UPLIFT AND DEPOSITION

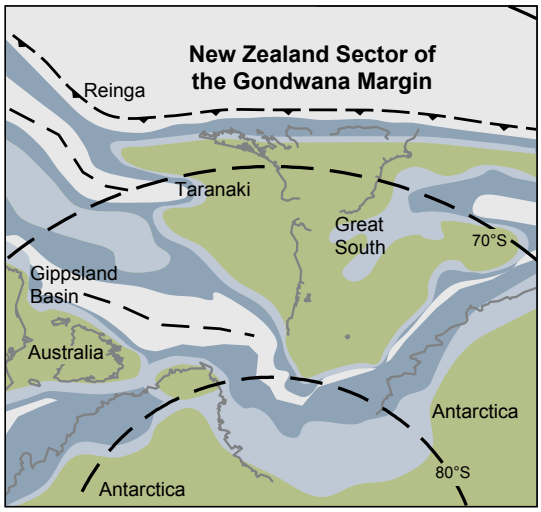
By earliest Miocene time, a southwest-dipping subduction zone was present in northern New Zealand. Large calc-alkaline stratovolcanoes erupted on and immediately west of what is now Northland, a part of the overriding Australian Plate. The Reinga Basin, originally a rift, became an intraplate back-arc basin. The thick Cretaceous to Oligocene passive margin sedimentary sequence which had accumulated northeast of the New Zealand landmass was obducted and emplaced part-way into the Reinga, East Coast and Raukumara basins as a series of thrust sheets (Northland and East Coast allochthons). Southwest-directed subduction in northernmost New Zealand was short-lived and calc-alkaline volcanism had all but ceased there by the end of the Early Miocene. Allochthon emplacement took place over about three million years.

Southwest-directed oblique convergence at the plate boundary east of North Island and Kermadec Ridge continues at the present day. The Kermadec Trench and its prolongation as Hikurangi Trough extend south to meet Chatham Rise just south of Cook Strait. In South Island, deformation on the Alpine Fault is mainly strike-slip, with about 480 km of dextral offset in the last 20 million years. The rate of plate boundary convergence accelerated from Middle Miocene time, resulting in increasingly rapid uplift, with erosion of Northland volcanoes, the axial ranges of the North Island, and the elevated mountain chain of the Southern Alps. The supply of vast amounts of sediment resulted in progressive infilling of marine depocentres and progradation of the continental shelves. Most New Zealand basins have thick Neogene successions of slope and basin floor mudstones, with intercalations of turbidite sandstones. Burial by thick Neogene sequences has raised the maturity of underlying Cretaceous and Paleogene rocks to levels sufficient to generate and expel hydrocarbons.

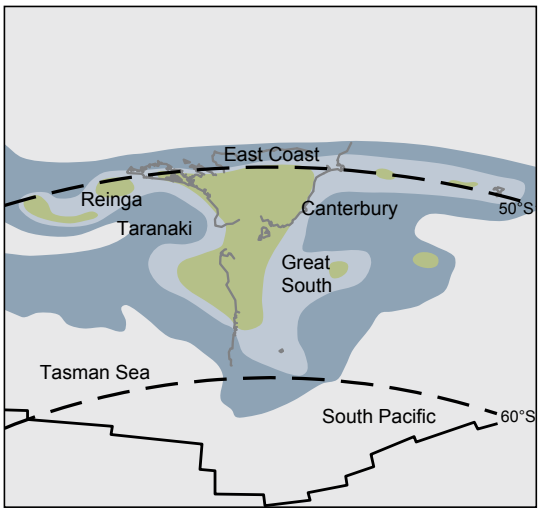
EARLY CRETACEOUS -APTIAN- 120 Ma



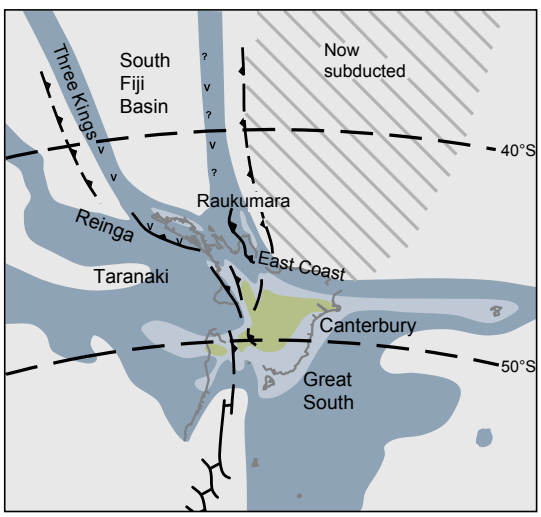
LATE CRETACEOUS - SANTONIAN- 85 Ma



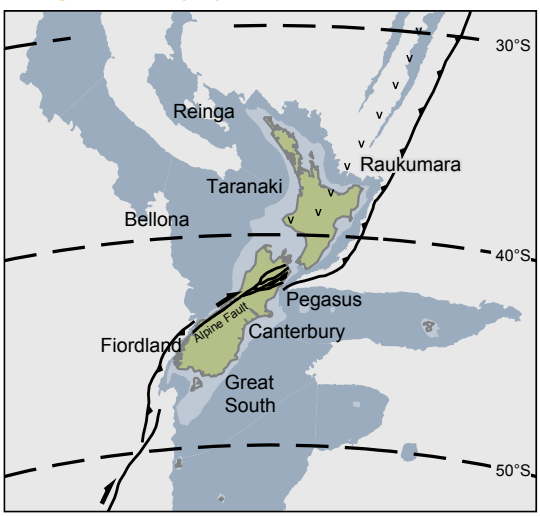
LATE PALEOCENE - 57 Ma

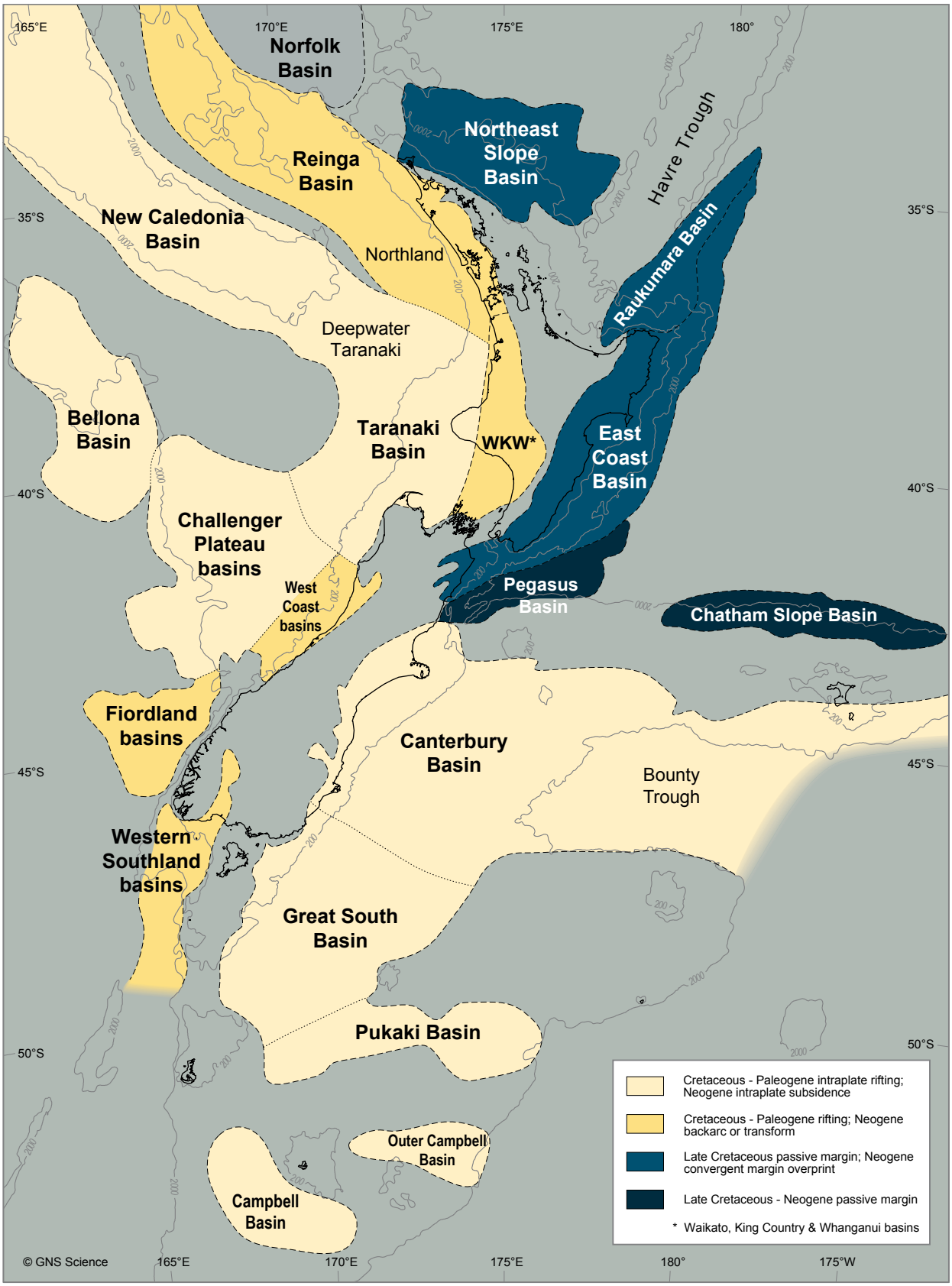


EARLY MIOCENE - 22 Ma



PRESENT DAY - 0 Ma




165°E
170°E
175°E
180°
175°W
35°S
40°S
45°S
50°S
Norfolk Basin
Reinga Basin
Northeast Slope Basin
Havre Trough
New Caledonia Basin
Northland
Raukumara Basin
Deepwater Taranaki
Bellona Basin
Taranaki Basin
WKW*
East Coast Basin
Challenger Plateau basins
West Coast basins
Pegasus Basin
Chatham Slope Basin
Fiordland basins
Canterbury Basin
Bounty Trough
Western Southland basins
Great South Basin
Pukaki Basin
Outer Campbell Basin
Campbell Basin
Cretaceous - Paleogene intraplate rifting; Neogene intraplate subsidence
Cretaceous - Paleogene rifting; Neogene backarc or transform
Late Cretaceous passive margin; Neogene convergent margin overprint
Late Cretaceous - Neogene passive margin
* Waikato, King Country & Whanganui basins
© GNS Science

SEDIMENTARY BASINS OF NEW ZEALAND

BASIN	AREA (km²)	LATE CRETACEOUS CHARACTER	NEOGENE CHARACTER
TARANAKI BASIN	330,000	Failed rift, subsidence and marine transgression	Intraplate to back-arc subsiding, minor compression
CANTERBURY BASIN	360,000	Failed rift, subsidence and marine transgression	Intraplate subsiding basin
GREAT SOUTH BASIN	130,000	Failed rift, subsidence and marine transgression	Intraplate subsiding basin
BELLONA BASIN	80,000	Failed rift, subsidence and marine transgression	Intraplate subsiding basin
CAMPBELL BASIN	40,000	Failed rift, subsidence and marine transgression	Intraplate subsiding basin
CHALLENGER BASINS	120,000	Failed rift, subsidence and marine transgression	Intraplate subsiding basin
OUTER CAMPBELL BASIN	20,000	Failed rift, subsidence and marine transgression	Intraplate subsiding basin
PUKAKI BASIN	60,000	Failed rift, subsidence and marine transgression	Intraplate subsiding basin
REINGA BASIN	170,000	Failed rift, subsidence and marine transgression	Back-arc subsiding basin, some compression
WAIKATO, KING COUNTRY, AND WHANGANUI BASINS	40,000		Back-arc subsiding basin, some compression
WEST COAST BASINS	25,000	Failed rift, subsidence and marine transgression	Transform
WESTERN SOUTHLAND BASINS	40,000	Failed rift, subsidence and marine transgression	Transform to back-arc
FIORDLAND BASIN	35,000		Transform
EAST COAST BASIN	120,000	Passive margin, subsidence and marine transgression	Convergent margin fore-arc
RAUKUMARA BASIN	36,000	Passive margin, subsidence and marine transgression	Convergent margin fore-arc
NORTHEAST SLOPE BASIN	80,000	Passive margin, subsidence and marine transgression	Convergent margin fore-arc
PEGASUS BASIN	25,000	Passive margin, subsidence and marine transgression	Passive margin, subsidence and marine transgression
CHATHAM SLOPE BASIN	40,000	Passive margin, subsidence and marine transgression	Passive margin, subsidence and marine transgression
NEW CALEDONIA BASIN	300,000	Failed rift, subsidence and marine transgression	Intraplate subsiding basin

New Zealand Petroleum Systems

The presence of commercial accumulations of oil and gas is critically dependent on key geological factors such as deposition and preservation of adequate source rocks, maturation, generation and expulsion of petroleum, followed by migration to porous and permeable reservoir rocks in structural or stratigraphic traps bounded by adequate seals.

SOURCE ROCKS

Most of New Zealand's known oil and gas accumulations are geochemically typed to coaly facies of Late Cretaceous and Paleogene ages. Lacustrine rocks are known from several periods and one Eocene formation has been mined as an oil shale. Oil from the Kora-1 discovery in the Taranaki Basin is typed to a Late Paleocene marine mudstone, which commonly occurs in many basins around New Zealand. Similarly, oil seeps in the East Coast Basin have been typed to the same unit as well as to an older Cretaceous to Paleocene marine mudstone.

As exploration moves out into deeper water, marine systems are being increasingly encountered leading to the suggestion that the deep-water basins around New Zealand may be more oil-prone than those onshore and in shallow water. In addition to in situ coaly rocks, turbidite successions may, in some instances, have entrained large volumes of coaly kerogen and may be effective "self-sourcing" reservoir facies. Another trend is that possible source rocks older than those now known to contribute to New Zealand petroleum systems are recognised. Finally, some basins, such as Raukumara and Pegasus, appear to have experienced very rapid rates of deposition in the Neogene, suggesting that older Neogene rocks with high Total Organic Carbon (TOC) may be effective source rocks in some regions. New Zealand's potential source rocks therefore range in age from Jurassic to Miocene.

The range of petroleum source rocks known to be present and effective in New Zealand basins, as well as those believed to be present, includes:

- in situ coaly rocks, coals and coaly mudstones
- lacustrine mudstones
- marine mudstones and black shales
- turbidites containing entrained coaly kerogen.

Coaly source rocks

In New Zealand basins, coal-bearing successions occur in most periods from Jurassic to Miocene. Most petroleum accumulations discovered to date were sourced from Late Cretaceous and Paleogene coals and coaly mudstones. The spread of age of these source rocks indicates that the important factors are facies and maturation history, not geological age.

Triassic to Early Cretaceous sedimentary rocks of western and southern New Zealand were deposited in a large fore-arc basin along the Gondwana margin. In outcrop, these rocks are commonly, but not always, metamorphosed to a low grade. As such, although they are known to include organic-rich marine and non-marine facies, including coal measures, they have been considered until recently to be economic basement. At Waka Nui-1 in the Reinga Basin, Jurassic coal measures with vitrinite reflectance values of around 0.7% were sampled, suggesting that these and similar beds elsewhere have source potential.

Mid-Cretaceous syn-rift terrestrial and marginal marine coals and coaly mudstones are present in the Taranaki, Reinga, West Coast and Great South basins and may be present at depth within the Northeast Slope, Raukumara, East Coast, Pegasus and Chatham Slope basins, as well as Campbell Plateau basins.

Late Cretaceous rift basin alluvial and coastal plain facies cover large areas. Late Cretaceous source rocks comprise variably carbonaceous mudstone and coal deposited in alluvial and coastal plain settings with marginal marine influence. These facies are widespread and economically important. In the Taranaki Basin they include the Rakopi Formation, the source rock for much of the oil discovered in New Zealand so far; equivalent units are present in other basins with an early rift history such as Canterbury, the West Coast and Great South Basin. The coal-forming vegetation was mainly woody gymnosperms such as podocarps and araucarians, and the resulting coals are dominated by hydrogen-rich vitrinite, with variable amounts of liptinite. Late Cretaceous coals and coaly mudstones have TOC and Hydrogen Index (HI) values typically in the ranges of 2 to 75% and 200 to 400 mg HC/g TOC, respectively, indicating mixed gas- and oil-prone to oil-prone kerogen.

As the New Zealand continent drifted away from Australia and Antarctica, basins became progressively inundated and coaly facies were restricted to shrinking coastal plain regions. Nevertheless, thick coal measures were deposited during the Paleogene and these contributed to hydrocarbon charge, particularly for the oil and gas-condensate fields of onshore Taranaki. Coal-bearing units as young as Late Miocene in age are known from onshore in Northland and some examples may have been buried deeply enough to expel hydrocarbons.

Lacustrine source rocks

Lacustrine rocks are recorded from Triassic and Jurassic outcrops onshore, where they are considered to be part of the economic basement. However in deep-water areas, such as around the Waka Nui-1 well, any examples of lacustrine organic-rich shales may be effective source rocks.

Mid- to Late Cretaceous lacustrine rocks are known from the Western Southland region and interpreted to be present in the Great South Basin, although no organic -rich examples have been described. Late Cretaceous to Paleocene lacustrine mudstones crop out and have been extensively drilled in the West Coast basins.

In Western Southland, Eocene lacustrine mudstones are known in the Waiau Basin and at Ohai and a correlative oil shale at Orepuki was mined for a short while around the start of the twentieth century. Although no discovered oils in New Zealand basins have been geochemically typed to lacustrine sources, organic-rich lacustrine rocks may make a significant contribution to undiscovered petroleum accumulations in some regions.

Marine source rocks

The Cretaceous period contains some of the world's richest marine source rocks, deposited during global anoxic events. As exploration progresses in the deep-water basins, more examples of Early Cretaceous marine deposition have been encountered. Marine shales deposited during known global anoxic events may include rich organic shales. These events occurred in the Early Aptian (about 120 Ma), the Early Albian (about 110 Ma), across the Albian/Cenomanian boundary (about 100 Ma) and the Cenomanian/Turonian boundary (about 93 Ma).

The best known New Zealand example of a marine black shale is the Late Paleocene Waipawa Formation, an organic-rich unit deposited in marginally marine (hyposaline), anoxic to oxic environments. It has a distinctive biomarker signature and its age is well known from northern Taranaki and East Coast basins, where it is seen in outcrop, and the Great South Basin, where it has been drilled by several wells. It is also present in the Reinga Basin, where it is found in seafloor dredging and was drilled by Waka Nui-1. It is also likely to be present in the Raukumara, Pegasus and other basins. The shale has an average thickness of about 30 m. TOC values are generally between 2 to 6% (up to 12.5%), with variable amounts of terrestrial-derived material as part of the organic matter assemblage. HI values vary widely from less than 200 to more than 300 mg HC/g TOC, indicating variably gas- to oil-prone kerogen.

MATURITY AND MIGRATION

The required depth of burial for source rocks to generate and expel oil and gas is variable, mainly due to differences in heat flow histories and source rock properties. Thermal modelling studies predict that, over most of New Zealand, the required present-day depth of burial for source rocks to start expelling oil is 4,000 to 4,500 m. However, where source rocks have been buried to moderate depths for long periods of time and recent sedimentation rates are low (as in Great South Basin), the required depth of burial may be as little as 3,000 m.

Many Taranaki oils, such as those from the Tui and Maari-Manaia fields, have been geochemically typed to coal measures of the Late Cretaceous Rakopi Formation. Recent work has shown that leaf cuticle forms much of the kerogen available for conversion to oil and that large volumes of oil may be expelled from cuticle-rich coals and coaly mudstones. Many of the coaly source rocks in New Zealand basins have a marine influence, which is beneficial for oil generation and expulsion.

In general, the deepest source rocks comprise Type III terrestrial coals, shaly coals and coaly mudstones, with variable syn-sedimentary marine influence. In the deep parts of some basins, petroleum may have been expelled as early as the Late Cretaceous, but in many basins the present-day mature kitchen areas are essentially confined to areas where there has been significant Neogene deposition. Paleocene Type II marine source rocks are generally close to maturity or have just entered the window for oil expulsion across the northern Taranaki, Reinga, East Coast and Great South basins.

Migration from source rock to trap is not well understood. Where there has been significant deformation, faults are considered as pathways for petroleum migration, as seals for trapping and development of structural compartmentalisation, or as leakage pathways that compromised integrity of top seals. Lateral migration distances are generally considered to have been short, although in one area there is some evidence for migration distances from source to accumulation of as much as 30 km.

In most basins, generation, migration and entrapment of petroleum are likely to have taken place in the last 10 million years. Due to the composite tectonic evolution of most of the basins there may have been more than one critical moment - the time when all prerequisite geological factors for the charging of traps were in place. In general the key critical moment is likely to have been in Neogene time for all basins except in Deepwater Taranaki and Great South basins, and the more remote southern basins, where it is likely to have occurred in the Paleogene.

RESERVOIRS

Potential reservoir rocks are present throughout the stratigraphic record. The reservoirs for the gas-condensate discoveries in Galleon-1 in Canterbury Basin and Kawau-1A in the Great South Basin are Late Cretaceous non-marine and shallow marine sandstones. The most productive Taranaki Basin reservoir rocks are in Paleogene transgressive shoreline systems, as in the Kupe, Maui and Pohokura fields, and a variety of facies belts within Tertiary Neogene clastic depocentres. Oligocene to Early Miocene limestone with high fracture permeability is the reservoir in the Waihapa Field. Petroleum is also produced from Miocene and younger, deep-water turbidite sandstones in the Maari, Kaimiro and Ngatoro fields. Oil is known from Miocene volcaniclastics in the Kora-1 well and gas is present in Pliocene sands at Karewa; similar Neogene slope and basin floor fans are present in many New Zealand basins and gas shows have been encountered in East Coast Basin Miocene turbidites at Titihaoa-1. In the East Coast region, Pliocene and Pleistocene coquina limestone has favourable reservoir characteristics.

SEALS

Seals are abundant in all potential petroleum basins and overpressures, indicative of fluid confinement within the stratigraphic sequence, are sometimes encountered during drilling. Seal rocks in most basins are marine mudstones deposited during both the passive margin transgressive phase and the regressive convergent margin phase. Those Oligocene and Early Miocene limestones which were not fractured during Neogene tectonism may also provide seals in many basins.

OVERBURDEN

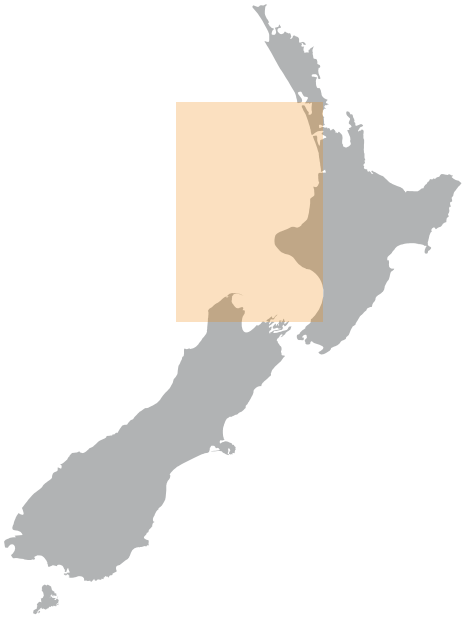
In general, two major periods of sediment deposition resulted in substantial burial. Late Cretaceous rifting produced grabens several kilometres deep that rapidly filled with sediments. Petroleum generation may have begun in the deepest of these basins during the Late Cretaceous and early Cenozoic. A second phase of rapid sedimentation occurred in many basins during the Neogene, when the development of the plate boundary through New Zealand caused rapid uplift and erosion. Very high Neogene sedimentation rates, in excess of 1,000 m per million years, have occurred in some basins. For example, deposition of the Giant Foresets Formation increased the overburden thickness by as much as 2,000 m over a large part of Taranaki Basin in just 3 million years. The volume of the Giant Foresets Formation is greater than the present day volume (above sea level) of the New Zealand landmass.

TRAPS

The commercial petroleum accumulations encountered in the Taranaki Basin to date are in structural traps formed in the Neogene, though the Kupe Field also has an element of stratigraphic trapping. Traps include four-way dip closures, three-way and fault closures, drape folds over basement horsts, thrust-controlled anticlines, and overthrusts. The same range of trap types is predicted to be present in other basins around New Zealand, which share this structural history. Dip-closed and thrust-bounded anticlines are common in the East Coast Basin as a result of the post-Early Miocene convergent margin overprint.



Taranaki Basin



BASIN SUMMARY

The Taranaki Basin covers an area of about 330,000 km^2 and is currently the only producing basin in New Zealand. This basin is contiguous with the Reinga Basin to the north and the West Coast basins to the south. Over 400 onshore and offshore exploration and production wells have been drilled in Taranaki to date. None have been drilled beyond the shelf edge. The basin remains under-explored compared to many comparable failed rift complex basins of its size and there remains considerable potential for further discoveries.

PLAYS

+ **Inversion structures**
+ **Thrust features**
+ **Extensional fault blocks**
+ **Volcanic edifices**
+ **Half-graben fills**
+ **Submarine fans**

EXPLORATION AND PRODUCTION

Wells

Over 400 onshore and offshore exploration and production wells drilled since 1950. No wells beyond the continental shelf.

Producing Fields

Offshore: Maui (gas-condensate and oil), Tui Area (oil), Pohokura (gas-condensate), Maari-Manaia (oil), Kupe (gas-condensate).

Onshore: Kapuni (gas-condensate), McKee (oil and gas), Tariki-Ahuroa (gas-condensate), Waihapa-Ngaere (oil and gas), Ngatoro (oil and gas), Kaimiro (oil and gas), Mangahewa (gas-condensate), Rimu-Kauri (oil and gas), Cheal (oil), Turangi (gas).

Recent Discoveries

Tui (2003, oil), Karewa (2003, gas), Turangi (2005, gas), Awakino South (2008, gas), Kowhai (2008, gas), Beluga (2010, gas), Copper Moki (2011, oil/gas), Sidewinder (2011, oil/gas)

GEOLOGY

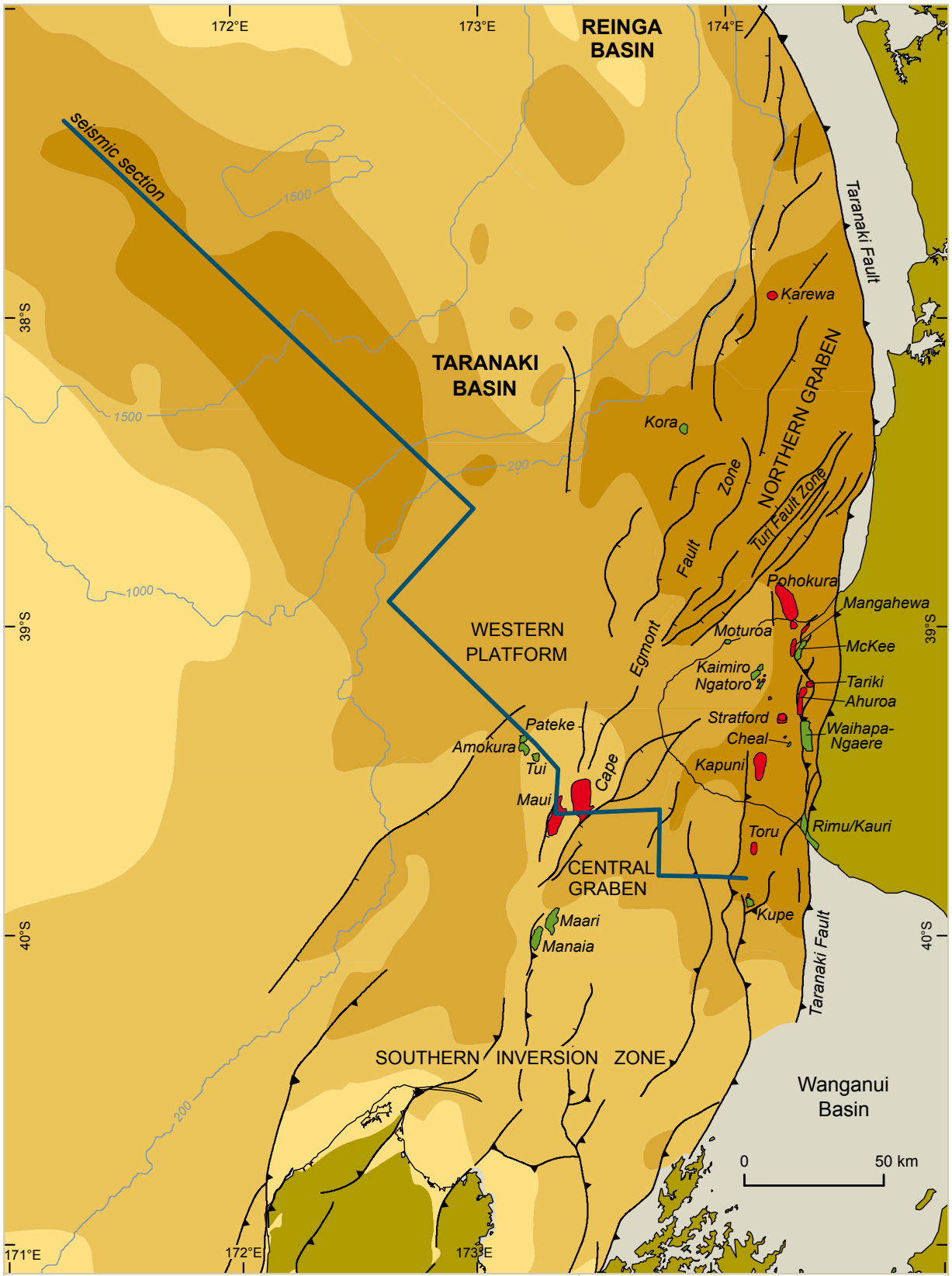
Jurassic Murihiku marine rocks and coal measures may be basement or earliest basin-fill. Early Cretaceous to Paleogene intraplate rift basin, with marine transgression and onlap. Large Cretaceous delta beyond the shelf edge. Back-arc subsidence and some compression in Neogene. Basin-fill up to 9,000 m.

Source Rocks

Jurassic, mid- to Late Cretaceous and Paleogene coaly rocks, Cretaceous and Paleocene marine shales.

Reservoir Rocks

Late Cretaceous to Eocene terrestrial, paralic and near-shore sandstones, Late Cretaceous to Pliocene turbidites, fractured Oligocene limestone, Miocene volcaniclastics, Pliocene shelf sands.


172°E
173°E
174°E
REINGA
BASIN
Seismic section
1500
38°S
TARANAKI
BASIN
1500
200
Kora
Karewa
Taranaki Fault
NORTHERN GRABEN
Zone
Turi Fault Zone
Fault
Egmont
1000
39°S
WESTERN
PLATFORM
Pohokura
Mangahewa
Moturoa
McKee
Kaimiro
Ngatoro
Tariki
Ahuroa
Stratford
Cheal
Waihapa-
Ngaere
Pateke
Amokura
Tui
Cape
Kapuni
Maui
Toru
Rimu/Kauri
CENTRAL
GRABEN
Kupe
Maari
Manaia
40°S
Taranaki Fault
SOUTHERN INVERSION ZONE
Wanganui
Basin
200
0
50 km
171°E
172°E
173°E
Legend
Oil fields
Gas-condensate fields
Subsurface fault
Basement outcrop
Sediment Thickness
0 - 2 km
2 - 4 km
4 - 6 km
6 + km
N

BASIN HISTORY

EXPLORATION AND PRODUCTION

The Taranaki Basin produces from about 20 fields, from the giant Maui gas-condensate field (original gas reserves 3.4 tcf), to a number of small oil and gas fields of about 10 mmboe. Some 418 mmbbl of oil and 6190 bcf of gas had been produced by the end of 2011. All wells drilled so far have been onshore or on the shelf. Exploration has focused on four-way dip closures, horsts, rotated fault blocks, foreland folds and detached thrust sheets along the overthrust eastern margin of the basin. Petroleum has been encountered on a commercial scale at every stratigraphic level from Pliocene to Paleocene. Structural, stratigraphic and diagenetic trapping mechanisms have yet to be fully evaluated. Cretaceous reservoirs, especially those in the deepwater province, have not been drilled, with the first well in the deepwater province scheduled to be drilled in 2013/14.

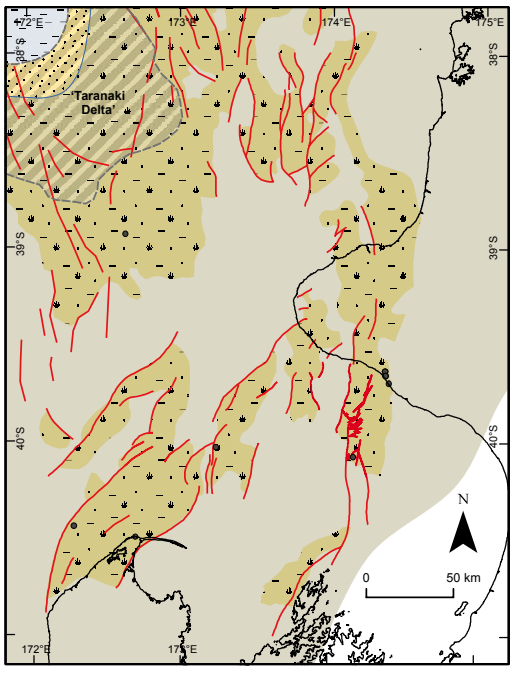
EXTENT

The Taranaki Basin was considered to extend only as far offshore as the continental shelf edge but seismic surveys over the last decade proved that thick sedimentary sequences extend into deep water beneath the modern bathymetric feature of New Caledonia Basin. Limits of the Taranaki Basin are now defined by the Taranaki Fault and associated Waimea-Flaxmore faults in the east, by a minimum sediment thickness along the flanks of Challenger Plateau and Lord Howe Rise to the southwest, and the West Norfolk Ridge to the northeast. It merges with the Reinga Basin to the north and the West Coast basins to the south. Northwest beyond New Zealand territory satellite gravity indicates that it may merge with the New Caledonia Basin. It is about 1,000 km long and up to 300 km wide and extends beyond the limit of New Zealand jurisdiction.

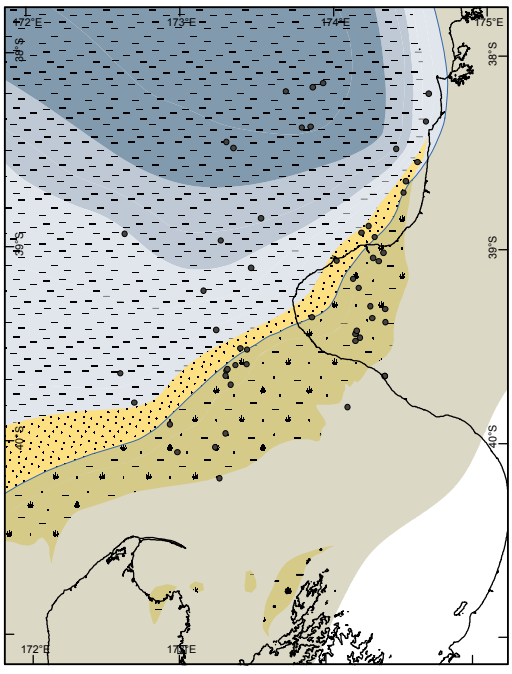
GEOLOGICAL HISTORY

The Taranaki Basin was the site of late Mesozoic extension on the landward side of the Gondwana margin. Jurassic to earliest Cretaceous Murihiku marine and non-marine rocks form either basement or the earliest basin-fill. Rift faulting diminished through the Late Cretaceous with sea floor spreading in the Tasman Sea. Terrestrial sediments deposited in early grabens and half-grabens were succeeded by thick marine rocks. A large delta built out to the northwest of the present-day shelf, fed by Late Cretaceous fluvial systems, which were transgressed in the latest Cretaceous. Eastern parts of the basin experienced north-south rifting in the Late Cretaceous and Paleocene as part of the breakup of Gondwana. Terrestrial and shallow marine deposits, including coal measures, were deposited throughout the Paleogene but, by the Oligocene,

Late Cretaceous (75 Ma) paleogeography



Middle Eocene (40 Ma) paleogeography



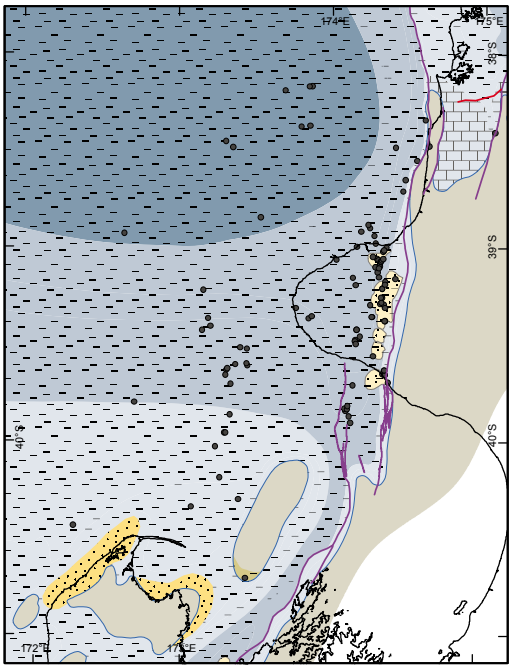
carbonates were dominant. In the Early Miocene, the first effects of the modern plate boundary were seen with the influx of an increasingly large volume of clastic sediments. Andesitic volcanism began in the Middle Miocene and continues to the present day. Turbidite systems developed from Late Eocene time (Tangaroa Fm.). Shelf progradation during the Neogene was mud-dominated, but punctuated by discrete turbidite depositional phases in the Late Oligocene (Tariki Sandstone), Middle Miocene (Moki Formation), Late Miocene (Mt Messenger Formation), and Pliocene (Mangaa Formation). The deep-water province contains a major channel and turbidite system. From the Pliocene, high rates of uplift and erosion in the hinterland led to the progradation of the shelfal clinoforms of the Giant Foresets Formation. Migration of the developing Neogene subduction complex resulted in rifting of the North Taranaki Graben during the Pliocene.

SUB-BASINS

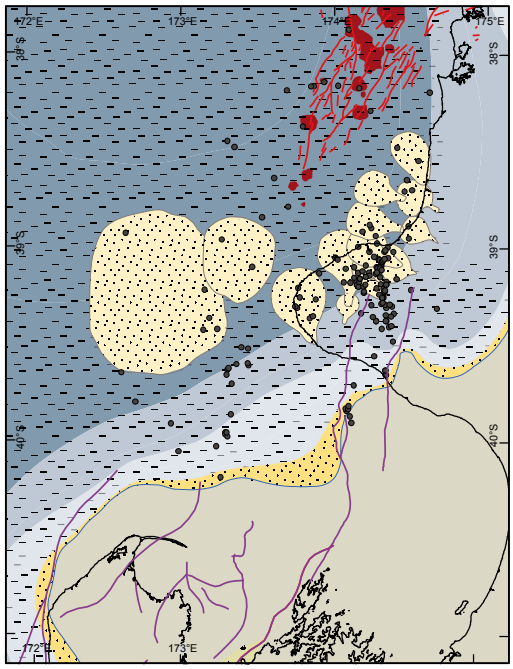
The Northern Graben is a 50 km wide Neogene depocentre, bounded by the Cape Egmont and Turi fault zones. Source rocks are mixed gas- and oil-prone and generally mature. Buried volcanoes and volcaniclastics are prospective as shown by the Kora oil discovery. Cenozoic submarine mass flow sand systems are present in stacked systems and prospective. Other significant leads include fault block plays along both margins. The Central Graben lies mostly over and south of the Taranaki Peninsula between the faulted Cape Egmont high in the west and the Manaia anticline in the east. This deep graben contains thick, mature source rocks. Possible plays include structural closures in the Cretaceous-Paleogene and stratigraphic traps in Neogene formations. The Taranaki Fault is the major reverse fault forming the eastern margin of the basin.

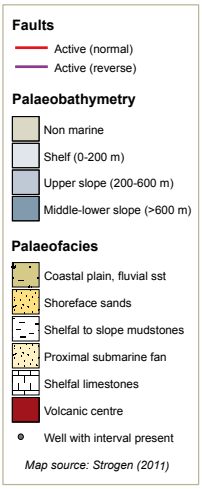
Neogene movement has deformed sedimentary rocks beyond the hanging wall in the complex Tarata Thrust Zone. Deformation has formed a wide range of traps at multiple levels. To the south, the zone of deformation continues in the comparable Waimea and Flaxmore faults. The Western Stable Platform has undergone limited or no Neogene deformation. It includes the Maui Field, and the recent Tui discoveries, which have confirmed generation and prospectivity. Paleogene shoreface sandstone are likely to be prospective elsewhere in the area. Beyond the present shelf edge, the Cretaceous Taranaki Delta is a thick unit of marginal to marine facies, likely to contain both source and reservoir facies. Northwest of the delta front, laterally equivalent facies of presumed mid-Cretaceous age are inferred to include marine shale source rocks and possible Cretaceous turbidites.

Oligocene (27 Ma) paleogeography



Late Miocene (10 Ma) paleogeography



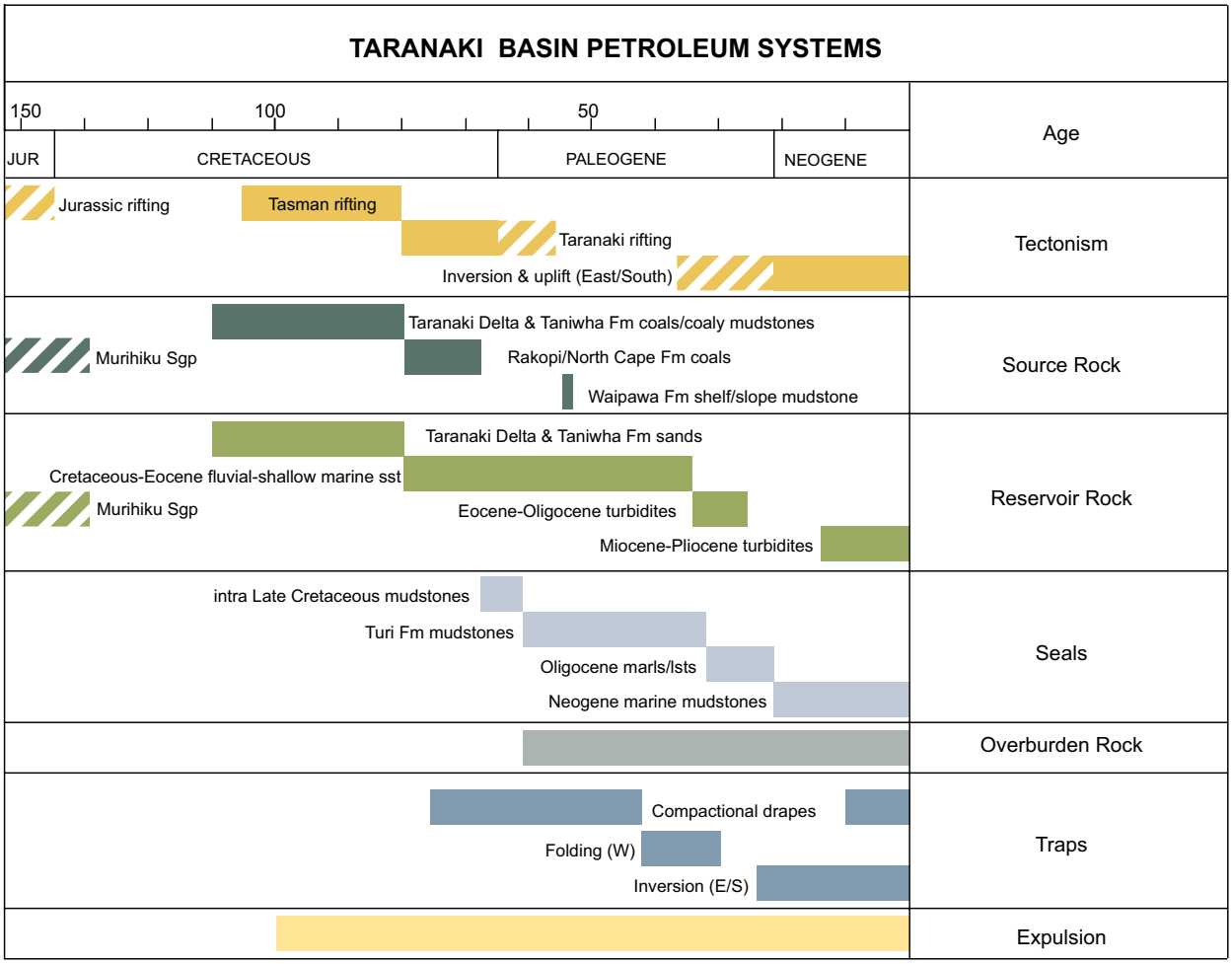


SOURCE ROCKS AND GENERATION

Modelling of source rocks in the shelf and onshore parts of the basin suggests that about 1,600 billion bbl of oil and 2,400 tcf of gas have been expelled by Cretaceous to Eocene source rocks. No modelling has yet been attempted for the sequence beyond the shelf edge. The full potential has yet to be assessed.

The majority of Taranaki oil, primarily waxy crude of about 45° API, has been geochemically typed to Late Cretaceous to Paleogene coaly source rocks. Paralic facies accumulated on a broad, landward-migrating shelf and coastal plain, where extensive coal swamps developed. Rapid Neogene burial has brought these rocks to depths where they are mature and expelling both oil and gas today. In the deep water province, much of the Rakopi Formation remains at, or just above, the present oil expulsion window. Paleocene and Eocene coaly source rocks are generally more gas-prone than those of the Late Cretaceous, but have sourced commercial accumulations of oil in McKee and other onshore fields.
They are modelled to be mature and expelling mostly gas in deep onshore and nearshore kitchens and may be the source of gas in Kapuni and Pohokura fields.

The Waipawa Formation, a thin Late Paleocene organic-rich marine shale, is present around much of New Zealand. Oil from the Kora discovery in the Northern Graben has been geochemically typed to the Waipawa Formation, suggesting the formation is both present and mature in this area. Older pre-Late Cretaceous source rocks may be present in parts of the basin, especially in the deep water province, but are poorly understood. These include Mid Jurassic to Early Cretaceous coal measures of the Murihiku Supergroup, and the mid Cretaceous Taniwha Formation. Early parts of the basin fill in the deep water province could also include Cretaceous marine source rocks.



TARANAKI BASIN PLAYS

Taranaki Basin is characterised by its variety of play types, mostly structural in nature. The majority of wildcats to date have targeted anticlinal or four-way dip closures. The mechanisms that produced these features can be radically different due to the basin's complex tectonic history. A Miocene compressional phase following Late Cretaceous rifting produced most of the structures currently trapping hydrocarbons at several stratigraphic levels with the multiple tectonic events leading to accumulations of oil above gas. Other play concepts are associated with half-graben fill, submarine fan systems, buried volcanic edifices and diagenetic traps.

THRUST FEATURES

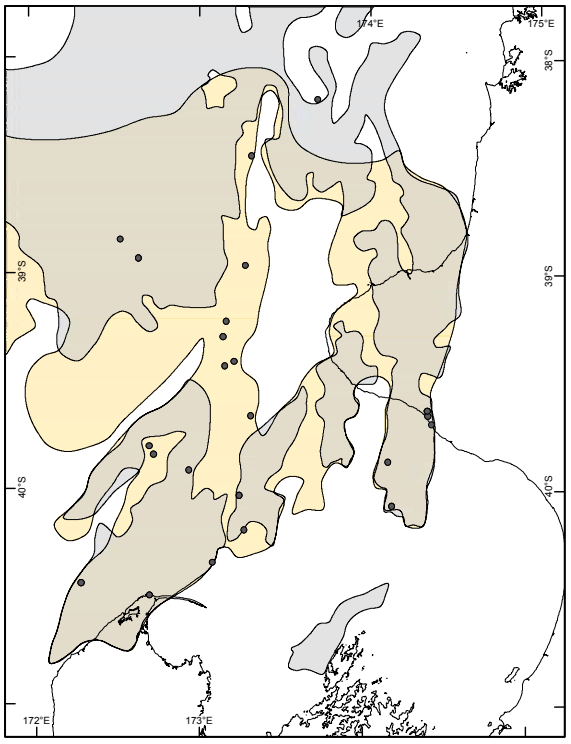
There are a number of plays along the eastern side of the Taranaki Basin associated with thrusting along the Taranaki Fault which has formed structures making up the Tarata Thrust Zone. Traps along the Tarata Thrust Zone are composed of relatively thin sheets of detached allochthonous material thrust in a westward direction. Mesozoic rocks have been thrust into Tertiary sequences forming anticlinal structures ahead of the older rocks, dislocation and repetition of Oligocene and Miocene rocks, or anticlines below the Taranaki Fault.

The producing McKee, Tariki, Ahuroa, Waihapa and Ngaere fields are assigned to this play category. Thrust traps have been the primary target of several recent successful exploration programs in the eastern Taranaki Basin, including wells drilled at Rimu-1, Huinga-1, Tuihu-1, Kauri-1 and Kahili-1. In these thrusts there are often repeated sequences below the thrust basement providing a separate play and accumulations.

INVERSION STRUCTURES

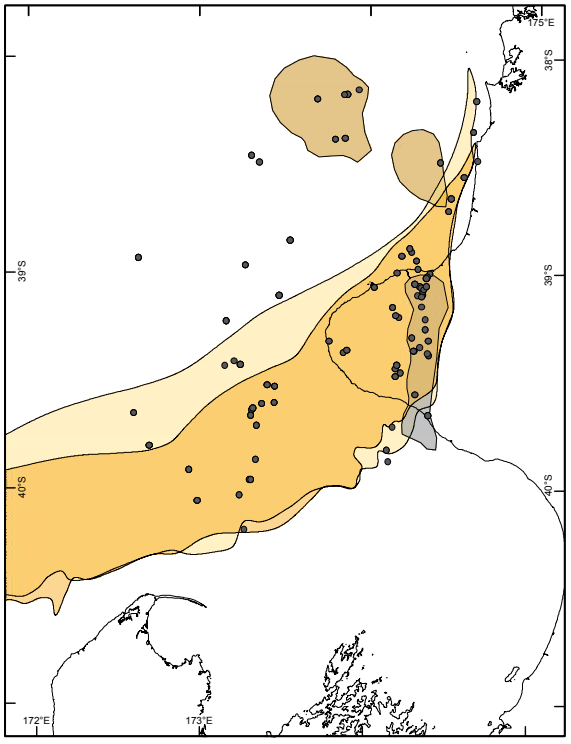
The largest fields discovered so far in the Taranaki Basin are trapped in inversion structures. Crustal shortening during the Miocene uplifted and inverted sub-basins in the eastern and southern parts of the Taranaki Basin, causing reactivation and reversal of movement along many of the basin's extensional faults. The resulting features are generally asymmetric with more steeply dipping beds to the west. The developed and undeveloped fields at Kapuni, Mangahewa, Kupe, Maari and Pohokura fall into this category.

Cretaceous play fairways



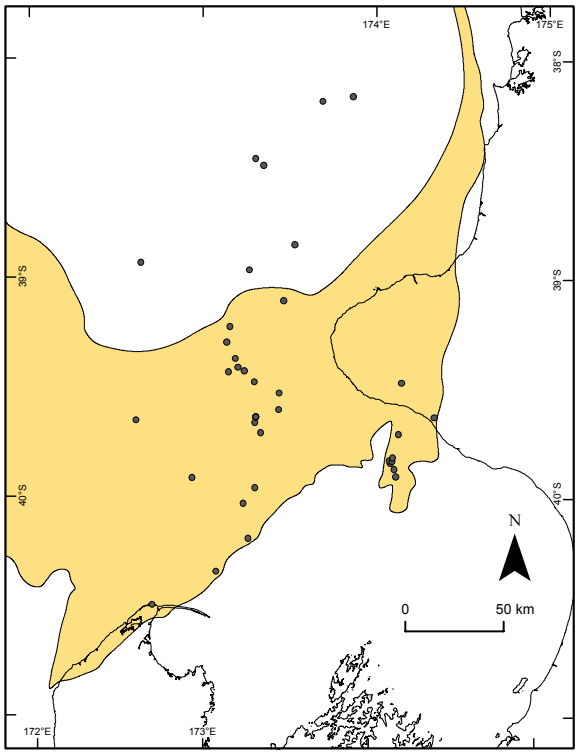
Map data source: Strogen (2011)

Eocene - Oligocene play fairway s

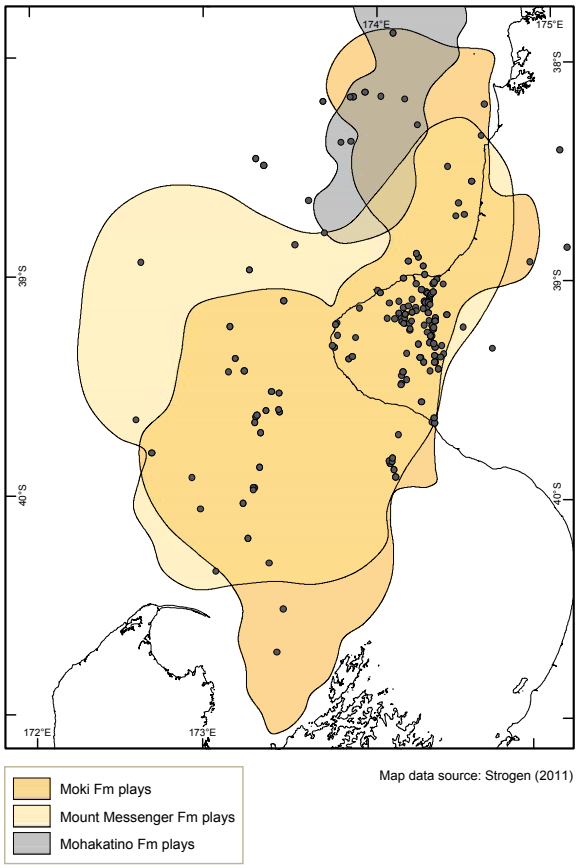


Map data source: Strogen (2011)

Paleocene play fairways



Miocene play fairways



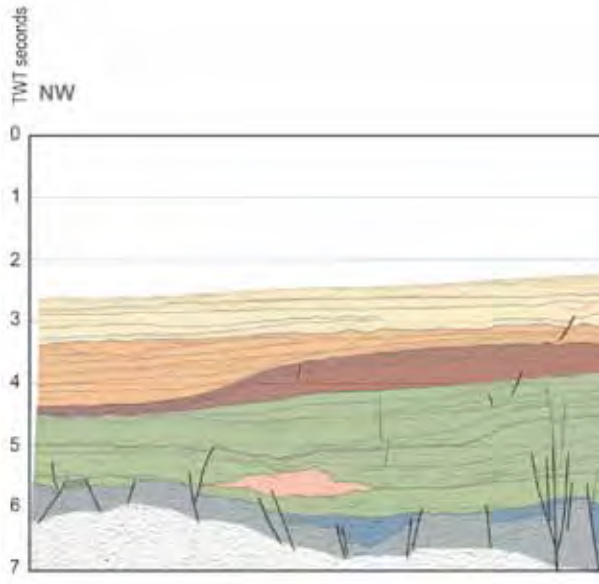
While the Maui structure is similar, it is mainly a drape structure over a fault block but with a steeper eastern limb.

Due to the early exploration successes in drilling this type of trap, most of the obvious inversion structures in the basin have been penetrated during the past five decades. Future exploration of this play concept will involve step-out wells on these features to delimit any traps with stratigraphic or diagenetic components along their margins.

EXTENSIONAL STRUCTURES

Post-compressional relaxation established a second extensional phase in the Taranaki Basin. Some normal faults that formed during Cretaceous and Paleocene rifting and experienced reverse movement in the Miocene were again subjected to a normal sense of movement in the Pliocene. Miocene uplift and subsequent Pliocene extensional tilting and block faulting created the traps at Kaimiro, Ngatoro and the recent Goldie and Windsor discoveries.

In the Northern Graben, where the inversion phase of regional tectonism had little effect, structures have been subjected primarily to extensional stresses since Late Cretaceous. Many basement-cored horst and tilted fault blocks adjacent or in the source rock kitchen depocentres have not been explored in this part of the basin. Compaction and drape of younger sediments over these highs may play an important role in trapping hydrocarbons in this part of the basin.



VOLCANIC EDIFICES

During the Miocene, a belt of submarine stratovolcanoes developed along the axis of the Northern Graben. The structures are primarily andesitic in composition and consist of volcanic and lapilli tuffs and volcanic slump breccias. Pore geometry is complex and highly variable and reservoir distribution is vertically and laterally discontinuous. Many stratovolcanoes are overlain by 1,500 to 2,000 m of Pliocene sediments and have been displaced by later normal faulting.

Kora-1, a well drilled in 1988 on the margin of the Northern Graben, penetrated the flank of the Kora stratovolcano en route to a deeper Eocene target. The volcaniclastics tested 668 bopd of 35° API gravity oil. Although not deemed economically viable at the time because of lack of adequate top seal, Kora-1 demonstrated volcaniclastic reservoir porosities up to 30% and permeabilities as high as 300 mD.

The stratovolcanoes along the axis of the Northern Graben are more deeply buried by Pliocene sediments than at Kora, and seal may be improved. In addition, the intrusive nature of the volcanic pipes that supplied magma to the stratovolcanoes has uplifted the underlying Eocene and Early Miocene reservoirs into large domes. Both of these parameters enhance the trapping capabilities in and beneath the volcanic edifices.

HALF-GRABEN FILL

Many half-grabens developed in the Taranaki Basin during the Cretaceous-Paleocene rifting episode. Although formed during extensional tectonism, half-graben traps generally differ from rotated fault blocks in the basin as they contain more unconformities and stratigraphic elements. Uplift and rotation of basement-cored blocks with syn-rift sedimentation and episodic erosional and non-depositional events produced wedges of Cretaceous fluvial and marginal marine lithologies on the dip slopes. Although potential reservoirs in Cretaceous rocks have yet to be explored, the proximity of source to reservoir in these sequences is promising. Rakopi Formation coals are believed to be the source rocks for Maui oil.

SUBMARINE FANS

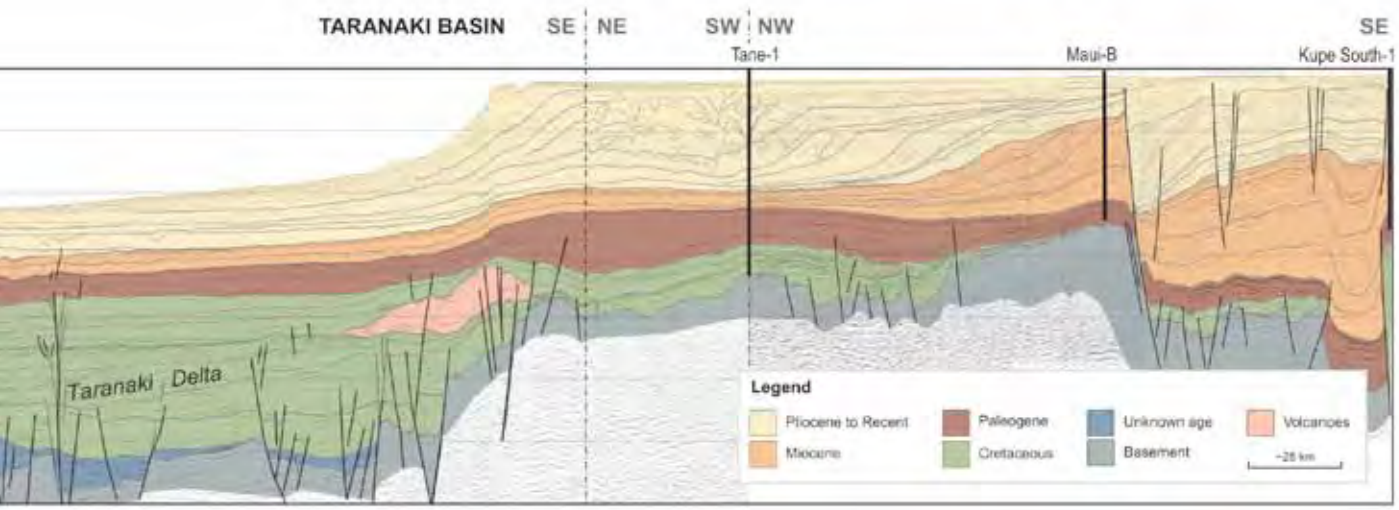
Historical exploration in Taranaki Basin concentrated on drilling anticlinal or fault block traps and the stratigraphic trapping capabilities of the Miocene turbidite sandstones were ancillary considerations. Neogene shelf and basin mass-emplaced sandstones outcrop along the Taranaki coastline and have now been penetrated by numerous wells. Miocene turbidites are the primary reservoirs in the Kaimiro and Ngatoro fields, and in Windsor-1 and Goldie-1.

Late Eocene turbidite systems have also been penetrated by several wells in the Northern Graben. Kora-1 recovered oil on test from the Tangaroa Formation, a relatively large fan complex north of the Kora structure. Thin turbidite sandstones of the Eocene Mangahewa Formation in Turi-1 exhibited good gas shows.

These wells indicate that slope and basinal fan sandstones are valid targets in the Northern Graben north of the fairway of productive Eocene paleocoast sediments that trend northeastwards through the Maui field to Pohokura. Post-rift deposition of Eocene sands is likely along the axis of the graben, but this depocentre has never been drilled. This play concept provides the most promise of large undiscovered commercial resources in the Taranaki Basin. The 2003 Karewa discovery is probably in Mangaa Formation turbidite sands of earliest Pliocene age.

DIAGENETIC TRAPS

Some of the most difficult plays to define in the Northern Graben are associated with diagenetically altered sandstones, where Miocene volcanism has altered the reservoir characteristics of Cenozoic rocks proximal to the intrusive centres. In Kora-1, located within 2 km of the igneous pipe complex that supplied magma to the Kora stratovolcano, the Eocene Tangaroa Formation exhibits much lower porosity and permeability values than penetrated by Kora-4 (20% porosity and 150 mD permeability) situated approximately 5 km from the pipe. This relationship demonstrates that a circlet of tight reservoir rock may exist near the igneous pipes that fed the stratovolcanoes, creating possible diagenetic traps for hydrocarbons in the more porous reservoir downdip. Determining the location of the transition between seal and reservoir in the same rock unit is difficult in the Northern Graben due the resolution of available seismic data, overlying volcanic edifices and lack of well control.



TARANAKI BASIN RESERVOIRS

The Taranaki Basin has producing reservoirs of Paleocene to Pliocene ages. Although there are no producing reservoirs of Cretaceous age, they remain prospective in parts of the basin. Gas-condensate and oil are found in Paleogene reservoirs, whereas Neogene reservoirs mainly trap oil. Stacked reservoirs are common in the Maui, Kapuni and Rimu/Kauri fields.

Rakopi Formation

Age: Late Cretaceous (85 to 75 Ma)

Facies: fluvial sandstones. Marginally marine-influenced in part

Distribution: Cretaceous valleys below the present shelf and onshore; coastal plain of 20,000 km^2 now below the present slope and basin floor

Porosity: up to 20%

Permeability: up to 18 mD

North Cape Formation

Age: Late Cretaceous (75 to 65 Ma)

Facies: dominantly marine, transgressive sandstones

Distribution: widespread across the basin except over the larger basement highs

Porosity: up to 26%

Permeability: over 500 mD

Farewell Formation

Age: Paleocene (65 to 55 Ma)

Facies: dominantly fluvial sandstones

Distribution: widespread across the basin except over the larger basement highs

Porosity: up to 20%

Permeability: several hundred mD

Fields: Kupe (gas), Maui (oil), Tui (oil)

Production rates: Maui (7,500 bopd), Tui (early production 45,000 bopd from horizontal wells)

Kaimiro Formation

Age: Early to Middle Eocene (55 to 45 Ma)

Facies: cyclic lower alluvial plain, coastal plain and marginal marine sandstones

Distribution: widespread across Taranaki shelf and onshore

Porosity: over 20%

Permeability: sub-mD to several Darcies

Fields: Maui (oil and gas-condensate)

Production rates: Maui (11,000 bopd)

Mangahewa Formation

Age: Middle and Late Eocene (45 to 34 Ma)

Facies: terrestrial and marginal marine sandstones

Distribution: widespread across Taranaki shelf and onshore

Porosity: 8 to 27%

Permeability: sub-mD to several Darcies

Fields: Kapuni (gas-condensate), Maui (gas-condensate), Mangahewa (gas-condensate), Pohokura (gas-condensate)

McKee Formation

Age: Late Eocene (38 to 33 Ma)

Facies: shallow marine transgressive sandstones

Distribution: Taranaki Peninsula

Porosity: average 17%

Permeability: 20 to several hundred mD

Fields: McKee (oil and gas)

Production Rates: McKee-1 (350 bopd)

Tangaroa Formation

Age: Late Eocene (38 to 33 Ma)

Facies: turbidite sandstones

Distribution: northwestern and central Taranaki shelf

Porosity: over 20%

Permeability: up to 100 mD

Fields: shows in Tangaroa-1 and Kora-1-4

Tariki and Matapo Sandstone members

Age: Oligocene (33 to 25 Ma)

Facies: submarine fan sandstones adjacent to upthrown blocks and marine glauconitic sandstones

Distribution: northern and eastern Taranaki Peninsula (Tariki); widespread across the basin (Matapo)

Porosity: up to 23%

Permeability: very variable, up to a few hundred mD

Fields: McKee (oil), Tariki-Ahuroa (gas), Rimu (oil and gas)

Tikorangi Formation

Age: Oligocene to earliest Miocene (33 to 23 Ma)

Facies: limestones and marls

Distribution: widespread across the entire basin

Porosity: fractures - variable

Permeability: highly variable

Fields: Waihapa-Ngaere (oil), Toko, Piakau, Kupara, Rimu

Production Rates: Waihapa (early production 10,000 bopd)

Moki Formation

Age: Middle Miocene (17 to 13 Ma)

Facies: basin floor fan turbidite sandstones

Distribution: southern and central Taranaki Shelf

Porosity: up to 26%

Permeability: variable, 100 mD in Maari

Fields: Maari (oil), Kaimiro, Ngatoro

Production Rates: Maari (40,000 bopd from 5 wells)

Mount Messenger Formation

Age: Late Miocene (11 to 9 Ma)

Facies: basin floor fan and slope fan turbidite sandstones

Distribution: northern coastal outcrops, central Taranaki Peninsula, and along the axis of the deep water part of the basin

Porosity: 14 to 25%

Permeability: 20 to 800 mD

Fields: Kaimiro (oil), Ngatoro (oil)

Urenui Formation

Age: Late Miocene (9 to 5 Ma)

Facies: slope sandstones

Distribution: inshore and onshore North Taranaki, offshore, south of Taranaki Peninsula

Porosity: more than 15%

Permeability: several hundred mD

Fields: Cheal (oil)

Mangaa Formation

Age: Latest Miocene to Early Pliocene (5 to 3 Ma)

Facies: seafloor turbidite fan sandstones

Distribution:within the North Taranaki Graben and extending along the axis of the deepwater part of the basin

Porosity: 15 to 25%

Permeability: up to 200 mD

Fields: Karewa (biogenic gas)

Matemateaonga Formation

Age: Latest Miocene to Early Pliocene (7 to 5 Ma)

Facies: shelf and marginal marine sandstones and conglomerates

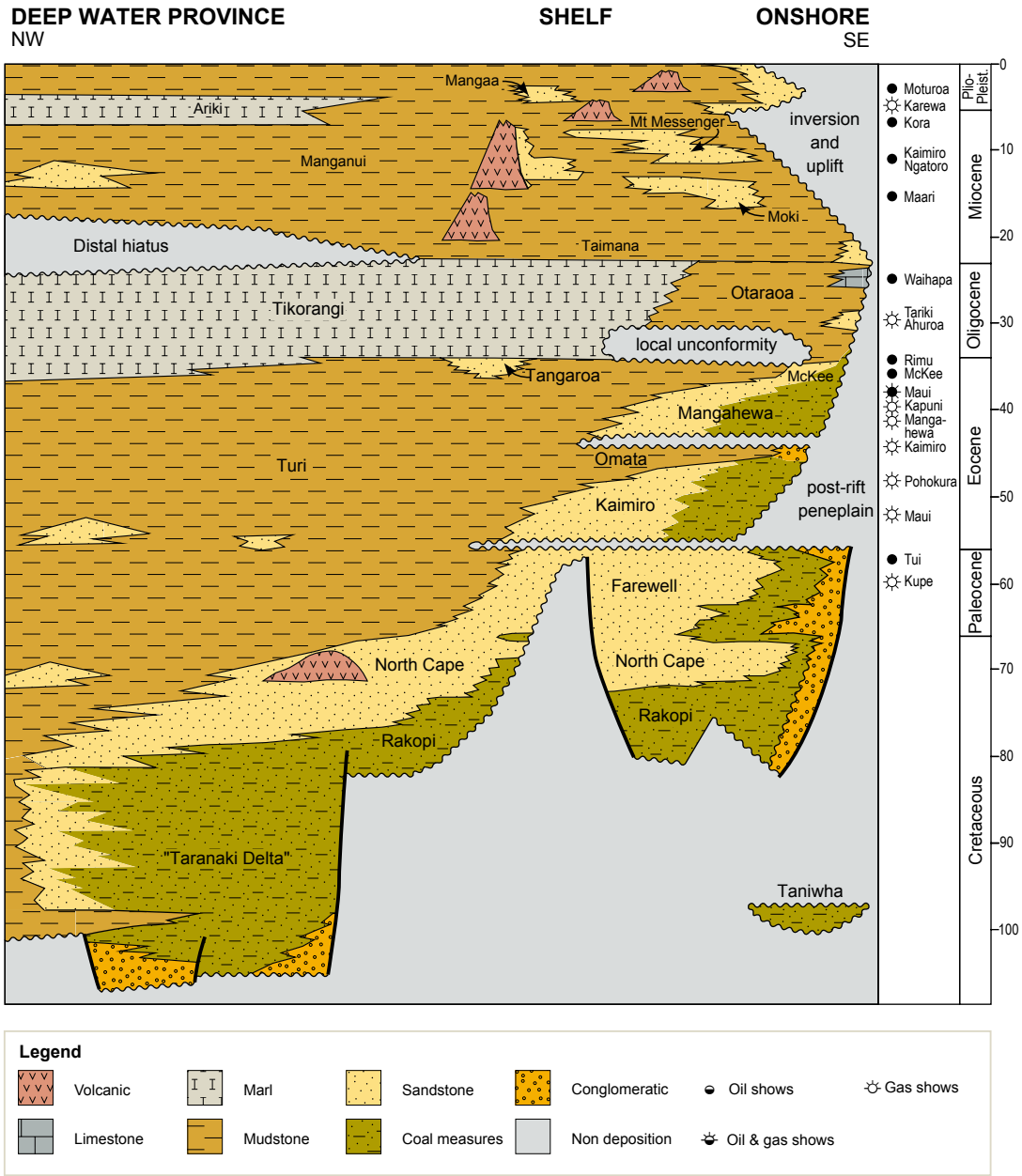
Distribution: Taranaki Peninsula and offshore to the south

Porosity: up to 25%

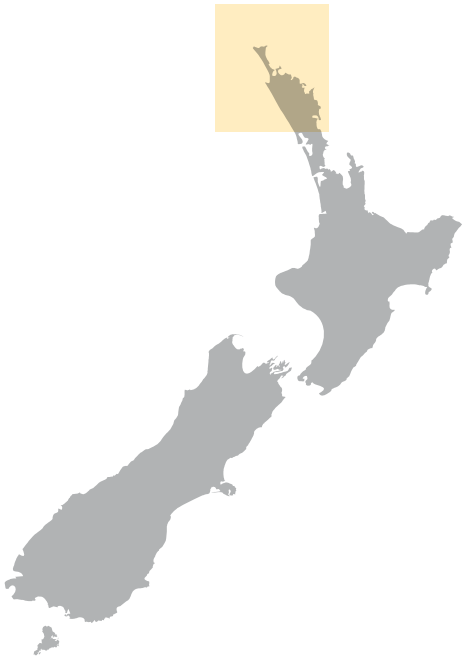
Permeability: up to 1,000 mD

Fields: Moturoa (oil)

Production Rates: Moturoa (at times up to 55 bopd)



Reinga-Northland Basin



BASIN SUMMARY

The Reinga Basin (including its Northland sector) is a large, little-explored basin that is geologically similar to the better known Taranaki Basin. In the Reinga Basin there is evidence of greater compression in the Eocene and Early Miocene, the presence of an allochthon close to the eastern margin, and some evidence of higher heat flow. The top of the oil expulsion window is modelled at 4.0 +/- 0.5 km below the seabed, implying a potential kitchen area of approximately 15,000 km^2 for Cretaceous source rocks, or a broader area if Jurassic source rocks are present. Oil and gas expulsion is modelled to occur from Late Cretaceous time, with significant expulsion predicted during and after the Eocene to Miocene folding and reverse faulting events that created structural traps.

PLAYS

- **Drape by younger sediments across upstanding basement blocks**
- **Eocene compressional structures**
- **Early Miocene compressional structures**
- **Transgressive on-lap against paleo-slopes**

EXPLORATION AND PRODUCTION

Wells

Six relevant wells: Waimamaku-1 and -2 onshore 1969 to 72; Waka Nui-1 offshore 1999; Karewa-1 (2003), Koromako-1 and Tarapunga-1 (2010) offshore at boundary with Taranaki Basin.

Hydrocarbons

Gas shows in Waimamaku-2 and reported in Koromako-1 and Tarapunga-1; small gas discovery at Karewa-1. Two gas seeps onshore. Satellite imagery shows three clusters of unassigned slicks offshore.

GEOLOGY

Jurassic marine beds and coal measures may be earliest basin-fill. Failed rift basin, transgression and on-lap from Early Cretaceous to Paleogene; Late Eocene contraction and erosion; Early Miocene back-arc subsidence, overthrusting and volcanism. Contiguous with Taranaki Basin. Up to 8,000 m of basin-fill.

Source Rocks

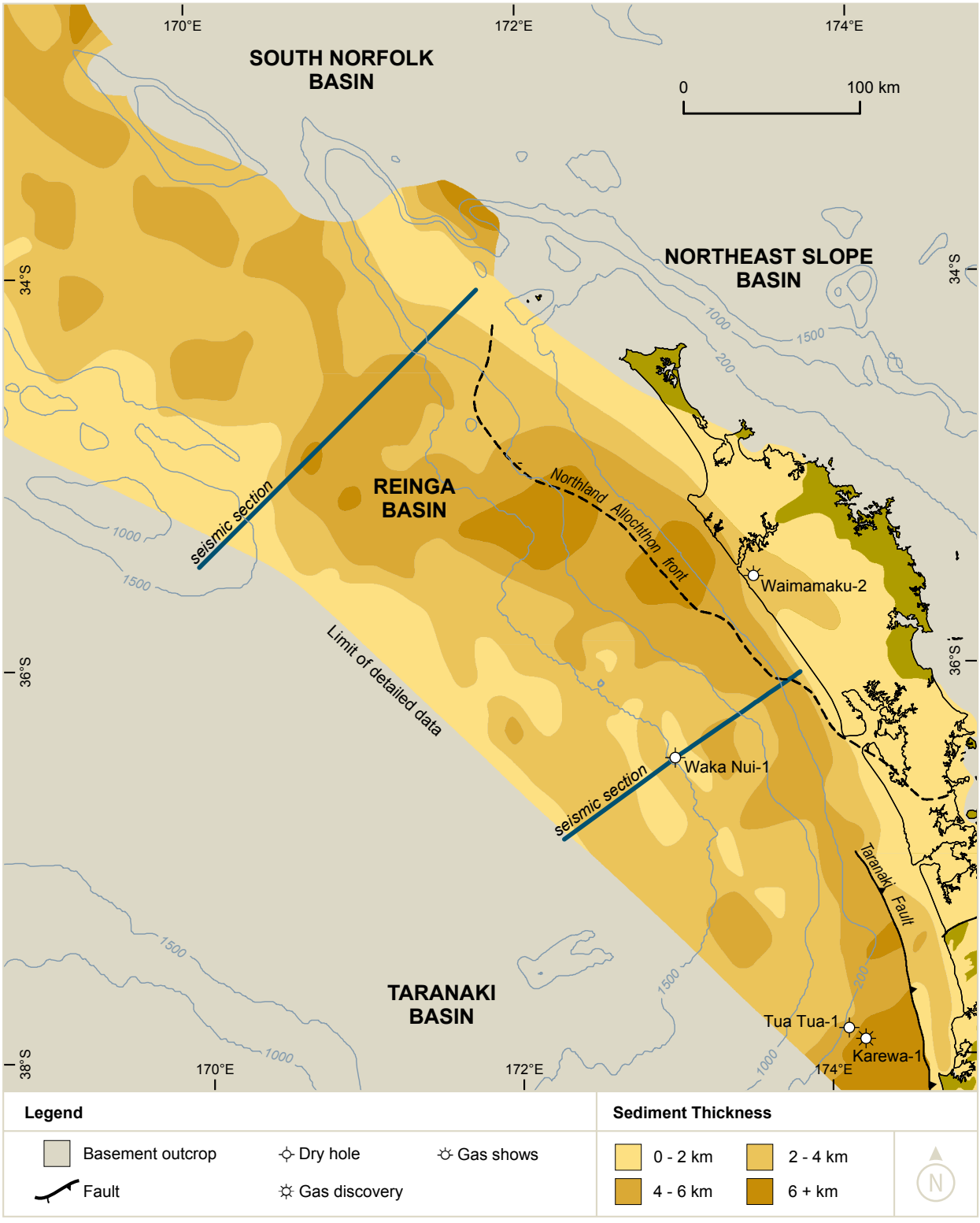
Jurassic to Eocene coal measures; Cretaceous to early Cenozoic marine mudstones including Waipawa Formation.

Reservoir Rocks

Late Cretaceous terrestrial, paralic and nearshore sandstones; Late Eocene turbidites related with erosion of the Reinga and West Norfolk ridges; Miocene volcaniclastics and turbidites.

Prospects

Strong geological similarities with the Taranaki Basin but virtually unexplored. Numerous anticlines in the north formed by Late Eocene contraction. Also includes a range of fault block highs and potential stratigraphic traps.


170°E
172°E
174°E
SOUTH NORFOLK
BASIN
0
100 km
NORTHEAST SLOPE
BASIN
34°S
1000
1500
200
REINGA
BASIN
seismic section
Northland Allochthon front
Waimamaku-2
36°S
Limit of detailed data
Waka Nui-1
seismic section
Taranaki Fault
TARANAKI
BASIN
Tua Tua-1
Karewa-1
38°S
Legend
Basement outcrop
Dry hole
Gas shows
Fault
Gas discovery
Sediment Thickness
0 - 2 km
2 - 4 km
4 - 6 km
6 + km
N

BASIN HISTORY

EXTENT

The Reinga Basin extends northwest for about 750 km, from basement outcrop in onshore Northland to the Norfolk Ridge, at the limit of New Zealand territory. It has an area of about 170,000 km^2, and is contiguous with Taranaki Basin in the south and South Norfolk Basin in the northeast. The Reinga Basin is bounded by the West Norfolk Ridge and its now-buried extension to the southwest.

EXPLORATION

The first offshore seismic data from the basin were acquired in the 1960s. In the 1970s, Mobil, Gulf and Shell each completed a regional reconnaissance survey around New Zealand, including lines across the Reinga Basin. In 1982 Geco acquired a good quality speculative survey southwest of the Northland Peninsula. Research surveys by GNS Science and Geoscience Australia included seismic acquisition and dredging, which in the early 1990s, showed that the basin extended at least as far northwest as the limit of New Zealand territory and indicated the presence of source and reservoir rocks. The basin is now covered at reconnaissance level by industry-standard seismic data, some only recently acquired. Conoco held a large permit off Northland in the 1990s and acquired about 8,000 km of 2D seismic. A 5,400 km 2D survey was acquired by Crown Minerals and CGGVeritas in 2009.

Four relevant wells have been drilled, two of which (Waimamaku-1 and -2) predated recognition that the onshore Cretaceous and Paleogene sequence is mainly allochthonous.

Waimamaku-2 drilled through thrust sheets of the Northland Allochthon before reaching thin autochthon overlying earliest Cretaceous Murihiku Supergroup rocks. Minor gas shows were encountered but the drillhole was not a valid test of a structural or stratigraphic trap.

Conoco drilled Waka Nui-1 in 1,490 m of water on the dip slope of a Northland Ridge tilted fault block. Coal measures that overlie non-reflective basement were anticipated to be Cretaceous and analogous to Taniwha Formation of Taranaki. However, they were found to be Middle Jurassic, and correlated with the onshore Murihiku Supergroup. At the well site they are overlain by Paleocene conglomerate and sandstone. The well, drilled some distance down dip from the crest on the western side away from the main depocentre, found no significant hydrocarbons.

Conoco, Inpex and Todd Energy drilled Karewa-1 in 2003 to a TD of 2,023 m, discovering dry gas in Mangaa Formation turbidite sands draped across a Miocene volcano. The gas appears to be biogenic, although some higher hydrocarbons were reported.

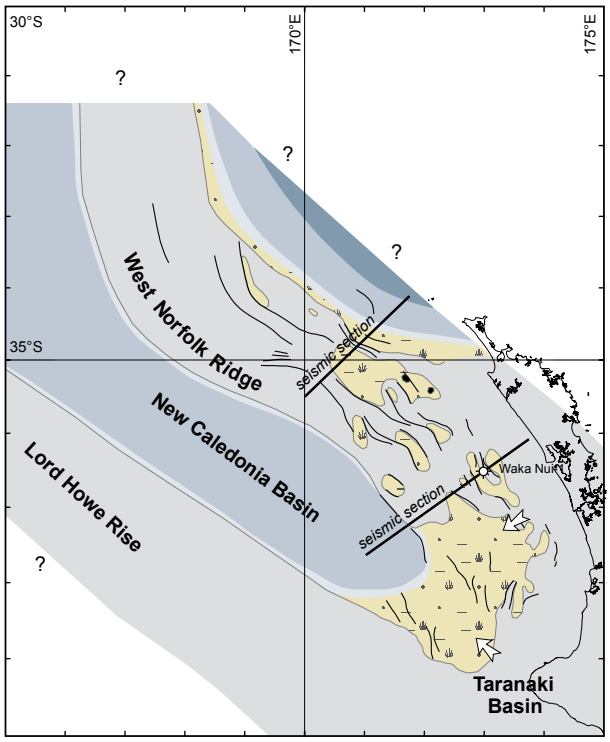
In 2010 Origin Energy drilled Koromako-1 and Tarapunga-1 wildcat wells to a TD of about 1,000 m targeting Late Miocene and Pliocene fan complexes. Both wells had minor gas shows, but no intervals of immediate economic significance were identified.

GEOLOGICAL HISTORY

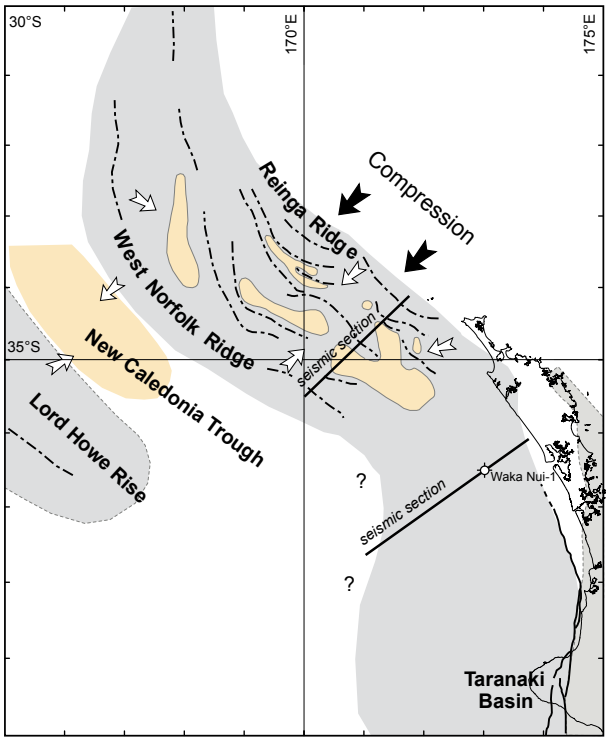
The Reinga Basin occupies a site of late Mesozoic back-arc extension close to the Gondwana margin. Jurassic and earliest Cretaceous Murihiku marine and non-marine rocks, as drilled in Waimamaku-2 and Waka Nui-1, may represent the earliest basin-fill. The Northland margin was a northwest-verging Mesozoic back-thrust to the Gondwana subduction zone, analogous to the Taranaki Fault, the margin of the modern Taranaki Basin.

Basin evolution of the Reinga Basin from the Cretaceous can be subdivided into six phases. (See the papers by Bache et al in the Suggestions for Further Reading for a more detailed explanation of the basin's evolution.)

Phase 1: End of rifting (late Cretaceous)



Phase 3: Late Eocene contraction



Legend

- Sedimentary transfers
- Major faults
- Taranaki Fault
- Emergent highs/Anticlines
- Land or non deposition
- Coastal plain/Fluvial
- Detritus (sand ?)
- Shallow marine/shelf
- Bathyal
- Abyssal

0 150 300 km

N

Phase 1

Cretaceous rifting and the formation of northeast-southwest oriented elongate sub-basins.

Phase 2

Late Cretaceous and Paleogene subsidence and marine transgression. Early rift sediments were buried beneath a marine sequence.

Phase 3

Late Eocene compression and folding related to initiation of subduction by reactivation of the ancient Gondwana margin to the northeast. The Reinga and West Norfolk ridges were uplifted and eroded while detrital products were deposited in the centre of the Reinga Basin. This phase characterizes the northwestern sector of the Reinga Basin and is not clearly observed farther south in the Northland or Taranaki basins.

Phase 4

Oligocene to Early Miocene tectonism associated with subduction included regional subsidence that led to the creation of accommodation space in the centre of the basins, southwest-directed emplacement of Northland Allochthon thrust sheets into the basin, and eruption of back-arc andesitic stratovolcanoes along and close to what is now the Northland Peninsula.

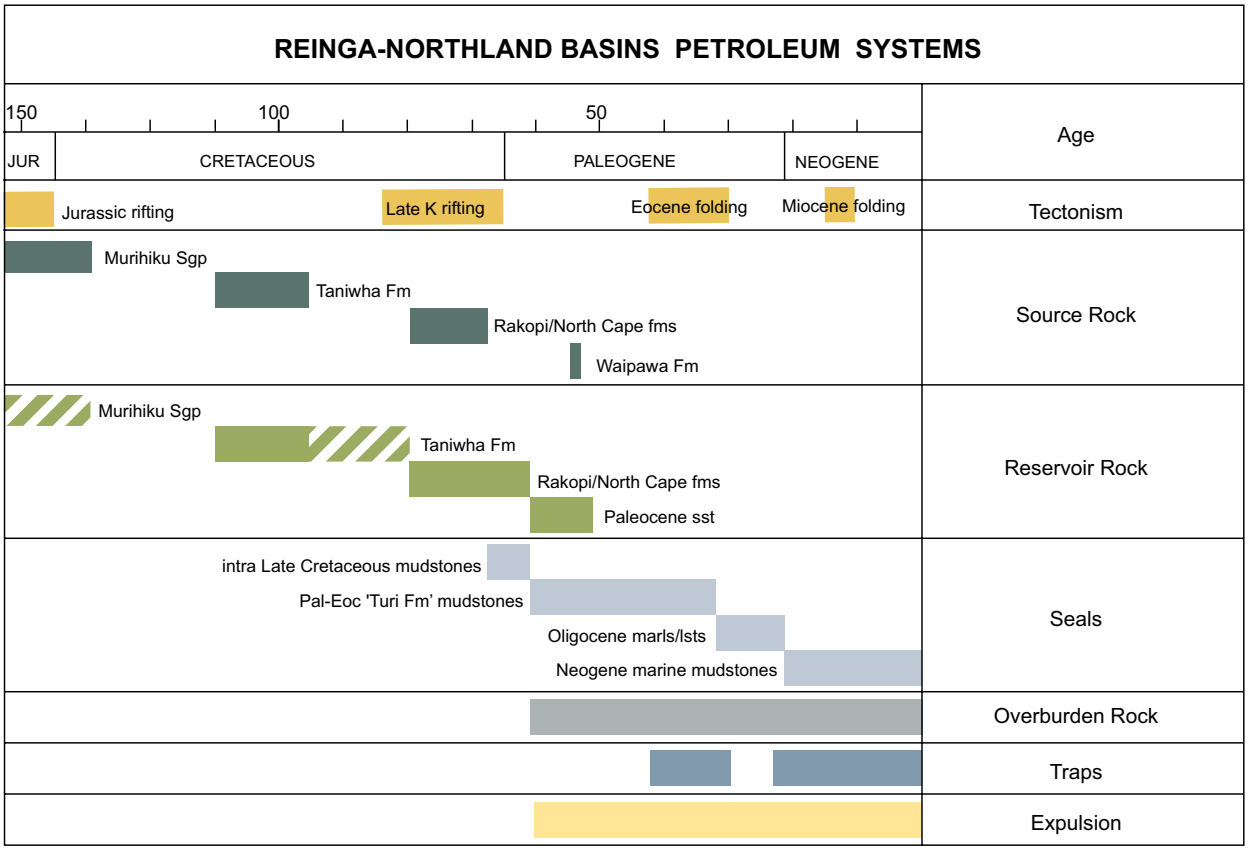
Phase 5

Uplift of the Wanganella Ridge occurred at the end of the early Miocene. In the southeastern part of the Northland Basin, the Mohakatino volcanic center was most active from middle to late Miocene. Phases 4 and 5 are interpreted to have occurred during the roll-back of the Kermadec Trench and the opening of the Norfolk and South Fiji basins. At this time dextral slip on the Veining-Meinesz Fracture Zone has been inferred to displace the Three Kings Ridge along the Reinga Ridge.

Phase 6

Late Miocene to present day quiescent period. During this time shelf and foreset beds developed in the Northland and Taranaki basins and hemipelagic ooze has been deposited in distal parts of the Reinga Basin.

There have been no petroleum exploration boreholes drilled in the Reinga Basin, and only one in the offshore Northland Basin (Waka Nui-1). As a consequence, our understanding of potential petroleum systems elements in the Reinga and Northland basins draws heavily on correlations with the Taranaki Basin.



SOURCE ROCKS AND GENERATION

Four sedimentary formations that may be present in the Reinga Basin contain potential source rocks. These are, from oldest to youngest: Murihiku Supergroup, Taniwha, Rakopi, and Waipawa formations.

High-amplitude seismic facies observed in the Reinga Basin near the base of grabens and the top of horst structures are interpreted as coaly facies. This high amplitude seismic unit is observed at the top of the horst penetrated in the Waka Nui-1 well, where Jurassic Murihiku Group mudstone, interbedded sandstone, siltstone, and coal were intersected between 3,544.5 and 3,682 m.

Seismic units in the Reinga and Northland basins are interpreted as lateral equivalents of the Taniwha, Rakopi and North Cape formations in the Taranaki Basin, and where deposited in similar settings may, contain coaly source rocks. Late Cretaceous carbonaceous sandstones and mudstones, typically associated with coal measures, have been dredged from two sites on the West Norfolk and Reinga ridges.

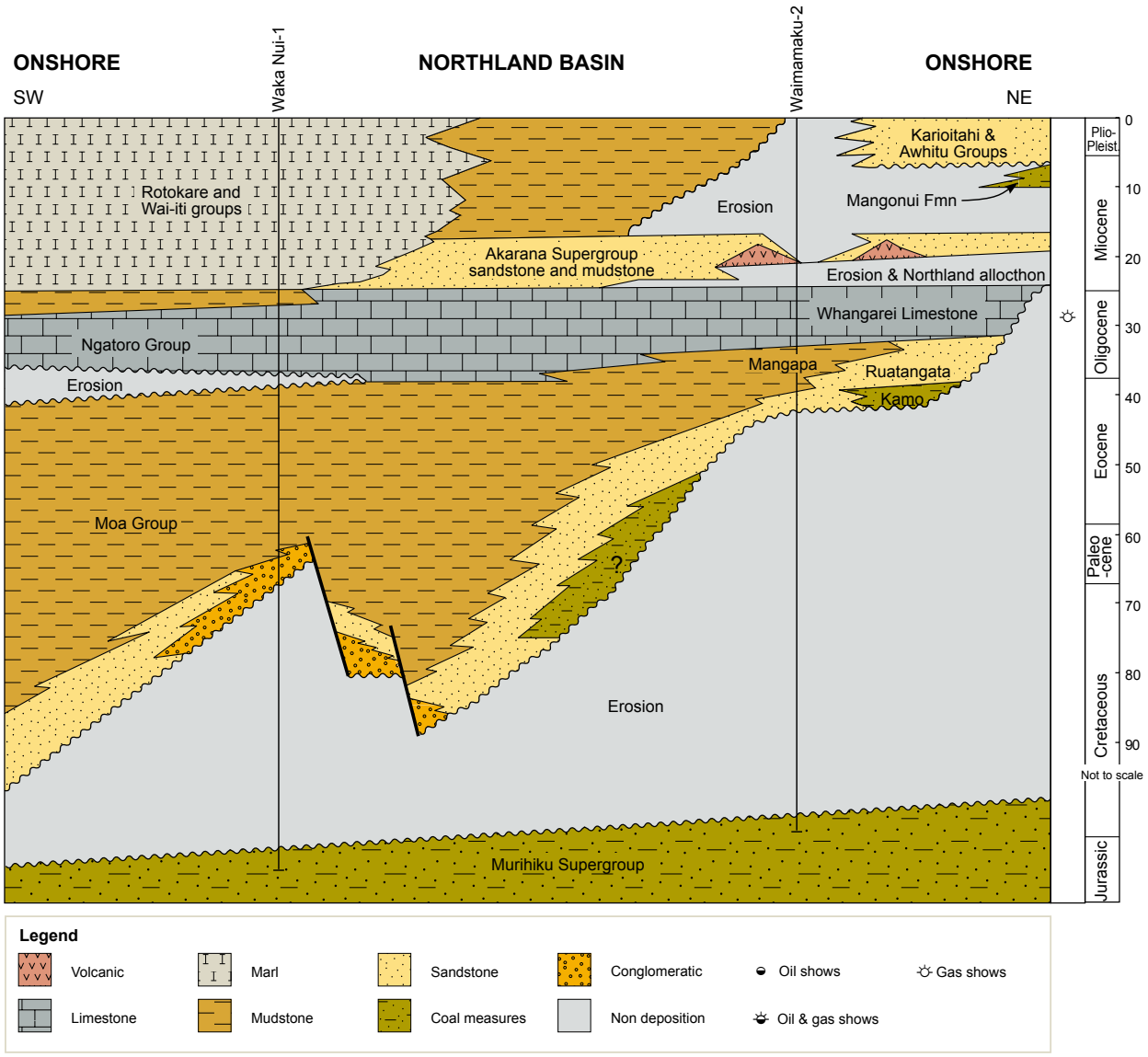
Waipawa Formation black shale of Paleocene age was drilled in Waka Nui-1. The characteristic high-amplitude signature corresponding to the formation diminishes to the north, so it is difficult to assess the presence of the Waipawa Formation in the Reinga Basin.

Maturation models predict that Cretaceous coaly source rocks within basal strata would begin to generate and expel petroleum in early Cenozoic time and expulsion would continue to the present day. The top of the oil expulsion window is modelled at 4.0 +/- 0.5 km below the seabed, implying a potential kitchen area of approximately 15,000 km^2 for Cretaceous source rocks, or a potentially broader area if Jurassic source rocks are present. Significant oil and gas expulsion is predicted to occur during and after the Eocene to Early Miocene folding and reverse faulting events that created structural traps.

REINGA-NORTHLAND BASIN PLAYS

A range of play types is recognised. In the Northland sector, where seismic coverage is relatively dense, about 50 four-way dip closures have been mapped at levels from basement to top-Miocene. Over the remainder of the basin the available seismic coverage is a reconnaissance grid with a line spacing of about 50 km, and is too coarse to map potential trap structures. However, a number of single line leads have been interpreted.

A common play type is drape by younger sediments across upstanding basement blocks. Differential compaction forms four-way dip closures. Some Eocene compressional structures are present in the central part of the basin but they are more significant in the northwest where major folds have been identified. Southwest directed convergence associated with Early Miocene subduction at the plate boundary may also have formed compressional structures. Emplacement of the Northland Allochthon resulted in deformation of the autochthonous rocks ahead of, and below, the thrust sheets.

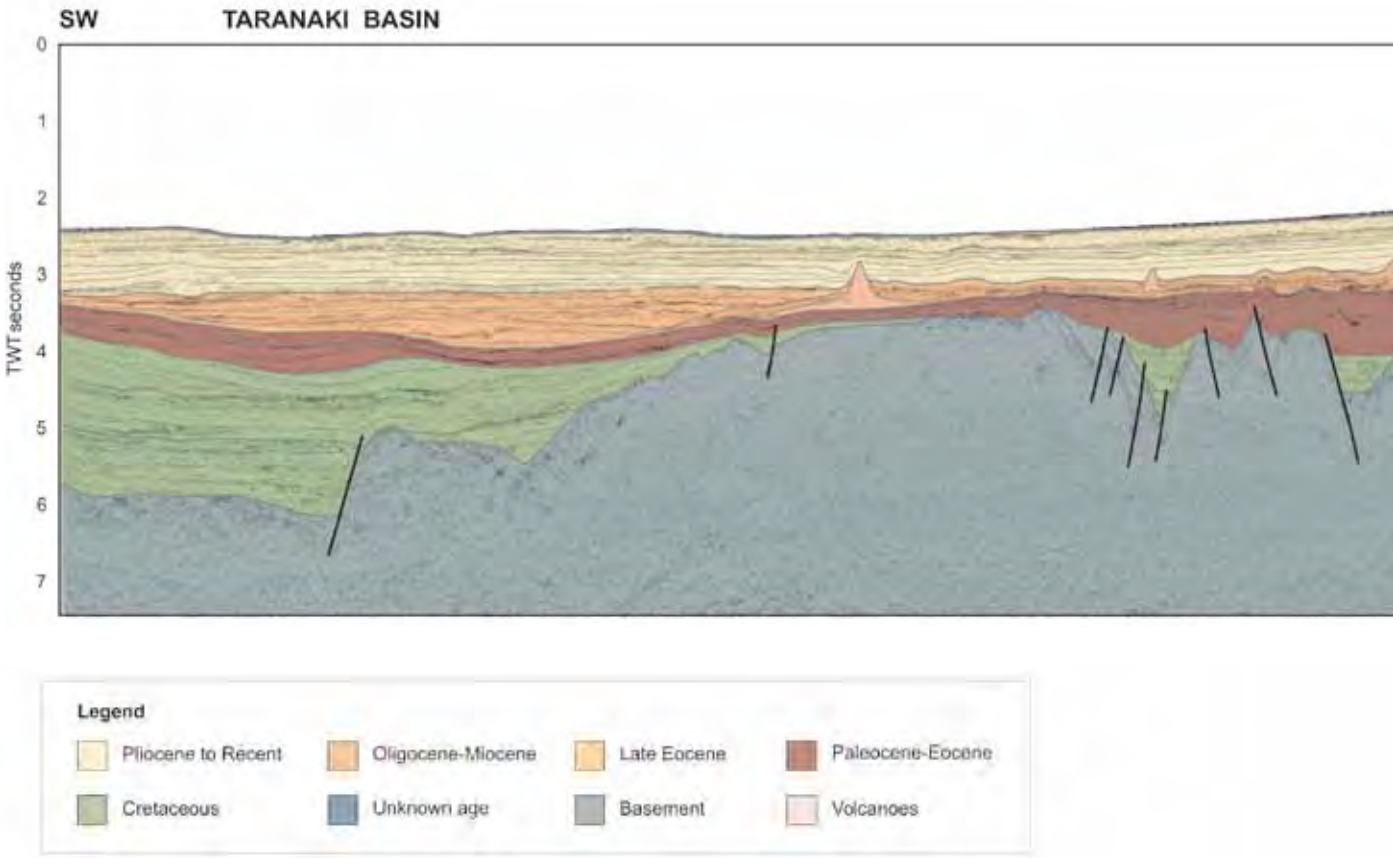


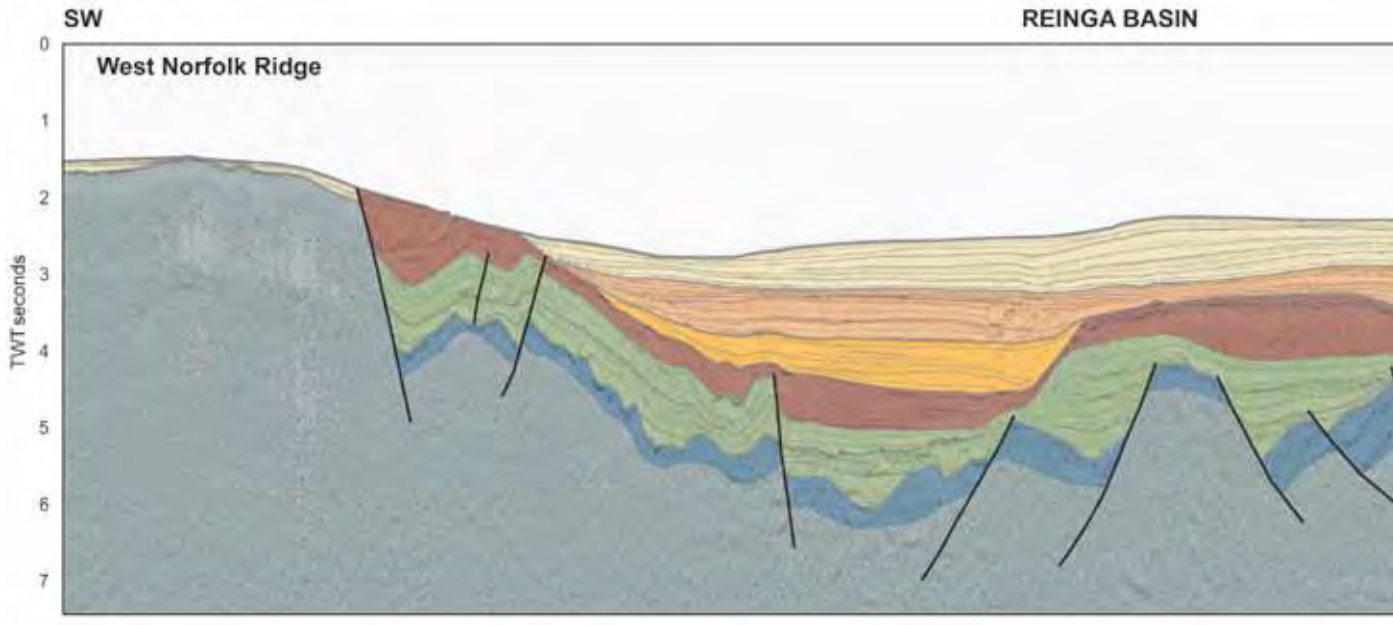
Loading by the thick thrust sheets is likely to have caused crustal flexure and roll-over, with the potential to form large traps. The area at the allochthon front is also the area of deepest burial of potential source rocks. The potential for stratigraphic trapping includes transgressive on-lap against paleoslopes, channels, and turbidite fan mounds.

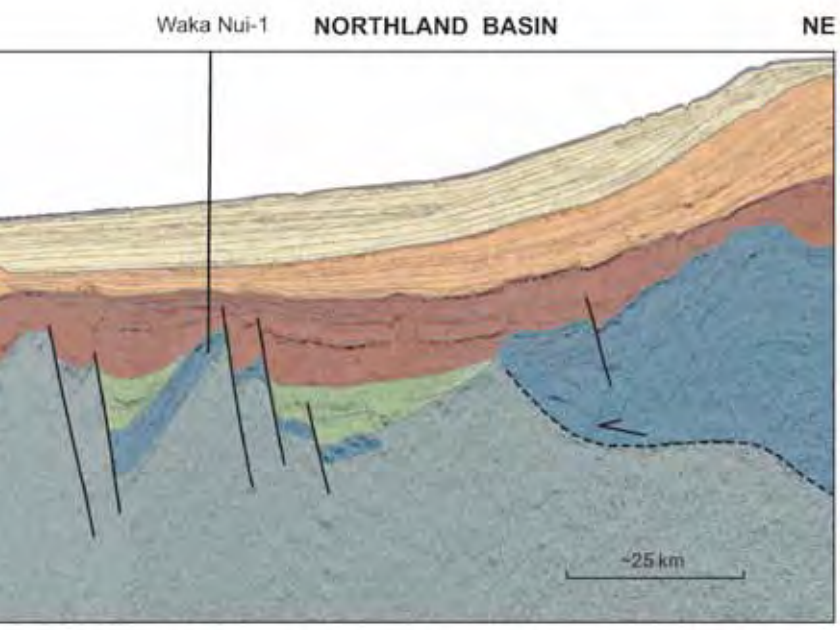
REINGA-NORTHLAND BASIN RESERVOIRS

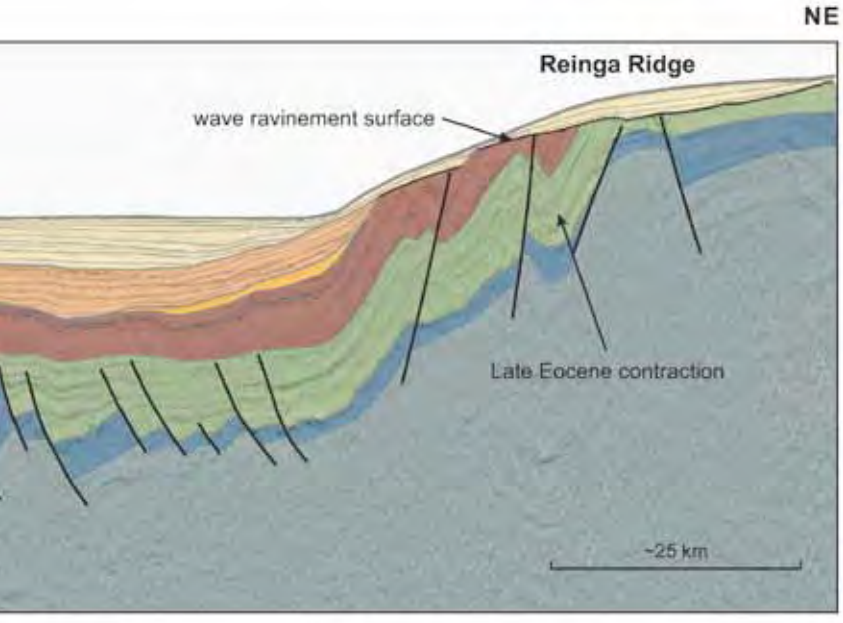
Paleocene transgressive sands were found in Waka Nui-1. The Karewa-1 reservoir is Pliocene turbidites. Both facies are likely to be present at other levels in the basin-fill. Late Cretaceous to Eocene rocks are inferred to include shallow marine and non-marine facies as parts of the overall transgressive succession. Other possible reservoir facies are Neogene turbidite sandstones, Early Miocene volcaniclastics (as in Kora-1), and Eocene to Miocene fractured carbonates.

Late Eocene contraction (Phase 3) is recorded in the Reinga Basin by folding of older seismic units, subaerial erosion of the West Norfolk and Reinga ridges and deposition of sediment in sub-basins between uplifted structures. In sub-basins high-amplitude seismic facies within the seismic unit could represent reservoir sandstones; however this seismic unit has never been sampled. Oligocene to Early Miocene deformation (Phase 4) also led to a supply of sediment from wave abrasion, which may have been deposited as reservoirs in deep-water fans. In summary, both contractional phases recorded in the Reinga Basin (Phases 3 and 5) may have resulted in contemporaneous deposition of reservoir facies, and a wide range of structural traps were created at various deeper stratigraphic levels.


SW
TARANAKI BASIN
TWT seconds
0
1
2
3
4
5
6
7
Legend
Pliocene to Recent
Oligocene-Miocene
Late Eocene
Paleocene-Eocene
Cretaceous
Unknown age
Basement
Volcanoes


SW
REINGA BASIN
West Norfolk Ridge
TWT seconds
0
1
2
3
4
5
6
7


Waka Nui-1
NORTHLAND BASIN
NE
~25 km


NE
Reinga Ridge
wave ravinement surface
Late Eocene contraction
~25 km

East Coast Province



BASIN SUMMARY

The East Coast Province covers at least 180,000 km^2, and incorporates the Raukumara, East Coast, and Pegasus basins. The geology of the province varies from structurally simple to complex. Prior to the Neogene, all three areas have had a common geological history. Since the Early Miocene, and the inception of Pacific-Australia plate subduction tectonics in the province, some regions have been deformed more greatly than others, particularly the East Coast Basin. Insights from the little deformed Pegasus and Raukumara basins provide insights into the geology of the East Coast Basin.

PLAYS

- **Fault-bounded anticlines**
- **Stratigraphic pinchouts**
- **Turbidite**
- **Fractured limestones**
- **Tight oil/gas resource plays - onshore**
- **Thrust anticlines**
- **Drape over thrust features**
- **Gas hydrates**

EXPLORATION AND PRODUCTION

Wells

Over 300 known onshore oil and gas seeps occur in the province, attesting to active petroleum systems. More than 40 wells have been drilled since 1955, all of which are located in the East Coast Basin. Although only three of these wells are located offshore – the most recent being drilled in 2007 – all encountered significant gas shows. The Pegasus and Raukumara basins remain untested. A sub-commercial high-pressure dry gas discovery was made in 1998 at Kauhauroa in northern Hawke's Bay. Small historical oil production has occurred north of Gisborne, near the Waitangi and Totangi oil seeps.

GEOLOGY

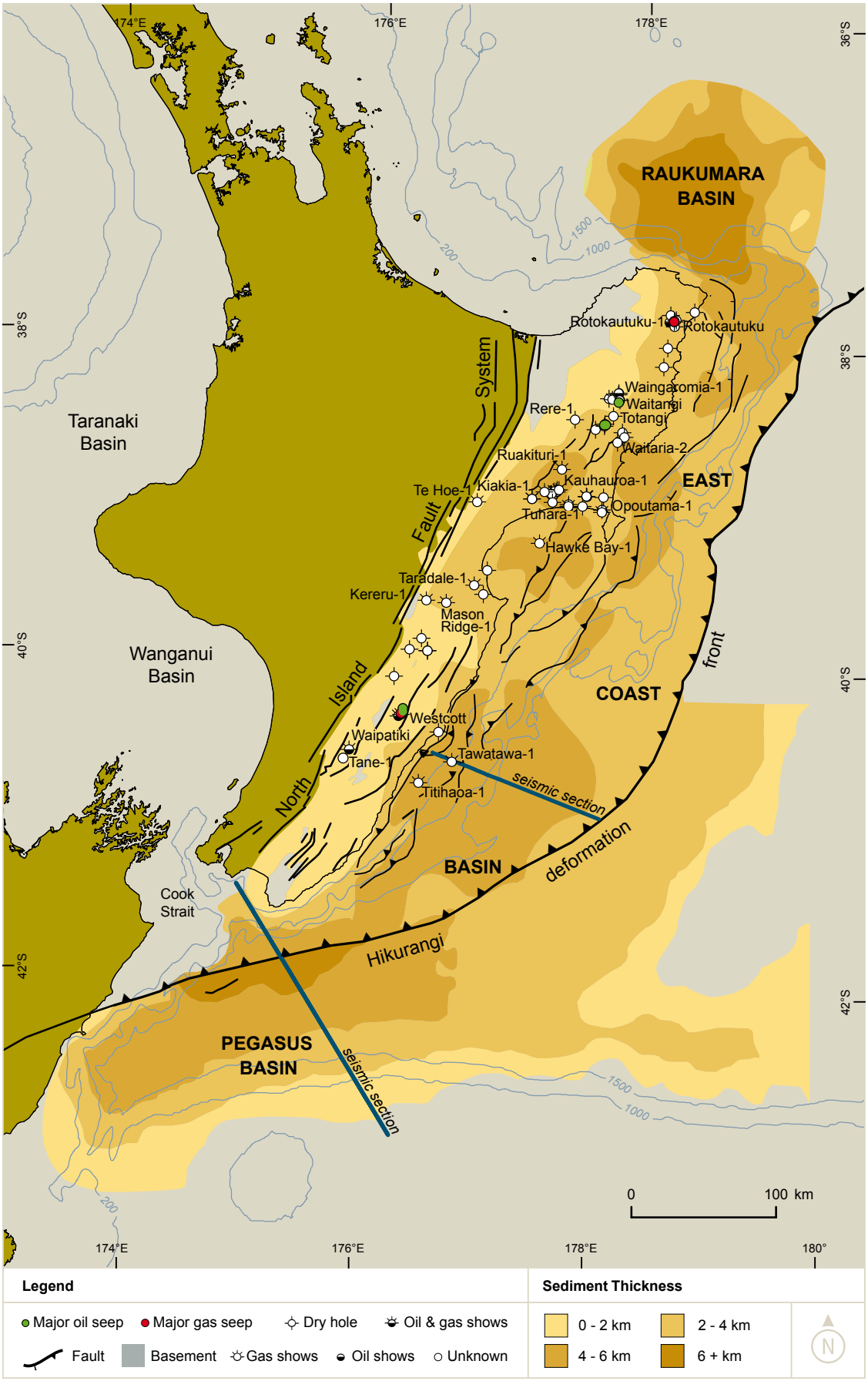
The sedimentary basins of the East Coast province are thought to have originally formed during the Early Cretaceous as a more-or-less continuous series of depocentres along the ancient Gondwana subduction margin. Seismic evidence from Pegasus Basin clearly shows a fossilised subduction margin and an associated thrust and fold belt underlying much of the north slope of the Chatham Rise.

Source Rocks

Source rocks in the province were deposited in marine environments, with varying proportions of terrestrial-derived organic material. The main known source rocks are the Whangai and Waipawa formations of Late Cretaceous–Paleocene age. There is a possibility of Early to mid-Cretaceous source rocks in parts of the province.

Reservoir Rocks

Include mid- to Late Cretaceous transgressive marine sandstones and turbidite sandstones, fractured Late Cretaceous–Paleocene mudstones, Eocene–Oligocene greensands, Neogene turbidite sandstones, shelf sandstones, and bioclastic limestones.


174°E
176°E
178°E
180°
36°S
38°S
40°S
42°S
RAUKUMARA BASIN
Rotokautuku-1
Rotokautuku
Waingaromia-1
Waitangi
Totangi
Rere-1
Waitaria-2
Ruakituri-1
Kauhauroa-1
Kiakia-1
Te Hoe-1
Opoutama-1
Tuhara-1
Hawke Bay-1
Taradale-1
Kereru-1
Mason Ridge-1
Westcott
Waipatiki
Tane-1
Tawatawa-1
Titihaoa-1
Taranaki Basin
Wanganui Basin
Cook Strait
System
Fault
Island
North
EAST COAST
front
deformation
BASIN
Hikurangi
PEGASUS BASIN
seismic section
seismic section
1500
1000
200
0
100 km
Legend
Major oil seep
Major gas seep
Dry hole
Oil & gas shows
Fault
Basement
Gas shows
Oil shows
Unknown
Sediment Thickness
0 - 2 km
2 - 4 km
4 - 6 km
6 + km
N

BASIN HISTORY

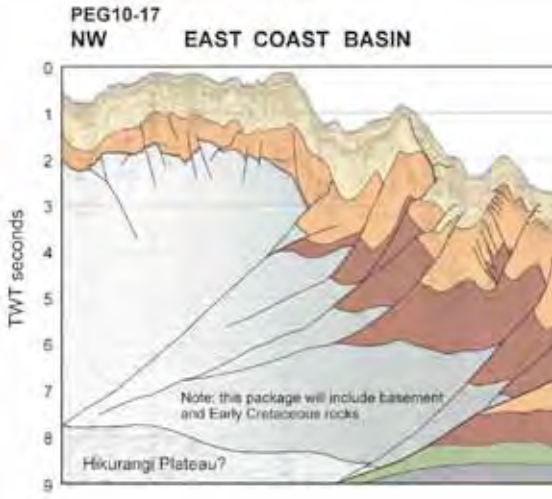
EXPLORATION

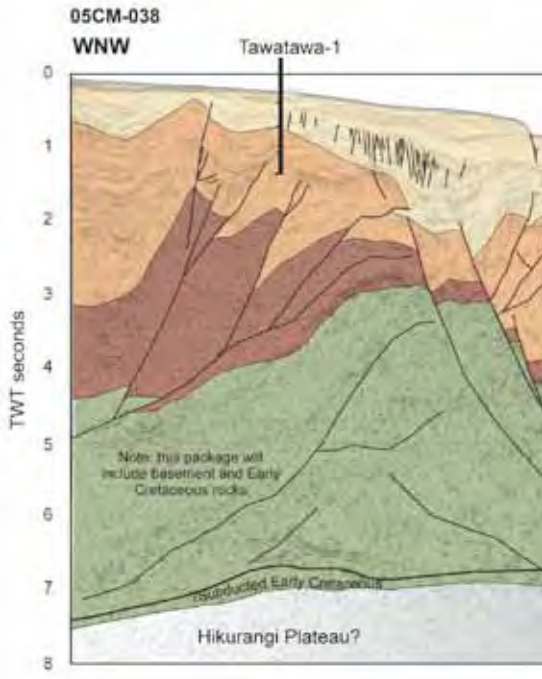
More than 40 wells have been drilled onshore in the East Coast Basin since the 1870s when pits and shallow wells were put down in and near oil seeps at Rotokautuku, Waitangi, and Totangi. Among the late 19th century successes were Rotokautuku-5, which produced 270 barrels of oil in its first month of production, Waitangi-1, which produced at a rate of 10 bopd, and the Waingaromia borehole (Waingaromia-1), which produced an estimated 20–50 barrels of oil per day from Miocene rocks until the rig burnt down in 1870. Further south, the Waipatiki-1 and -2 wells (Waipatiki Oilwells Ltd), and Tane-1 (Mangaone Oilfields Ltd) were drilled between 1912 and 1914 and, although each gave strong gas and/or traces of oil at various depths, none produced commercial quantities. It is unclear what the target was in these wells. Several more wells were drilled between 1919 and 1941, including additional shallow wells near the Waitangi oil seep. None discovered commercial quantities of hydrocarbons.

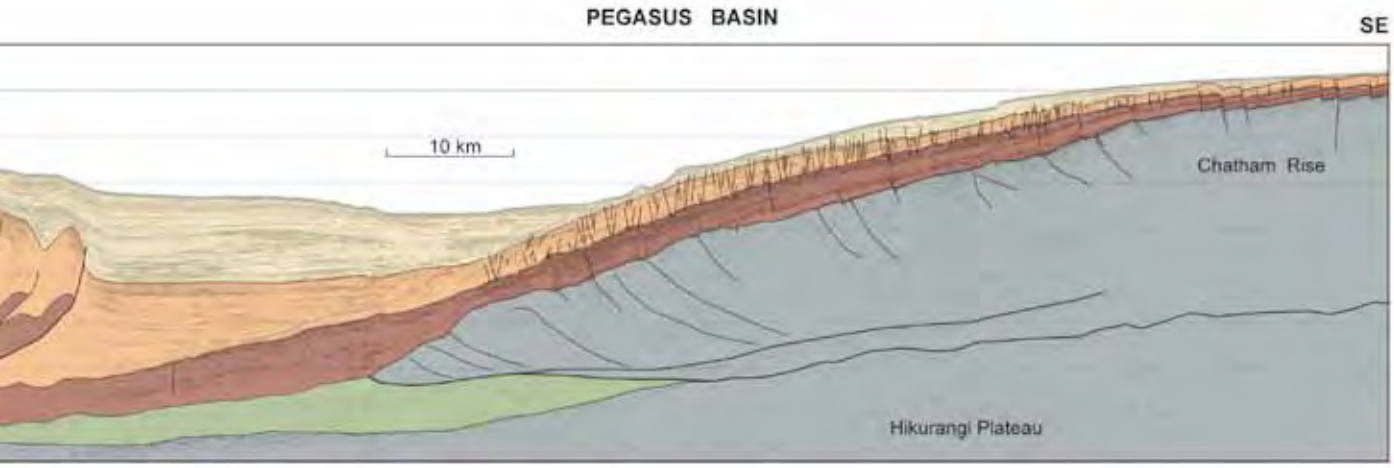
Most wells drilled since then have had shows of oil and/or gas. From the mid-1980s a renewed period of exploration began, with substantial seismic surveys being carried out across the region. Petrocorp Exploration drilled Te Hoe-1 in 1990. This well proved the reservoir potential of Late Cretaceous sandstones, and was the first well in the East Coast region to flow hydrocarbons (gas) under testing from a sealed reservoir when it flared gas briefly. A concerted drilling campaign by Westech in the 1990s discovered a sub-commercial high-pressure dry gas field at Kauhauroa-1. Following the discovery of gas at Kauhauroa, Westech drilled other nearby structural closures. Although gas shows were recorded from several wells, none were deemed significant. In 2007, Westech drilled Waitahora-1 as a direct offset to the Kauhauroa-1 discovery well. Waitahora-1 encountered high reservoir pressures and gas shows over a 70 m along-hole interval, probably in fractured limestone whilst drilling to the 1,352 m target depth. The well remains suspended awaiting appraisal.

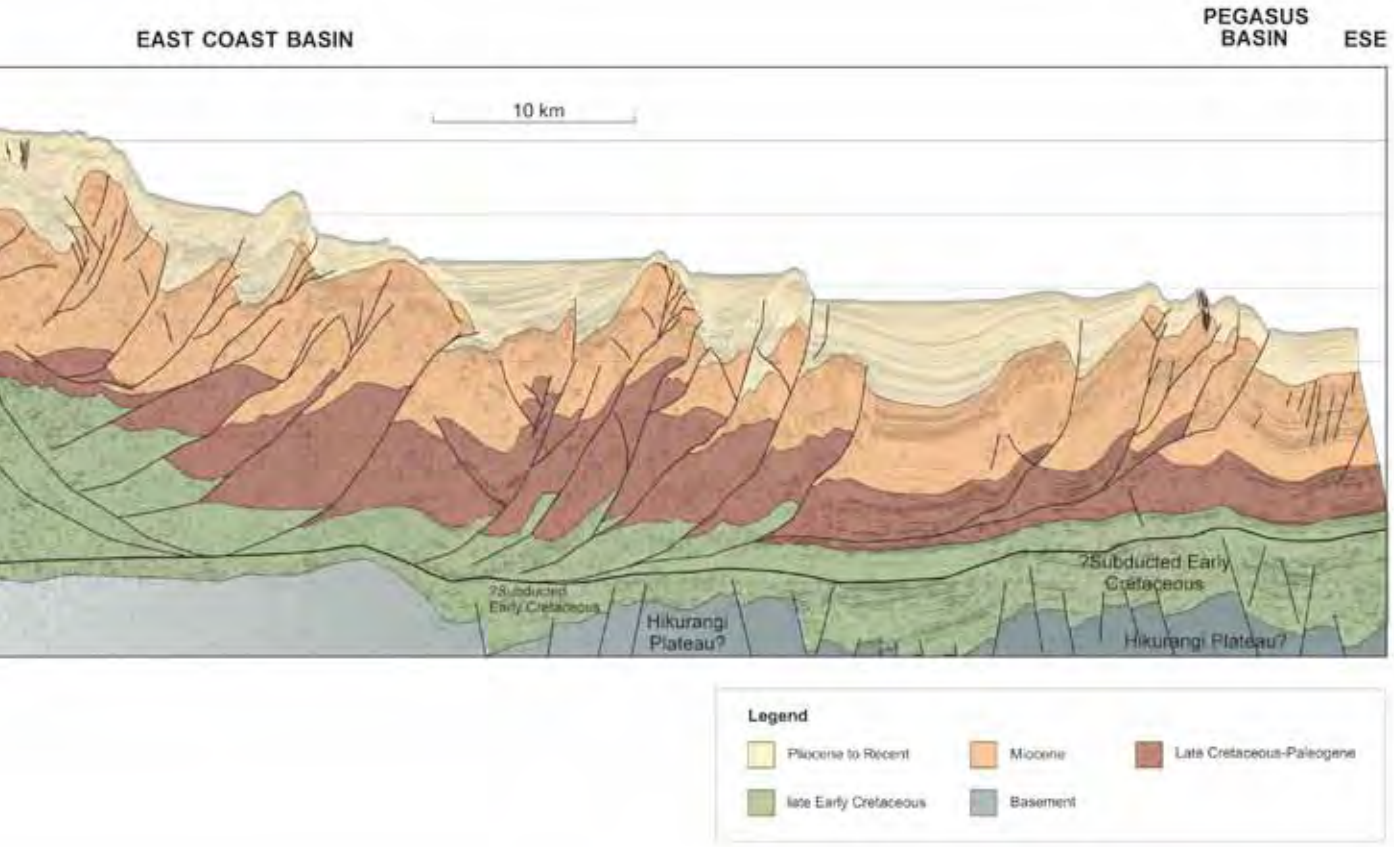
The first offshore well, Hawke Bay-1, was drilled to a total depth of 2,372 m in 1976 by BP, Shell, Aquitaine, and Todd Energy. The structure drilled is a large anticline and the intended target a high-amplitude reflector thought to be a Pliocene limestone. Gas shows were encountered in Middle Miocene marly limestone and thin sandstone. Titihaoa-1 was drilled to 2,727 m by an Amoco-led consortium in 1994, east of Castlepoint, Wairarapa. Gas shows were found in Miocene turbidite sandstones. In 2004, TAP Oil drilled Tawatawa-1, 35 km northeast of Titihaoa-1 to a total depth of 1,560 m. High-pressure gas shows were encountered, but there was no effective reservoir.

No wells have yet been drilled in the Raukumara or Pegasus basins, and their petroleum systems remain untested. The first permit in Pegasus Basin was awarded in late 2012. A moderate level of exploration activity continues in the onshore East Coast Basin at present.






PEGASUS BASIN
SE
10 km
Chatham Rise
Hikurangi Plateau


EAST COAST BASIN
PEGASUS BASIN
ESE
10 km
?Subducted Early Cretaceous
Hikurangi Plateau?
?Subducted Early Cretaceous
Hikurangi Plateau?
Legend
Pliocene to Recent
Miocene
Late Cretaceous-Paleogene
late Early Cretaceous
Basement

EXTENT

The East Coast province extends for at least 180,000 km^2, incorporating the Raukumara, East Coast, and Pegasus basins east of the North Island of New Zealand.

GEOLOGICAL HISTORY

Late Cretaceous-Paleogene Passive Margin

Subduction of the relatively buoyant Hikurangi Plateau, a region of thickened oceanic crust, below the Chatham Rise and Pegasus Basin part of the Gondwana margin from c. 110 Ma is inferred to have choked the subduction system along this part of the margin by c. 100 Ma. Late Cretaceous and Paleogene successions are relatively thin in this part of the East Coast province, suggesting a minimal sediment supply. Subduction may have continued until the mid-Late Cretaceous beneath the East Coast and Raukumara basins, though it had most likely ceased here by c. 85 Ma. From the mid-Cretaceous the East Coast province had transitioned from a convergent subduction margin in a prograding, passive continental margin. Passive margin deposition continued relatively uninterrupted until c. 23 Ma.

Up to 4 km of Early and Late Cretaceous cover strata crop out onshore in the East Coast province and several discrete, transgressive, westward-stepping cycles are recognised. Close to the basin margin, shallow marine sandstone facies are common. Away from the margin the predominant lithology is mudstone, with subsidiary intercalations of turbidite sandstones. In general the sequence is upward-fining. Late Cretaceous to Paleocene mudstone and marl of the Whangai and Waipawa formations are capped by Paleocene and Eocene smectitic and calcareous mudstone; maximum subsidence was in the Oligocene.

Neogene convergent tectonics

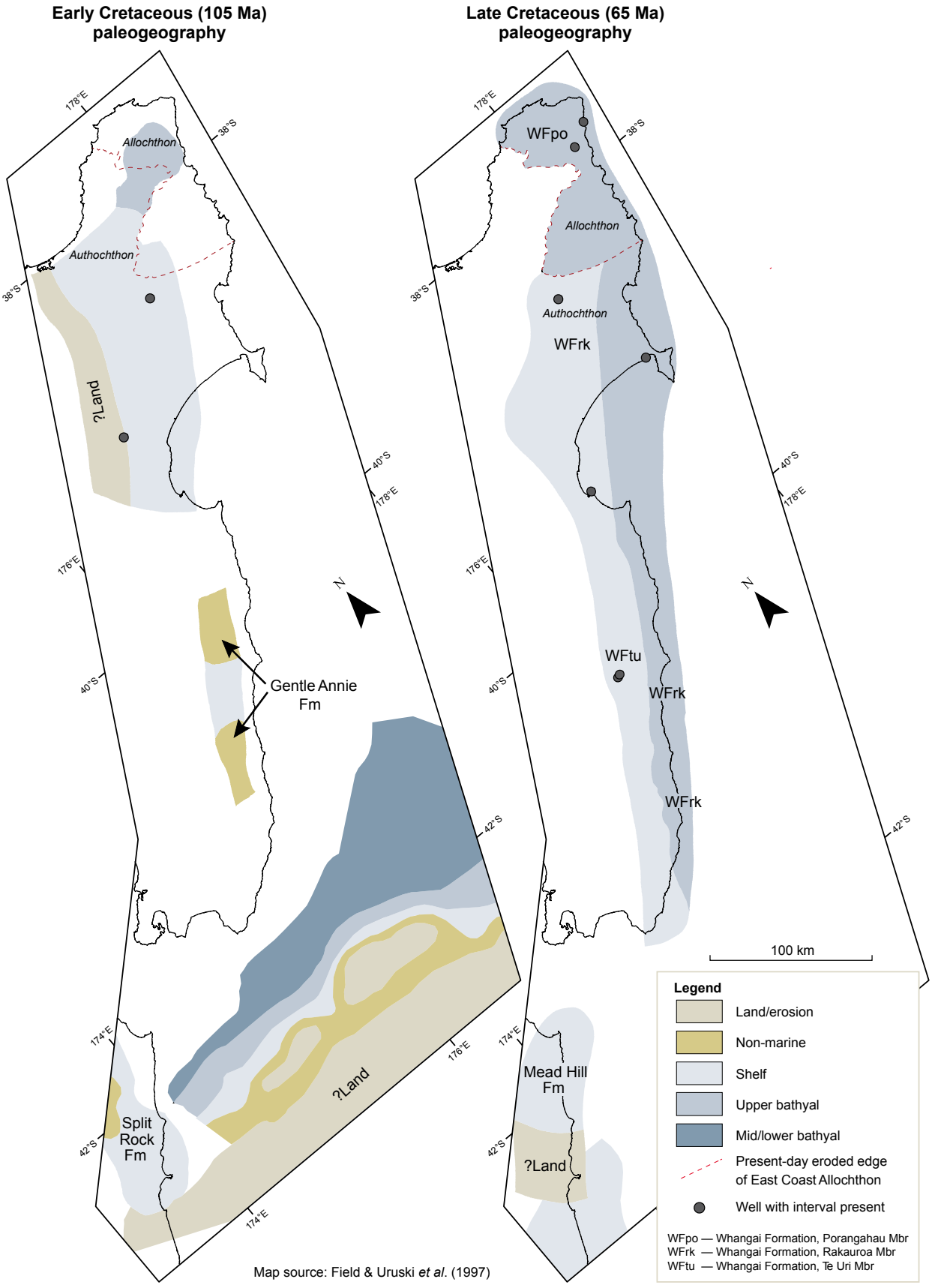
Establishment of the modern plate boundary through the New Zealand sub-continent transformed the passive margin setting to a convergent margin forearc in Early Miocene time. A thick package of Late Cretaceous to Oligocene rocks including Cretaceous and Paleogene ophiolite – the East Coast Allochthon, an equivalent of the Northland Allochthon – was obducted and emplaced into the East Coast Basin over part of what is now the Raukumara Peninsula. Similarly, in the Raukumara Basin the eastern margin, now the East Cape Ridge, was uplifted and the East Coast Allochthon was emplaced as a series of thrust sheets by gravity sliding, reaching the basin axis.

Since the Early Miocene ongoing oblique convergence associated with Pacific plate subduction has formed has a complex series of sedimentary depocentres in the East Coast Basin. These mostly comprise a highly faulted and folded accretionary prism with associated slope basins and, to seaward, a less deformed, more distal accumulation. Continued compression has resulted in imbrication and the formation of southwest-northeast oriented elongate anticlines, many bounded to the southeast by steeply dipping reverse faults and thrusts.

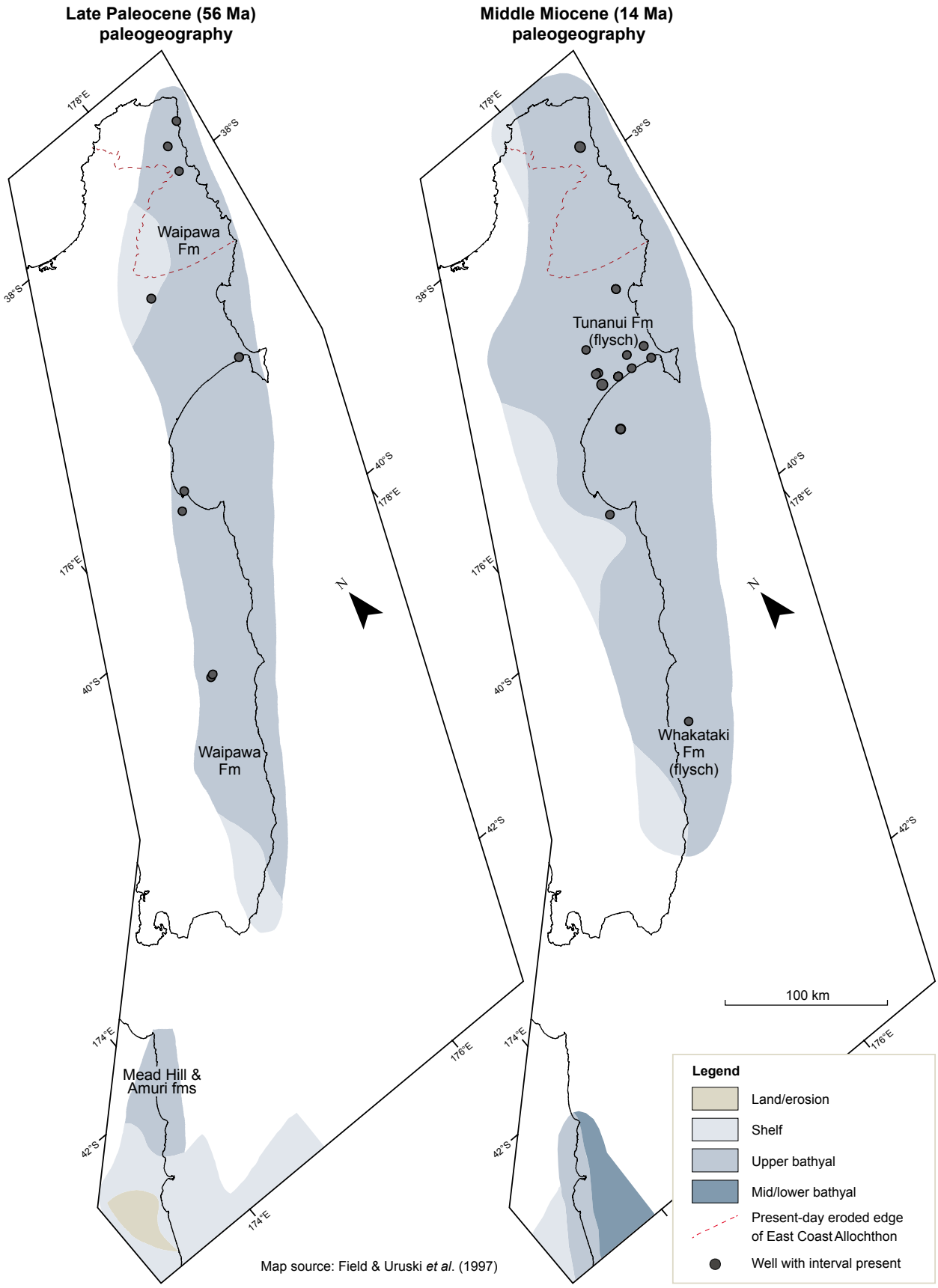
SUB-BASINS

The Raukumara Basin is also remarkably little deformed by Neogene tectonics even though it lies west of the modern subduction thrust. The East Coast Ridge may be partitioning plate convergence and therefore isolating the Raukumara Basin from the rest of the subduction margin. The broad synclinal form of the Raukumara Basin has been partly filled during the Neogene by what are thought to be mudstones and turbidite sandstones derived mainly from the uplifted and rapidly eroding Raukumara Peninsula.

By contrast the area of Pegasus Basin has been much-less deformed than the adjacent East Coast Basin due to the former's position entirely east of the modern subduction thrust. The basin was a depression into which sediments were deposited from the south and west, mainly as turbidite flows. The Neogene succession contrasts with older units in that it is very thick, with more than 6 km of Neogene sediments estimated to fill the axis of the basin. The central part of Pegasus Basin also contains several large anticlines created by blind thrusts rooted to the subduction/strike-slip plate interface.



Map source: Field & Uruski *et al.* (1997)



Map source: Field & Uruski *et al.* (1997)

EAST COAST PROVINCE PLAYS

Numerous prospects have been identified throughout the East Coast province, with the largest closures occurring offshore. Most plays involve fault-bounded anticlines, some of which require fault seal for effective closure. Stratigraphic pinch-outs are another significant play type throughout the stratigraphic record. Overpressure in many East Coast wells, particularly those in the onshore Wairoa area, indicates seal capacity of intervening mudstones is good.

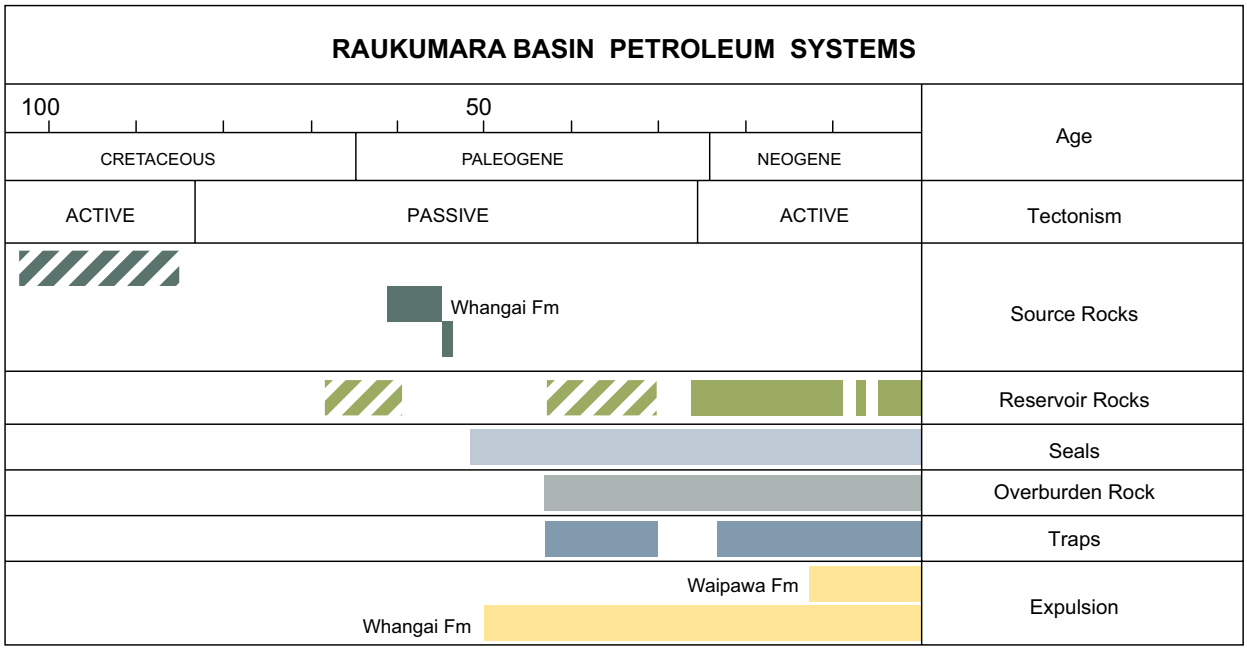
Seismic data suggest that Neogene turbidite sandstones may be thicker and better developed away from the structural highs, suggesting reservoir potential may be better away from the culminations in stratigraphically controlled structures. Fractured reservoirs, particularly Miocene limestone and sandstone, offer other play types. Pliocene limestone reservoirs, an early target in onshore Hawke's Bay exploration, have proven to be discontinuous in the subsurface.

Raukumara Basin play types include reactivated Gondwana margin fault anticlines, drapes across topographic features (e.g. the top of the East Coast Allochthon), and stratigraphic plays such as mounded turbidite bodies. Play types identified in the Pegasus Basin include: thrust anticline plays along the margin with the East Coast Basin, similar to those seen throughout the East Coast Basin; in the Pegasus Basin area, a thrust anticline play along the fossil Gondwana margin, now the northern slope of the Chatham Rise; blind thrusts in Neogene turbidites in front of the East Coast margin, and stratigraphic pinch-outs of Neogene turbidite sands against the north Chatham slope.

TIGHT GAS

Extensive fine-grained rock formations that could potentially yield tight gas occur in many parts of New Zealand, particularly in the onshore East Coast province where exploration and resource assessment are underway. The focus is on the Late Cretaceous–Paleocene Whangai and Waipawa formations, which are typically low to moderate TOC mudstones that compare favourably with the Bakken and Barnett shales of North America. In outcrop, the Whangai Formation shows significant fracture permeability, which may enhance its reservoir potential.

The Whangai and Waipawa formations are known to occur in northern and eastern Wairarapa, southeastern, central and coastal Hawke's Bay, and the Gisborne-Raukumara areas. They also occur both within the East Coast Allochthon and as autochthonous



rocks beneath the allochthon. The distribution of autochthonous Whangai and Waipawa formations beneath the East Coast Allochthon is very poorly known. However, it is apparent that mature source rocks must exist in places within this structural block as there are numerous active oil and gas seeps – such as the Waitangi, Totangi, and Rotokautuku seeps – that can be geochemically typed to the Whangai Formation. Whether these seeps are sourced from allochthonous or autochthonous Whangai Formation, or a combination of both, remains uncertain.

Overall, the Whangai Formation, which may be up to 600 m thick, has low to moderate source rock potential, although individual members have distinct differences with respect to source rock quality. The Te Uri and Porangahau Members have low source potential, the Rakauroa Member has higher potential – particularly in western areas – and the Upper Calcareous Member has the highest source rock potential, particularly in eastern areas.

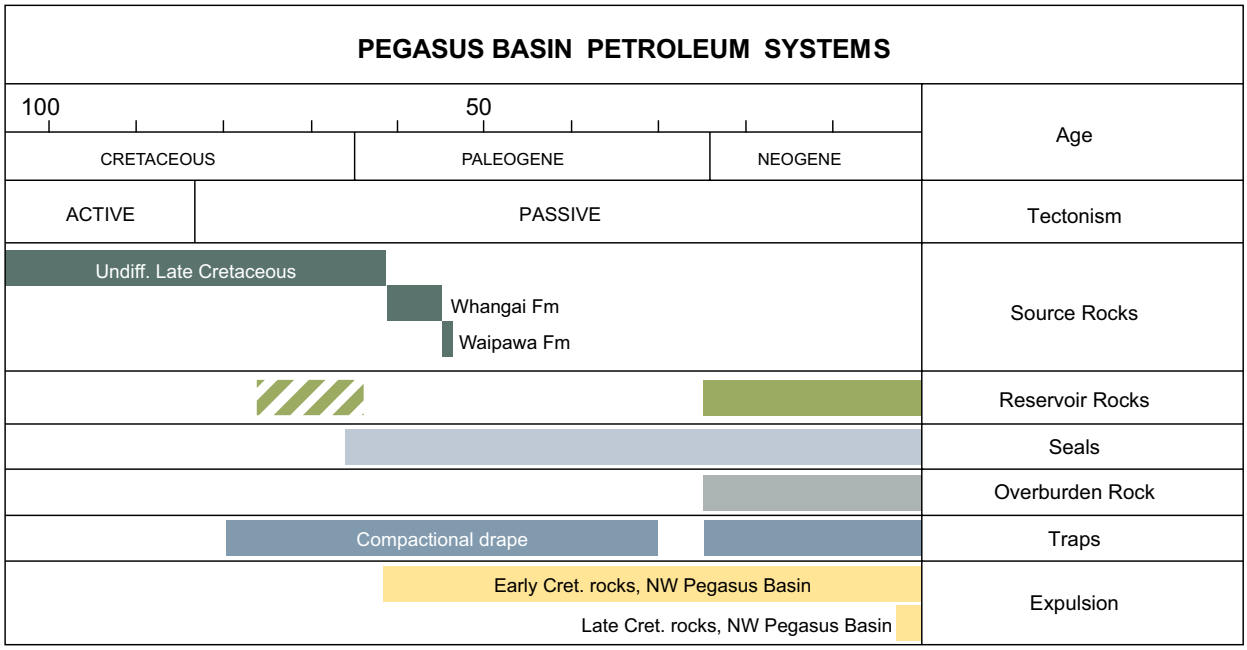
The Waipawa Formation is considered to be the most promising source rock in the province, although it has a patchy distribution and is generally fairly thin (2–50 m thick, average 17 m). Source rock properties indicate that this formation has high generative capacity but it is immature or marginally mature in outcrop sections throughout the East Coast region. Identification of Waipawa-sourced oils in the Raukumara and Wairarapa regions indicates that the formation must reach maturity at depth.

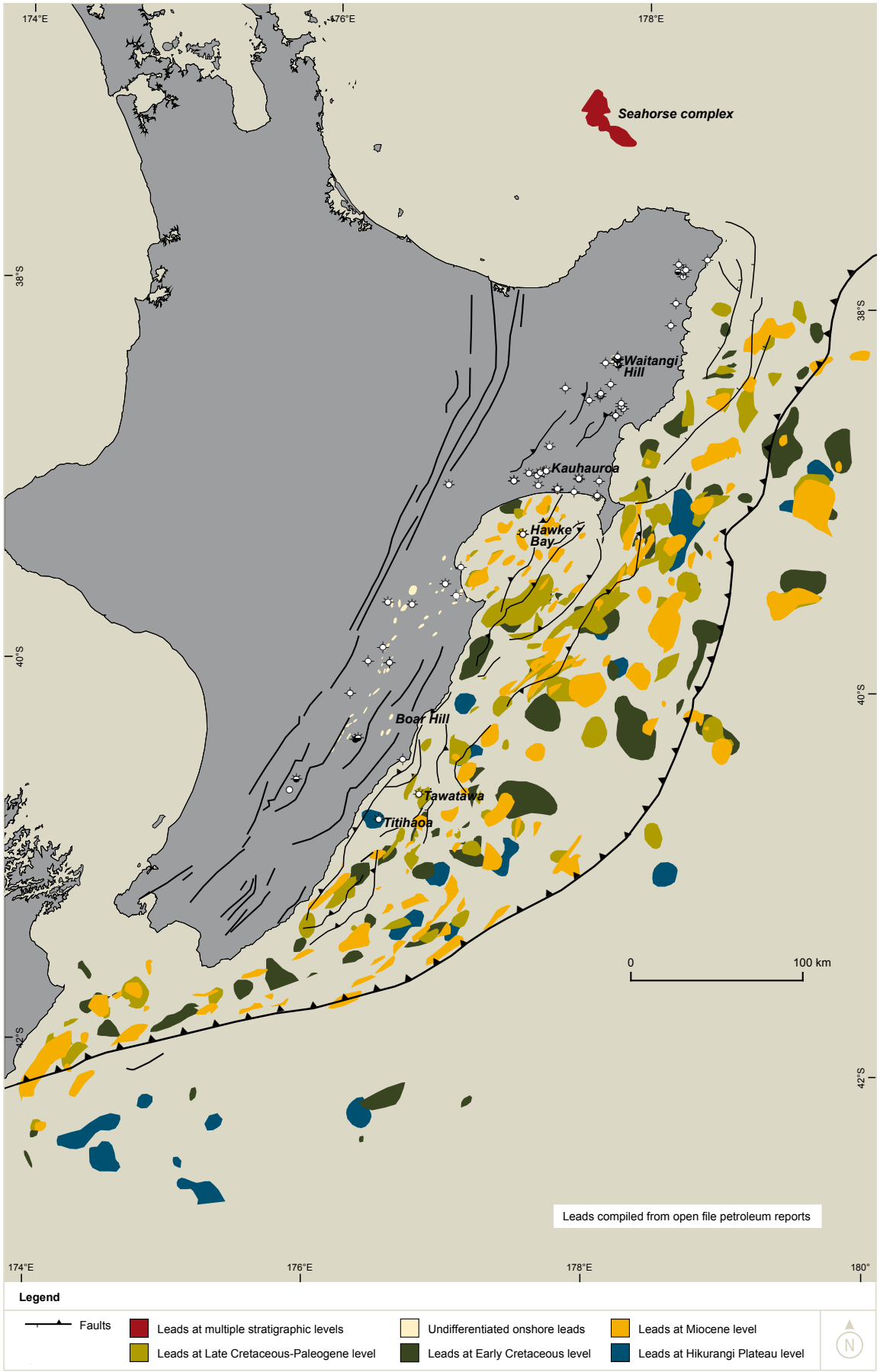
North American experience suggests the costs of assessment and development of tight gas plays may be high. Although the mainly frontier East Coast province currently lacks much petroleum infrastructure, the quantity of tight gas in the basin is potentially large.

GAS HYDRATES

Gas hydrates are crystalline solids consisting of gas molecules – usually methane – surrounded by a cage of water molecules. They offer a potential future source of natural gas. An established gas hydrate province occurs within the Hikurangi margin of the East Coast province. The hydrate accumulations were discovered from seismic data, based on the presence of bottom-simulating reflections (BSRs) produced at the base of the gas hydrate stability zone in sediments. Though gas hydrates occur in many parts of New Zealand's offshore territory the Hikurangi margin is thought to be New Zealand's most economically promising gas hydrate province, partly because of its proximity to major population centres, and partly because of its favourable geologic setting. To date, the Hikurangi margin is the only region in New Zealand from which gas hydrates have been recovered in shallow sediment cores.

Initial estimates of gas reserves within the 50,000 km^2 Hikurangi margin hydrate province are up to approximately 800 TCF. However, most of the gas hydrates are thought to be dispersed and are therefore not of economic interest. Initial estimates of gas volumes in highly concentrated hydrate zones are ~20 TCF, although this figure does not account for the quality of the reservoir rock. Research on the Hikurangi Margin is now focussing on identifying high-quality gas hydrate reservoirs with high permeability and connectivity.




174°E
176°E
178°E
180°
38°S
40°S
42°S
Seahorse complex
Waitangi Hill
Kauhauroa
Hawke Bay
Boar Hill
Tawatawa
Titihaoa
0
100 km
Leads compiled from open file petroleum reports
Legend
Faults
Leads at multiple stratigraphic levels
Leads at Late Cretaceous-Paleogene level
Undifferentiated onshore leads
Leads at Early Cretaceous level
Leads at Miocene level
Leads at Hikurangi Plateau level
N

EAST COAST PROVINCE RESERVOIRS

In the East Coast Basin, the most likely reservoir facies are mid- and Late Cretaceous transgressive marine sandstones and turbidite sandstones, fractured Whangai Formation shale and marl, Neogene turbidites, and Neogene shallow water bioclastic limestones.

CRETACEOUS RESERVOIRS

Mid- to Late Cretaceous shelf sandstones and turbidite packets were investigated by Rere-1 and Te Hoe-1, the latter being the first successfully first production of hydrocarbons on test from a sealed reservoir in the East Coast province. Mid-Cretaceous transgressive sandstones of the Tahora Formation have measured porosities of 8.3–17.1% and permeability 0.04–20 mD. Though similar facies are unlikely in offshore East Coast Basin equivalent transgressive sandstones are inferred to be present in Pegasus Basin, onlapping the Chatham Rise. The laterally equivalent though deeper water turbidite sandstones of the Tapuwaeroa Formation have measured porosities of ~9.3–28.5%. Permeability is typically <1 mD but is, in places, up to 198 mD. There is a high probability that equivalent strata of this age are present offshore.

MIOCENE RESERVOIRS

Miocene turbidite sandstone units offer perhaps the best reservoir potential in the province. The Early Miocene deep-marine Rere Sandstone Member yielded gas shows in Kauhauroa-1 although, on test, flow rates were erratic due to sediment blocking the choke. Maximum flow rates were 537 Mscf/day of gas and 550 bbl/day water. Middle Miocene Tunanui Formation turbidite sandstone beds have measured porosities of 8–30% and permeabilities of up to ~150 mD. Through-going fractures parallel to the principal horizontal stress direction suggest potential as a hybrid, clastic-fractured reservoir. Middle Miocene turbidite sandstones of the Whakataki Formation may offer reservoir potential in the Wairarapa area, particularly offshore. Middle Miocene turbidites had prominent gas shows in Titihaoa-1. Middle and Late Miocene turbidites in Tawatawa-1 and also had gas significant shows.

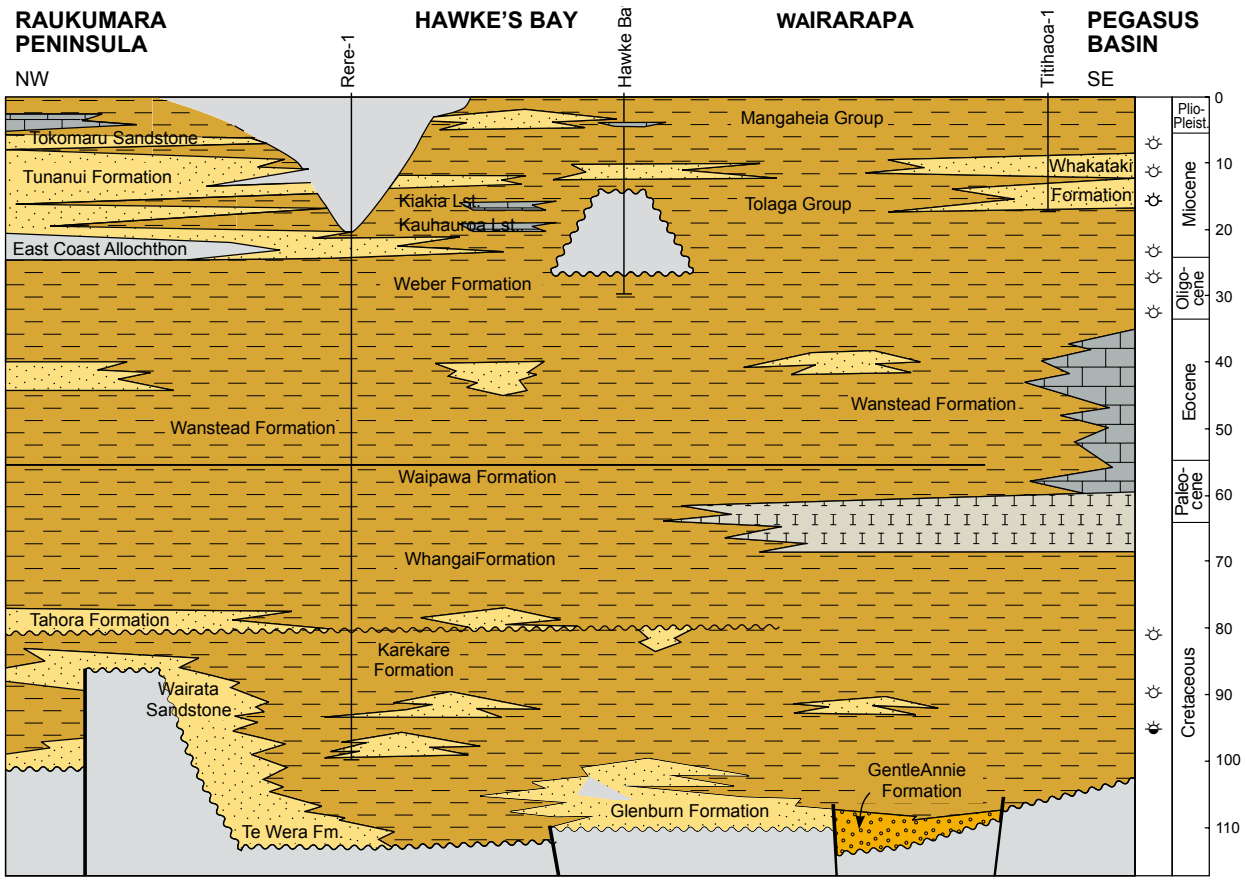
Early Miocene shelf Kauhauroa Limestone has been intersected by at least seven exploration wells in the Wairoa area. Virtually all porosity is thought to be from northeast-striking fracture systems. Initial testing in Kauhauroa-1 flowed gas at 11.5 MMcf/day; in a subsequent 220 hour-long test, flows were recorded as up to 6.2 MMcf/day and up to 2300 bbl/day of water. The similar Late Miocene Kiakia Limestone produced small quantities of gas and water on test in Kiakia-1/1A. Middle Miocene calcareous sandstone and limestone were gas-bearing in Hawke Bay-1, although the unit was not tested. Outcropping coarse-grained shelly limestones of Plio–Pleistocene age have high porosity (commonly 25–32%) and high permeability. The extent of similar bioclastic limestones in the offshore East Coast province is virtually unknown.

RAUKUMARA BASIN

By analogy with the onshore Raukumara Peninsula, the main reservoir facies in the Raukumara Basin are likely to be Cretaceous to Neogene turbidite sandstone bodies. Some limestone units with good reservoir properties may also be present. On land, Whangai Formation mudstones have fracture permeability. Transgressive sandstones of Cretaceous and Paleogene ages may also be present, particularly in the western part of the basin.

PEGASUS BASIN

In Pegasus Basin, Cretaceous, Paleogene and Neogene transgressive and turbidite sandstones, and fractured Late Cretaceous to Paleocene mudstones, are considered to be the most likely reservoir facies. The Chatham Rise crest is thought to have remained emergent until Eocene time. Transgressive sandstones of Late Cretaceous and Paleogene ages are therefore likely to be present across the northern flank. Much of the Neogene succession in the centre of the basin is inferred to have been deposited by turbidite flows, with turbidite sandstones having reservoir potential. Several large incised channels have been identified along the Chatham Rise margin and these may contain good quality reservoir sands.



Legend

Volcanic
Limestone
Marl
Mudstone
Sandstone
Coal measures
Conglomeratic
Non deposition
Oil shows
Oil & gas shows
Gas shows

Great South-Canterbury Province



BASIN SUMMARY

This Province incorporates the Great South and Canterbury Basins, southeast New Zealand. It is a large, little-explored area covering nearly 400,000 km^2. Basins originated as Cretaceous failed rifts, evolving into subsiding basins during the Late Cretaceous and Paleocene.

The province has several sub-basins. The Canterbury Basin contains up to 6 km of strata, thickest in the Clipper Sub-basin. The succession is up to 9 km thick in the Great South Basin, being thickest in the Central Sub-Basin.

PLAYS

- **Stratigraphic drape over basement highs**
- **Differential compaction**
- **Faulted anticlines**
- **Folds**
- **Turbidites**
- **Basin-floor fans and feeder channels**
- **Current commitment to drill Carrack-Caravel in 2013/2014**
- **Potential stratigraphic traps - Late Cretaceous transgressive sands, prograding deltas, and deep-water fans**

EXPLORATION AND PRODUCTION

Wells

Thirteen offshore exploration wells, with sub-commercial gas-condensate discoveries in both basins. Galleon-1 tested 10 mmcf/d of gas and 2,300 bbl/d of condensate. Cutter-1 discovered a 56 m gross hydrocarbon column in poor quality sandstones. A sub-commercial gas-condensate discovery in Kawau-1A. which tested 6.7 mmcf/day gas (7% CO_2) and 24 bbl/d condensate. Shows in three other wells include Tara-1, which reported "fizz-gas", Toroa-1, which reported oil staining and strong gas shows associated with oil, and Clipper-1. There is an oil seep onshore at Stewart Island. The most recent well is Horseshoe-1, drilled in 2012 from onshore Stewart Island out into the Basin. Results of this well remain unreported.

GEOLOGY

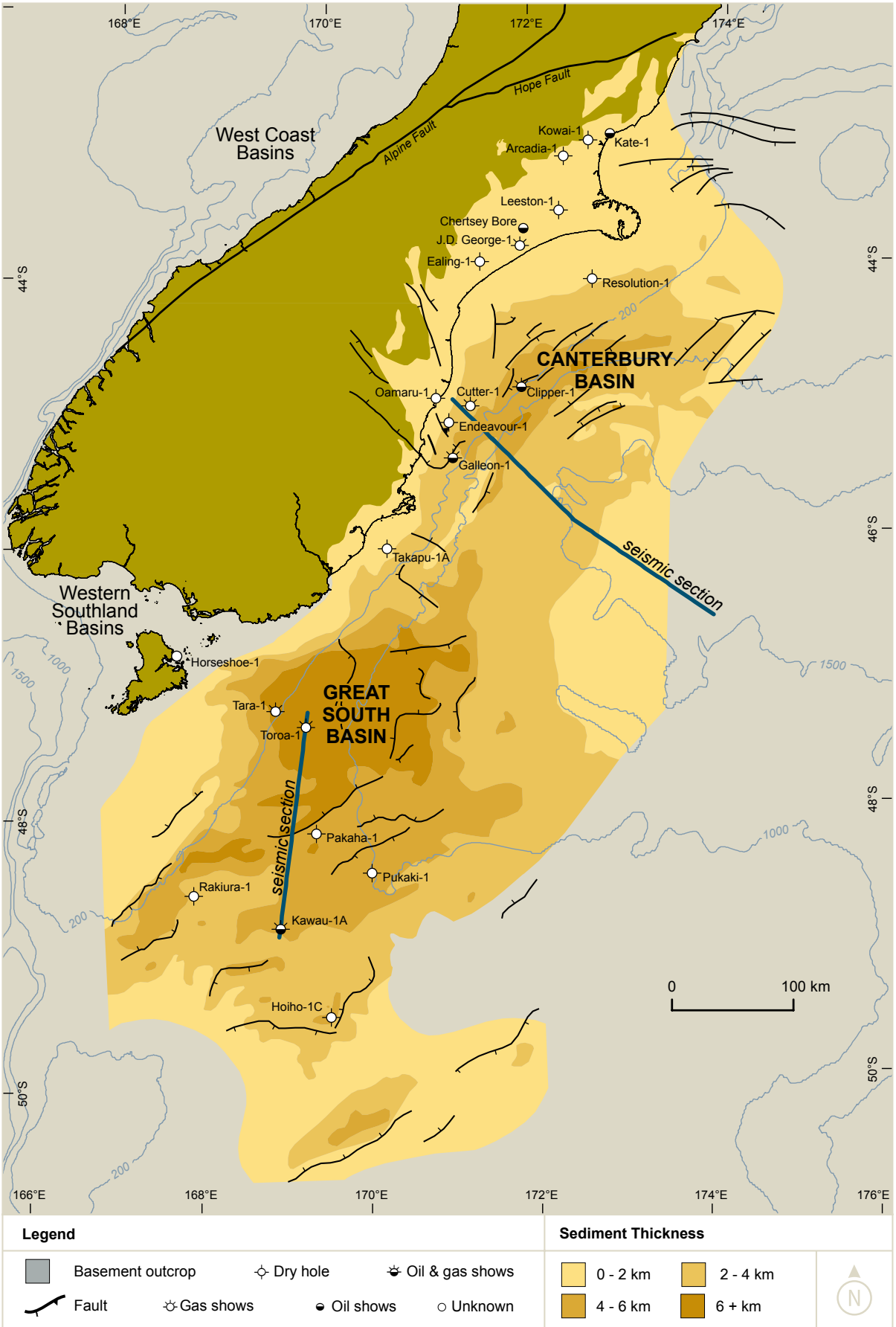
The Canterbury basin contains 5 km, or more, of Cretaceous-Cenozoic sediment and covers more than 40,000 km^2. It extends east into deeper water of the Bounty Trough and is contiguous to the south with the larger Great South Basin. The Great South Basin is a complex intracontinental rift formed during the mid-Cretaceous, with several distinct sub-basins containing up to 8.6 km of sedimentary fill. The extent of the basin is often defined by the 2,000 m total sediment thickness isopach.

Source Rocks

Middle Jurassic fluvial sediments (mostly in the Great South Basin), mid-Cretaceous syn-rift fluvial sediments, Late Cretaceous-earliest Paleocene fluvial/marine coaly sediments, and Late Cretaceous-Eocene marine mudstones and shales (e.g., Tartan Formation, equivalent to the Waipawa Formation black shale). Potential source rocks are dominated by coaly facies, c.f. Taranaki Basin. Potential for marine shale source rocks, although maturity is an issue.

Reservoir Rocks

In the Canterbury Basin reservoirs are primarily sandstones, generally quartzose, of Late Cretaceous to Miocene age, ranging from fluvial and paralic to shelf and turbidite depositional settings. In the Great South Basin reservoirs are mostly Cretaceous to Eocene fluvio-deltaic and transgressive marine sandstones. Mid-Cretaceous fluvial facies in Great South Basin, and mid- to Late Cretaceous transgressive to Eocene marine/paralic sands occur in both Canterbury and Great South Basins. Cretaceous–Paleogene turbidite sands are possible in the Great South Basin.


168°E
170°E
172°E
174°E
176°E
44°S
46°S
48°S
50°S
West Coast Basins
Alpine Fault
Hope Fault
Kowai-1
Arcadia-1
Kate-1
Leeston-1
Chertsey Bore
J.D. George-1
Ealing-1
Resolution-1
CANTERBURY BASIN
Oamaru-1
Cutter-1
Clipper-1
Endeavour-1
Galleon-1
seismic section
Takapu-1A
Western Southland Basins
Horseshoe-1
GREAT SOUTH BASIN
Tara-1
Toroa-1
seismic section
Pakaha-1
Pukaki-1
Rakiura-1
Kawau-1A
Hoiho-1C
200
1000
1500
0
100 km
Legend
Basement outcrop
Dry hole
Oil & gas shows
Fault
Gas shows
Oil shows
Unknown
Sediment Thickness
0 - 2 km
2 - 4 km
4 - 6 km
6 + km
N

BASIN HISTORY

EXPLORATION

Onshore wells were drilled as early as 1914 in the Canterbury Basin. The focus is now on the offshore part of the basin, with the most recent well in 2012 from onshore Stewart Island out into the basin. Following the acquisition of seismic surveys, Shell, BP and Todd Oil Services drilled four offshore wells, Endeavor-1, Resolution-1, Clipper-1 and Galleon-1 in the continental shelf in the 1970s and 1980s. Takapu-1A was drilled for Petrocorp at the boundary with Great South Basin in 1978. TAP Oil drilled Cutter-1 in 2006. Both Galleon-1 and Clipper-1 had significant hydrocarbon shows in Cretaceous coal measure sands. Modern reconnaissance seismic surveys extend over the continental shelf and some way beyond the 200 m isobath and there are detailed grids over some areas, including the large Carrack-Caravel prospect, which has been covered by a recent 3D seismic dataset.

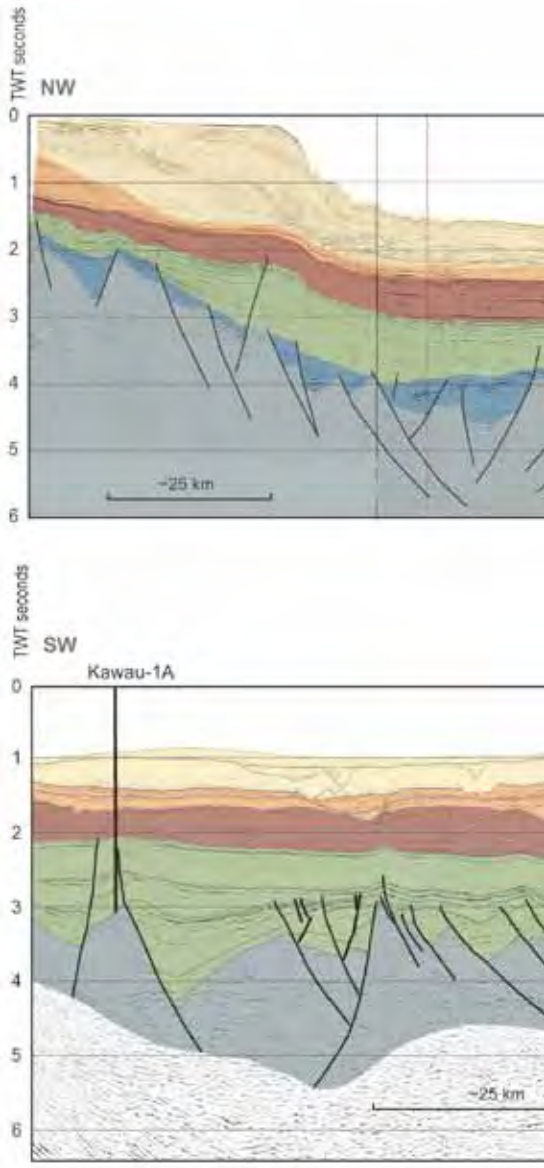
Exploration in the Great South Basin began in the early 1970s, when Hunt International Petroleum Company undertook a large regional seismic acquisition programme across a large (402,000 km^2) exploration permit. Over 13 years, more than 30,000 km of 2D seismic data were acquired. Hunt went on to drill four wells between 1976 and 1978 (Toroa-1, Pakaha-1, Kawau-1A, Hoiho-1C), with partners Phillips Petroleum and the New Zealand government. Petrocorp, operating for Crown interests, drilled two sole risk wells (Tara-1 and Takapu-1A) in 1978. Placid Oil Company took over as operator in 1982 and drilled Rakiura-1 and Pukaki-1. Toroa-1 drilled more than 300 m of oil shows but the well was plugged and abandoned due to mechanical issues before these could be tested and Kawau-1A was a sub-commercial gas-condensate discovery originally estimated at 461 bcf. Tara-1 had moderate gas-condensate shows and Pakaha-1 drilled gas-bearing sands. The permit was relinquished in 1985.

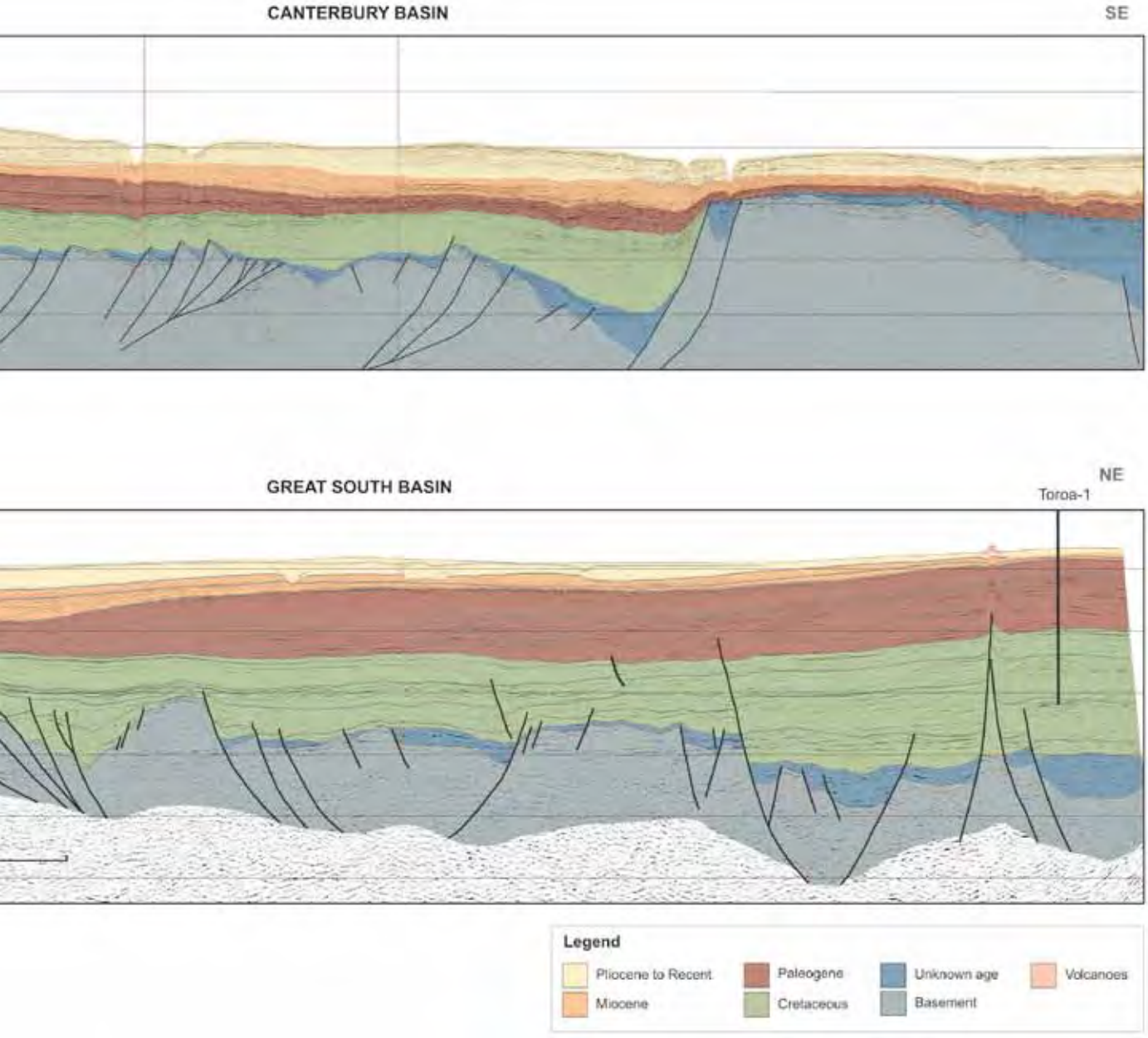
The western margin of the basin was licensed separately in 1982. More seismic data were acquired and a geochemical survey conducted but there was no drilling. Later permit holders acquired additional seismic data and reprocessed the earlier data and, in 2005, Crown Minerals acquired a 3,000 km 2D seismic data set across the northeast part of the basin. In the licensing round which followed, a large area was awarded to ExxonMobil and Todd Energy, three blocks were taken up by OMV, PTTEP and Mitsui, and one block was awarded to Greymouth Petroleum. These companies have since acquired more than 20,000 km of new 2D, and 1,500 km^2 of 3D seismic data in advance of drilling commitments. Shell were also allocated one block as part of the 2012 Block Offer round.

EXTENT

The Great South and Canterbury Basins of southeast New Zealand occupy a region on the landward side of the Gondwana margin of late Mesozoic extension and post-rift passive margin drift. The main basin-fill succession in the region, which may be up to 9 km thick, is of Cretaceous to Recent age.

During the early to middle Mesozoic the region encompassed backarc and forearc positions within the ancient Gondwana subduction margin. Latest Jurassic and earliest Cretaceous rocks of the Clent Hills Group, consisting of conglomerate, sandstone, mudstone and coals, and presently preserved mainly in fault-angle depressions, were deposited in terrestrial to deltaic settings within Canterbury Basin. Although traditionally considered as basement, these rocks may be locally significant to the petroleum geology of the basin. Similarly, Triassic and Jurassic marine and non-marine rocks of the Murihiku terrane, present in the Great South Basin, have traditionally been regarded as basement, though they may also have been the earliest basin-fill. Great South Basin seismic surveys show these rocks are well bedded and gently folded.




CANTERBURY BASIN
SE
GREAT SOUTH BASIN
NE
Toroa-1
Legend
Pliocene to Recent
Miocene
Paleogene
Cretaceous
Unknown age
Basement
Volcanoes

GEOLOGICAL HISTORY

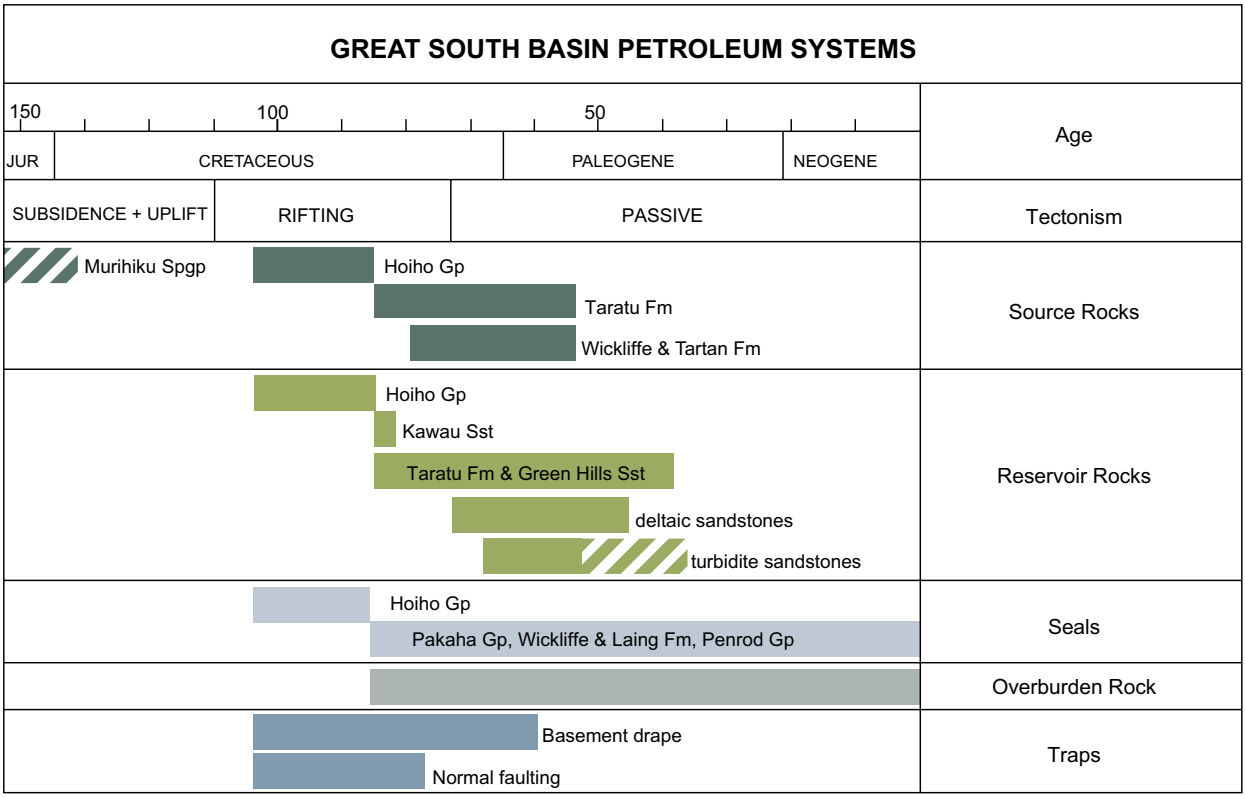
Cretaceous rifting

Rift faulting was active in the Early Cretaceous, and by the Late Cretaceous widespread fluvial systems generally flowing to the northeast had developed along the axes of the major grabens. Whether or not these fluvial systems contributed to a major delta beyond the present limits of the province basin in the Bounty Trough remains unknown. Early basin-fill deposition was in small, rifted basins formed during extension prior to, and in the early stages of, break-up at the eastern Gondwana margin. The rocks include conglomerate, sandstone, mudstone, and in places coal measures of the Hoiho Group in the Great South Basin. The syn-rift basins were sites of active rifting until about 85 Ma, during which time significant thicknesses of rock were deposited; over 3 km of syn-rift terrestrial sediment was deposited in the Clipper Sub-basin in the Canterbury area.

Late Cretaceous-Neogene passive subsidence

With continuing subsidence in Late Cretaceous to Paleogene time, the region was gradually inundated. Rocks deposited during this post-rift phase are more widespread and include terrestrial to shallow marine conglomerate, sandstone and mudstone and minor coal measures. Fluvio-deltaic systems including coaly swamps persisted at the western margin of the Great South Basin, with a source apparently through what is now Foveaux Strait. Paleogene turbidites were deposited across the basin floor in front of the delta. Further transgression during a passive drift phase laid down a thick transgressive marine succession, including quartzose sandstone, greensand, and mudstone up to 4400 m thick. In far offshore regions, background deposition was dominated by fine open-water carbonates. Deposition was mainly in shelf and upper slope settings.

By the late Paleogene sediment source areas were significantly reduced as clastic supply diminished, and carbonate deposition became increasingly widespread across the province.

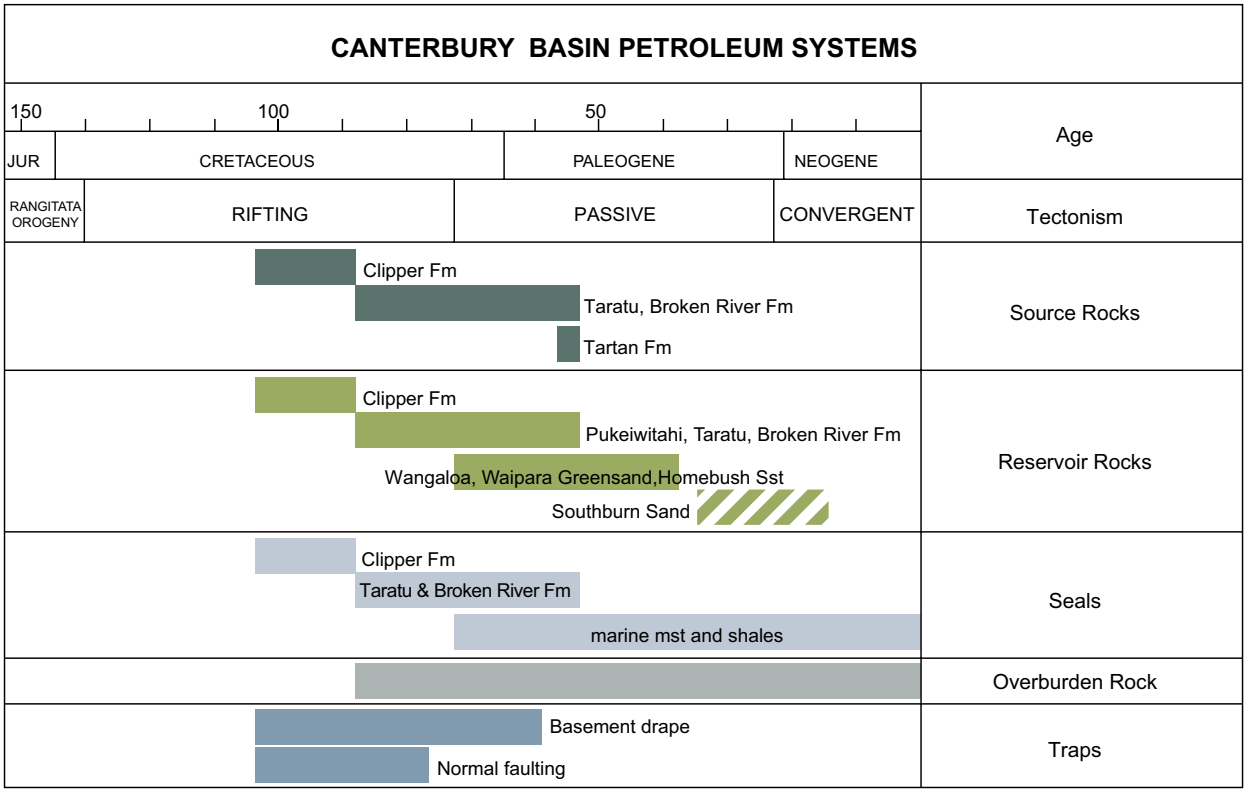


Late Neogene deformation

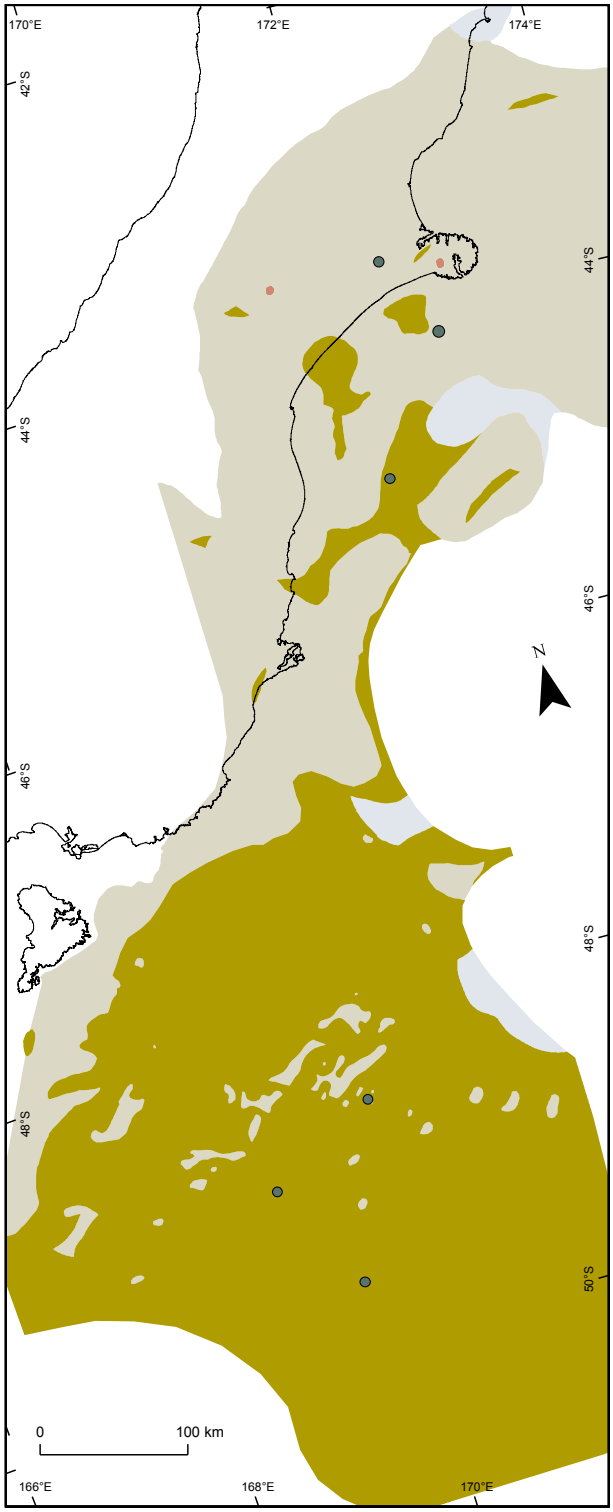
The Canterbury and Great South Basins have not been greatly deformed by the onset of deformation associated with propagation of the modern plate boundary. In Neogene to Recent time Canterbury Basin has mostly remained a moderate distance away from the active plate boundary, and there has been only minor structural inversion and faulting along the northern and western margins of the basin. Only the western margin of the Great South Basin has been affected by the plate boundary deformation. This area has been gently folded into a series of subtle anticlines and synclines.

From the late Neogene to Recent there has been rapid uplift and much erosion of the Southern Alps and hinterland to the west of the province. The Neogene succession is up to 2,200 m thick in Canterbury Basin, and generally thinner in the Great South Basin. Although much of the Neogene succession offshore is mudstone-dominated, conglomerate and sandstone were deposited in more proximal areas in the west of the Canterbury Basin. The modern Great South Basin continental shelf is underlain by a mix of clastic and cold-water carbonate sediments that have prograded across older strata. Variations in the thickness of the Neogene succession in the region are important for source rock maturation.

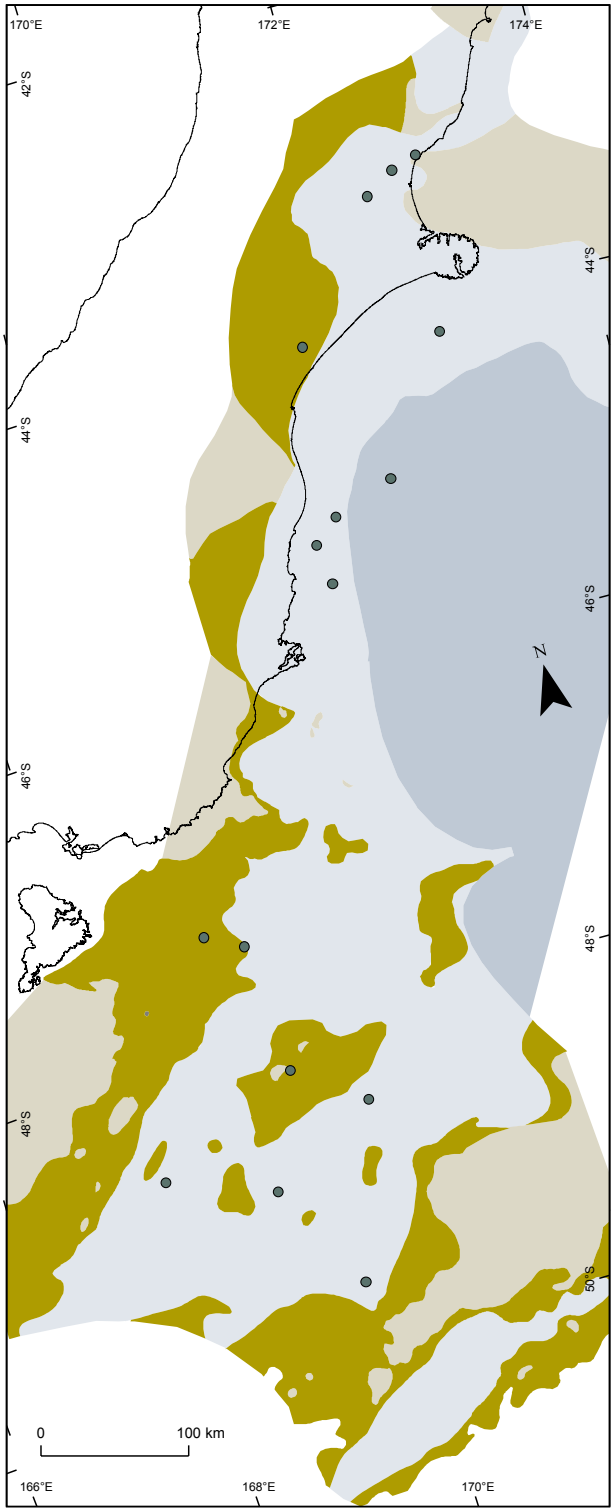
There have been several phases of volcanism in the region. Mid-Cretaceous rhyolitic volcanics and ignimbrite were extruded in central and southern parts of the Canterbury Basin. Paleogene basaltic volcanics include the Endeavour Volcanics in Endeavour-1 and Clipper-1, and Late Eocene to Oligocene rocks at Oamaru and north Canterbury. Banks Peninsula near Christchurch is a Middle Miocene to Pliocene volcano. A Middle Miocene igneous sill occurs in the Resolution-1 well.



Mid Cretaceous (105 Ma) paleogeography



Late Cretaceous (65 Ma) paleogeography

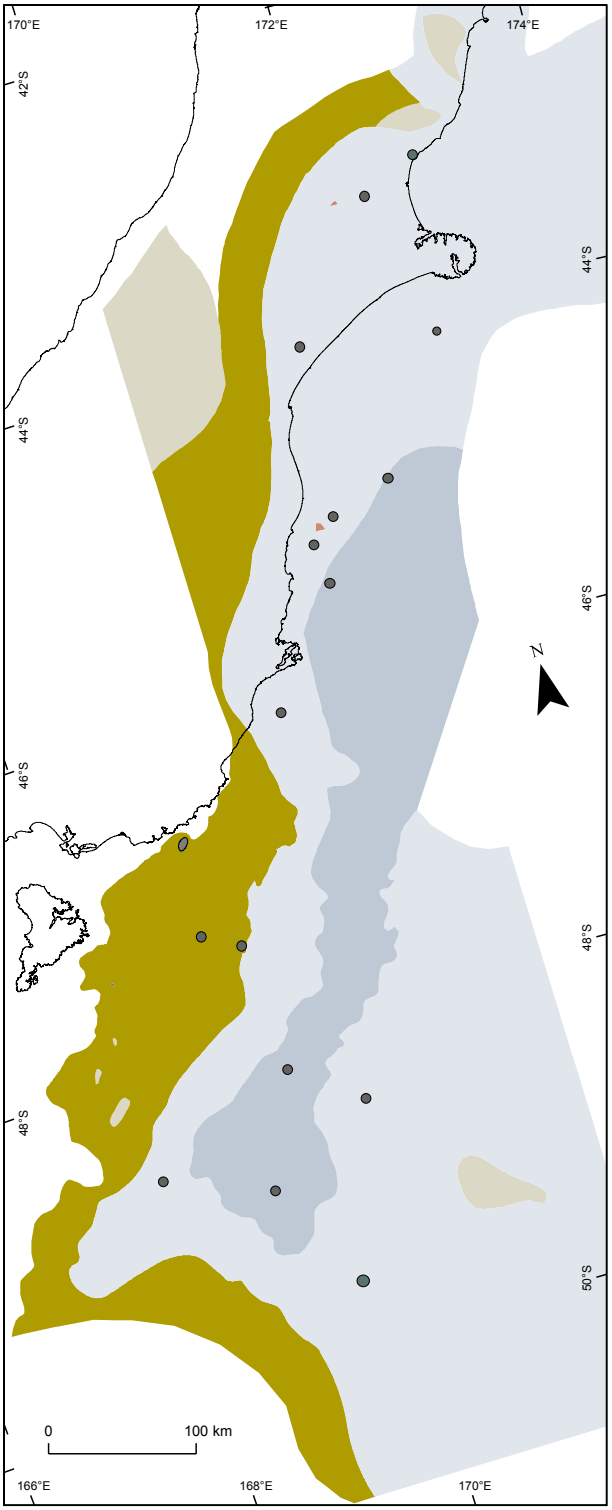


Legend

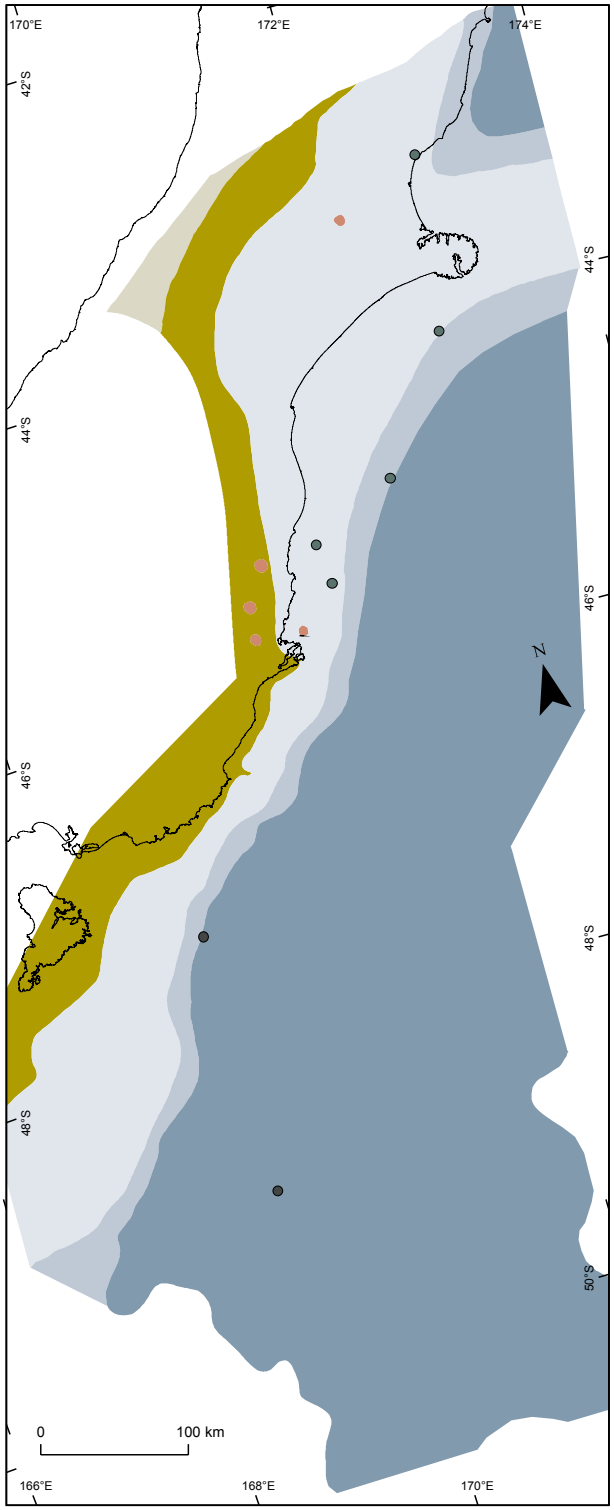
Erosion | Terrestrial | Shelf | Upper bathyal | Mid-Lower bathyal | Volcanics | well with interval present

Map sources: Field *et al.* (1989) Cook *et al.* (1999)

End Paleocene (56 Ma) paleogeography



Middle Miocene (14 Ma) paleogeography



Legend

Erosion | Terrestrial | Shelf | Upper bathyal | Mid-Lower bathyal | Volcanics | well with interval present

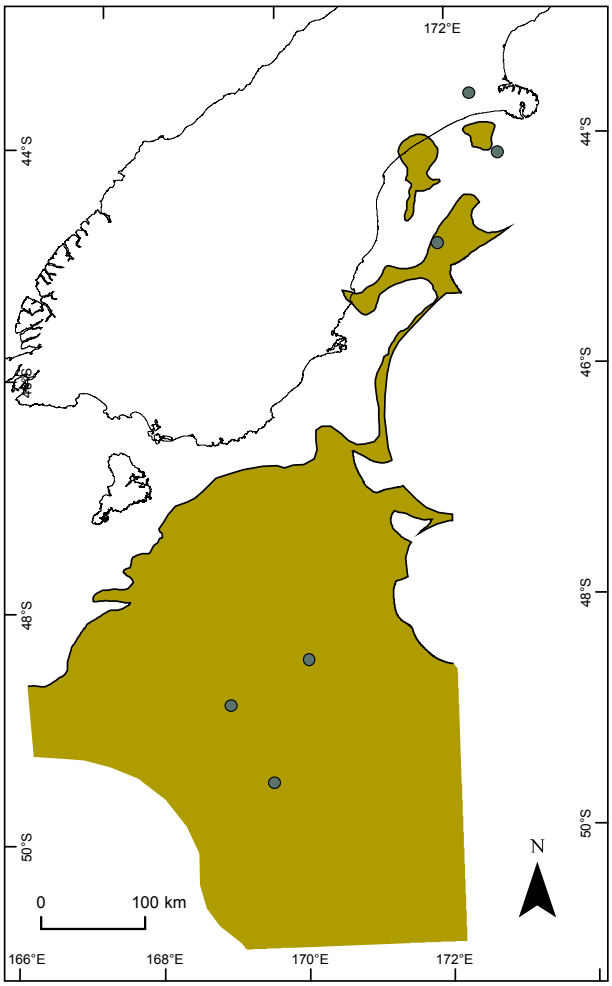
Map sources: Field *et al.* (1989) Cook *et al.* (1999)

GREAT SOUTH-CANTERBURY PROVINCE PLAYS

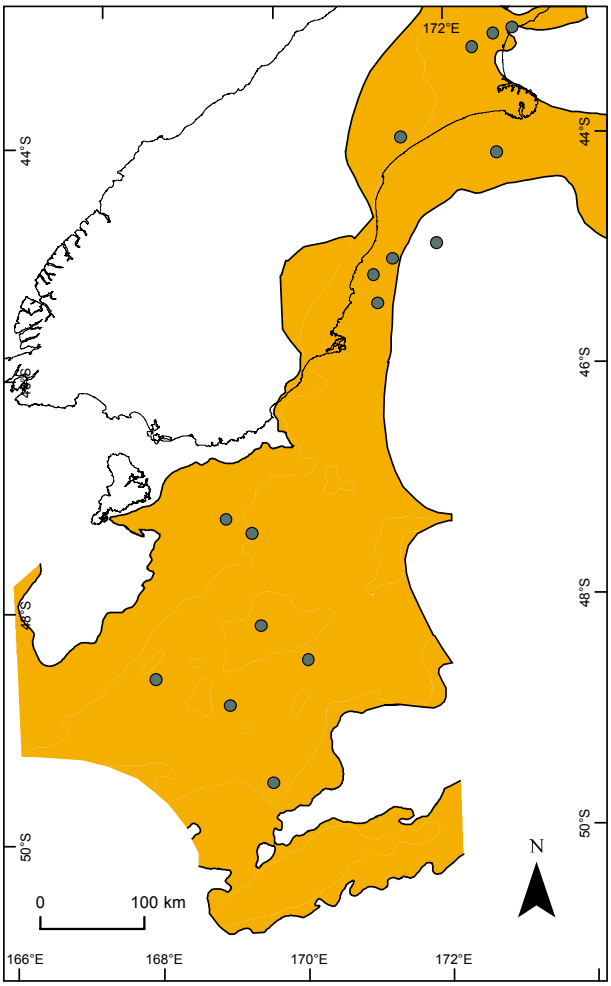
Most wells drilled in the province have targeted closures resulting from differential compaction of sediment draped across basement highs. These basement horsts mostly formed during mid- to Late Cretaceous normal faulting, though multiple stratigraphic levels of trap are present above the structures. Examples include the structures drilled by Resolution-1, Endeavour-1, and the Clipper-1, Kawau-1A and Galleon-1 discoveries. Kawau-1A, a sub-commercial gas-condensate discovery, remains the only successful test to date of hydrocarbons in the Great South Basin. At the time of discovery the overall reserve was estimated to be 461 BCF of gas. The Galleon-1 well encountered a 21 m-high gas-condensate column deemed non-commercial at the time. Clipper-1 was also a sub-commercial gas-condensate discovery, and is significant in that it intercepted a previously unknown mid-Cretaceous source rock succession.

Faulted anticlines of Miocene to Recent age, such as that drilled by Kowai-1 in Canterbury Basin, are a less common play type. Primary reservoir targets are Late Cretaceous to Eocene transgressive sandstones.

Mid Cretaceous play fairways



Late Cretaceous play fairways

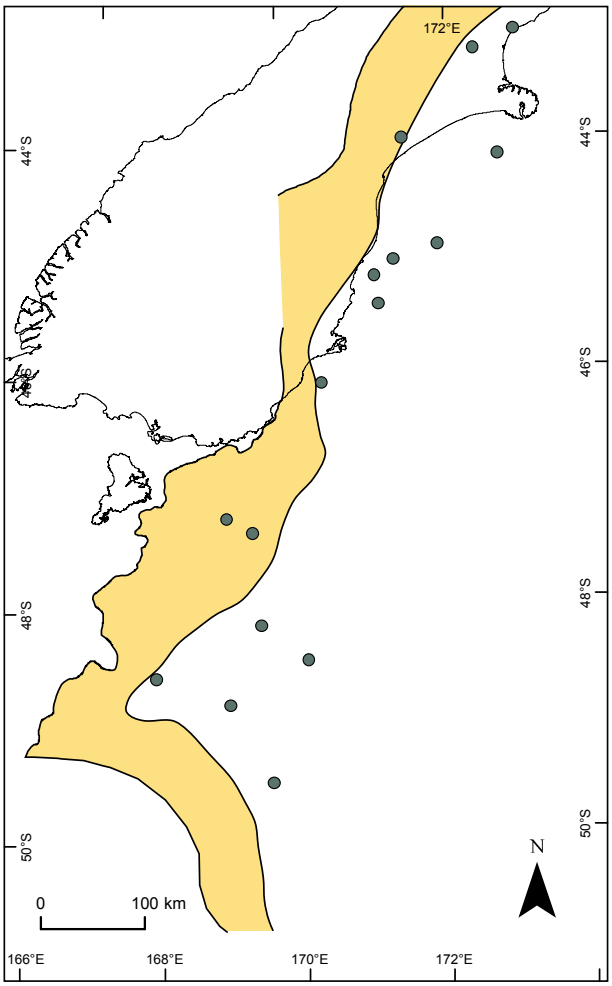


Transgressive sandstones may also be present in stratigraphic plays.

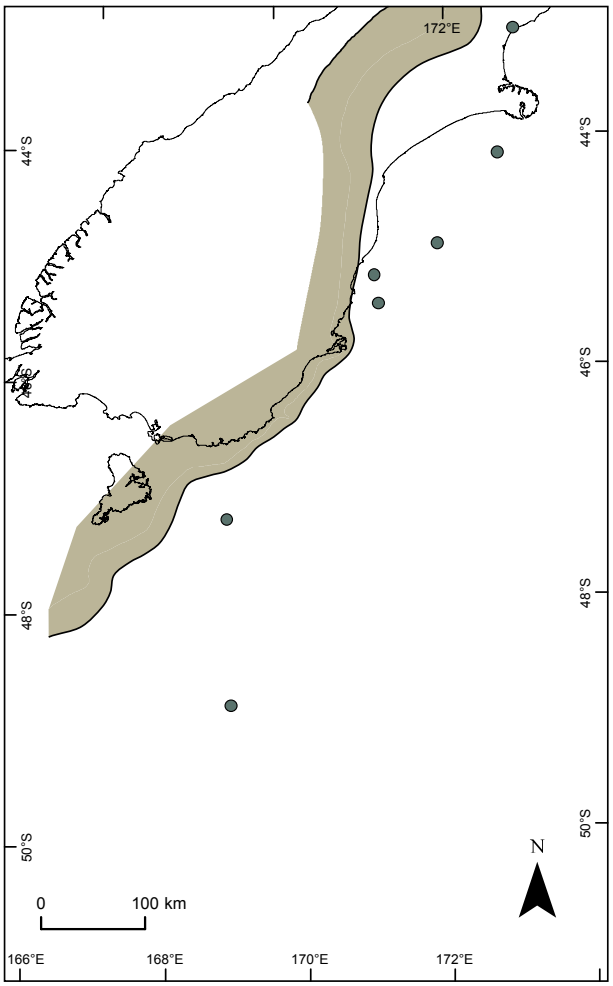
A series of broad, low-relief folds affecting relatively young basin-fill sediments lies below the present shelf and slope in the Great South Basin. Two were drilled by Toroa-1 and Tara-1 and both had good shows of hydrocarbons. Toroa-1 encountered oil staining in cuttings, and strong gas shows associated with oil. However the well was not tested due to mechanical failure. Tara-1 encountered 10–40% hydrocarbon saturation, gas bleeding from cuttings, and C4 and C5 hydrocarbons were detected.

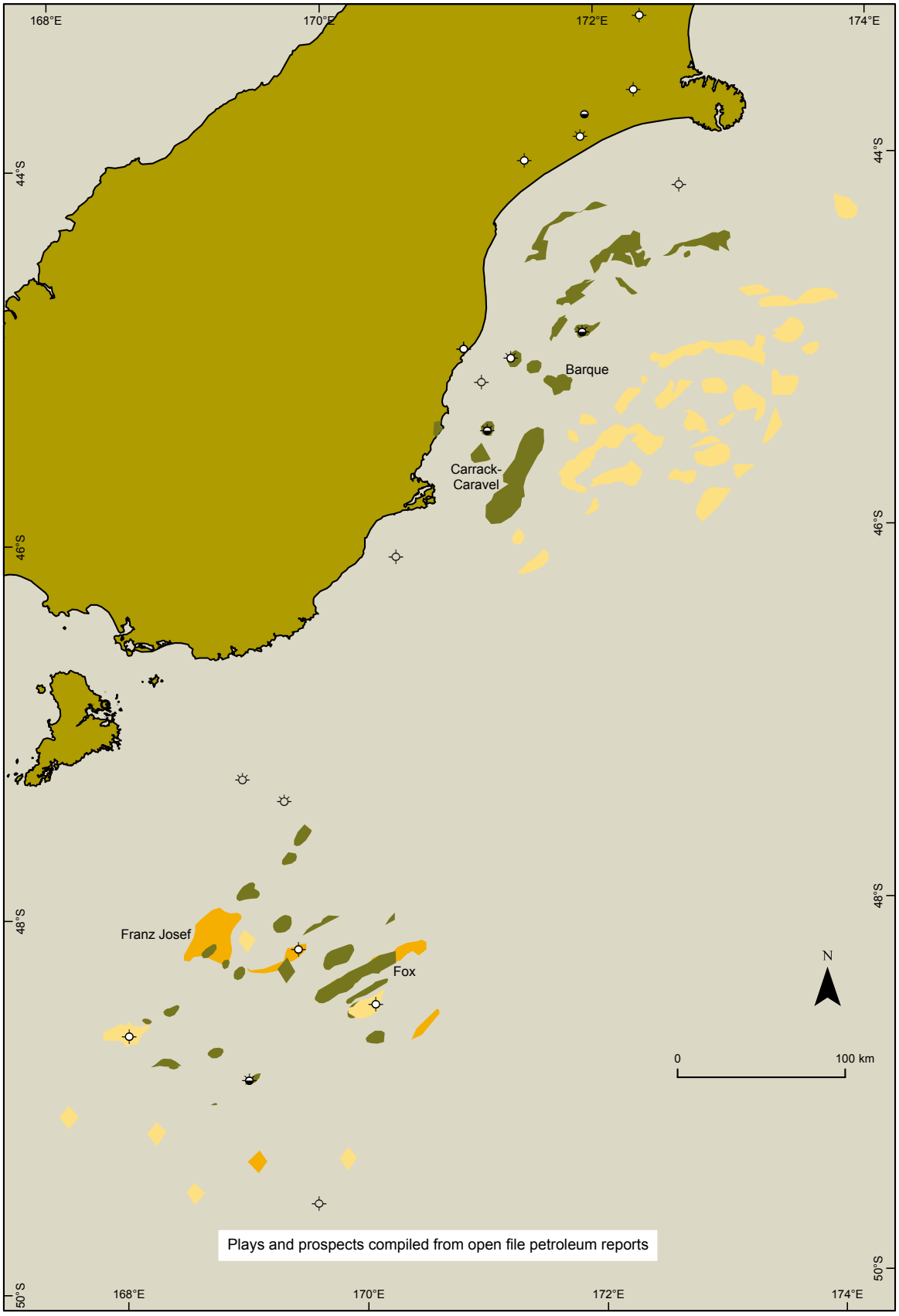
Turbidite fans may provide reservoirs for both structural and stratigraphic traps. Stratigraphic traps may be formed by depositional mounding and by up-dip pinch-out. Up-dip pinch-outs of Cretaceous to Neogene sandstones are likely along the northwest margin of the Great South Basin. Migration of oil into this area is suggested by the oil seep at Stewart Island.

Paleocene play fairways



Middle Miocene play fairways




168°E
170°E
172°E
174°E
44°S
46°S
48°S
50°S
Barque
Carrack-
Caravel
Franz Josef
Fox
N
0
100 km
Plays and prospects compiled from open file petroleum reports
Legend
Cretaceous
Paleogene
Unknown age

GREAT SOUTH-CANTERBURY PROVINCE RESERVOIRS

Potential reservoirs in the province are dominated by Late Cretaceous to Miocene sandstones, which are generally quartzose, and were deposited in environments ranging from fluvial and paralic to shelf and turbidite settings. In the Great South Basin there is also reservoir potential within Cretaceous to Eocene-aged fluvio-deltaic and transgressive marine sandstones, and Paleogene turbidite sandstones. Provenance of reservoir sands within the Canterbury Basin is likely to be largely from the Otago schist and its less metamorphosed equivalent, the Torlesse terrane. Other source areas locally include mid Cretaceous volcanics.

CRETACEOUS SYN-RIFT RESERVOIRS

The most promising reservoir rocks within the province are terrestrial sandstones of the Cretaceous syn-rift succession and fluvial, estuarine, and shallow-marine sandstones within the mid-Cretaceous to Eocene succession. Sandstones of mid Cretaceous age were primarily deposited in fluvial depositional settings. They comprise fining-up alluvial fan to fluvio-deltaic quartz-rich sands with interbedded lacustrine/overbank mudstone and coal. They record progressive onlap of basement rocks during initiation of post-rift subsidence (i.e. post 105 Ma). Earliest deposition appears to have been in Cretaceous half-grabens. Although prone to the effects of burial diagenesis, especially where buried to considerable depths, they could be effective reservoirs. In Canterbury Basin such rocks include the Kyeburn, Horse Range, and Clipper formations. Mid-Cretaceous rocks have been drilled by the Clipper-1 and Endeavour-1 wells. These rocks are 480 m thick at Clipper-1, where porosities of 1 to 16% and permeabilities of 0.07 to 11.5 mD were measured. Mid-Cretaceous rocks of the Hoiho Group are volumetrically the most significant and the most widespread reservoir unit in the Great South Basin, comprising conglomeratic and sandy lithofacies intercalated with fine-grained lacustrine, overbank and channel abandonment facies that include siltstones, mudstones, and coal measures. Reservoir quality is expected to reflect the nature of the provenance areas as well as depositional and diagenetic processes. The group is dominated by quartzose sands reflecting provenance both from granitoid, metaquartzites, and schistose basement terranes. Six wells, each above a basement high, penetrate the Hoiho Group in the Great South Basin, so not all facies variants of mid-Cretaceous will have been encountered. Reservoir quality of Hoiho Group sandstones may be degraded by the alteration of volcanogenic grains in areas adjacent to Murihiku or Brook Street Terrane basement rocks, as seen at Tara-1.

LATE CRETACEOUS RESERVOIRS

Latest Cretaceous Broken River formation sandstones crop out in the west of the Canterbury Basin and were penetrated in Resolution-1. The formation has measured porosities of 15 to 35% but there is little permeability data. They are generally fine- to medium-grained and interbedded locally with coals and glauconitic sands. The Pukeiwitahi and Taratu formations were intersected in Galleon-1 and Endeavour-1. These are generally quartz-rich sands that locally offer excellent reservoir properties. These paralic, sometimes coaly units are in places overlain by thick, mainly shoreface transgressive quartz-sands of the Late Cretaceous Wangaloa and Herbert formations (correlative of the Kawau Sandstone in the Great South Basin).

Late Cretaceous terrestrial sandstones were drilled in most wells within the Great South Basin, and some have excellent reservoir properties, with porosities up to 25%. They host oil and gas shows in Toroa-1, Pakaha-1 and Tara-1. The early Late Cretaceous Kawau Sandstone is potentially the best quality reservoir sand in the Great South Basin. It was deposited diachronously over much of the basin, in a transgressive shallow-marine setting. The formation is the reservoir for the gas-condensate accumulation in the Kawau-1A well. Late Cretaceous shelf sandstones of the Herbert Formation at Galleon-1 have porosities of 17% and permeabilities of 10 to 100 mD. High flow rates were obtained during testing of Galleon-1 in Canterbury Basin; flow rates of 2,300 bbl/d of condensate and 10 mmscf/d of gas were achieved, indicating the presence of good reservoir facies in the Late Cretaceous succession in offshore parts of the basin.

PALEOGENE RESERVOIRS

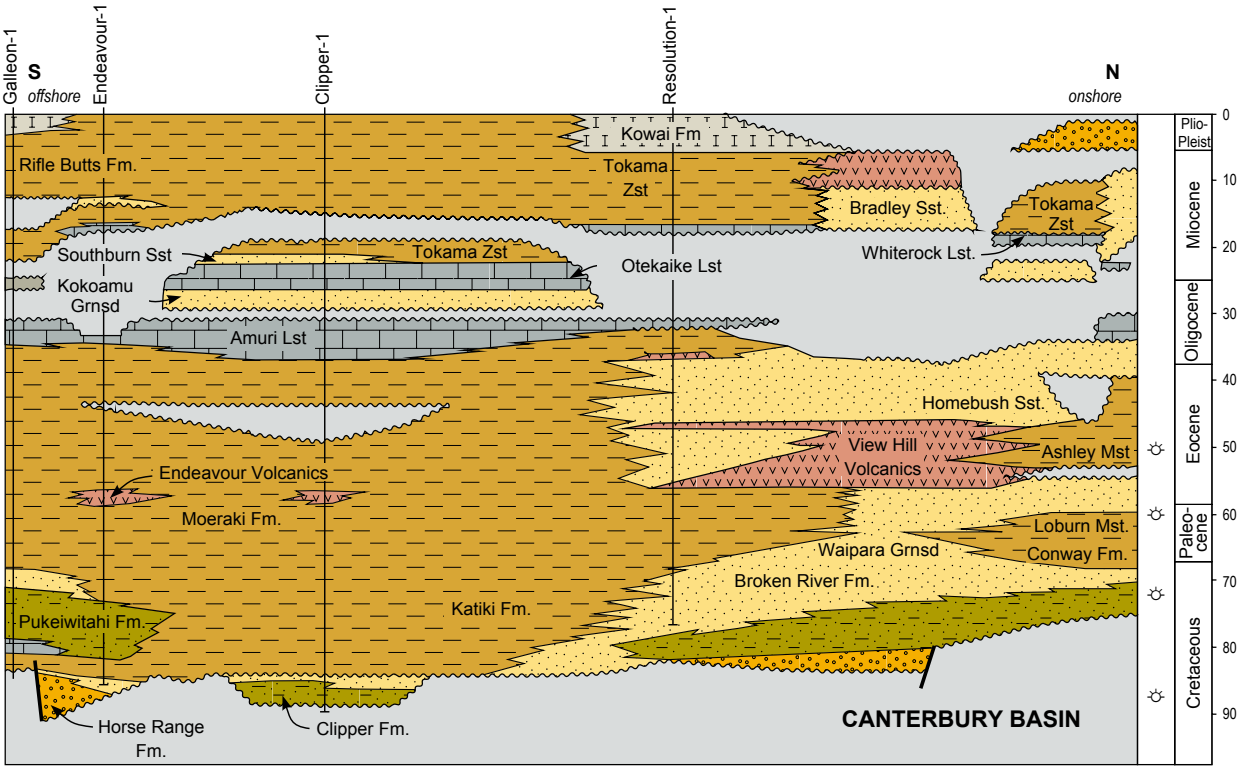
Late Cretaceous to Eocene turbidite sandstones and Paleocene–Eocene shallow marine sandstones deposited adjacent to highs also have reservoir potential. Late Cretaceous-Paleocene submarine fans have been recognised from seismic mapping in the deeper parts of the Great South Basin, but there is no drillhole data. The quartz-rich Paleocene aged shelfal Charteris Bay Sandstone, intersected at Resolution 1, can be hundreds of metres thick and has good porosity and permeability. Various Late Cretaceous to Eocene non-marine to shallow-marine sandstones can be expected around the western flank of the Great South Basin, adjacent to paleogeographic highs in central parts of the basin, and potentially along the basin's eastern margin. They were derived from the inundation and erosion of adjacent hinterland areas. Reservoir quality will reflect variability in provenance and depositional setting. Eocene deltaic deposits, interpreted to be related to increased tectonic activity associated with oceanic spreading between the Campbell Plateau and Antarctica, are another potential reservoir target. Such deposits may include beach and shoreface sands or lowstand deltas as potential reservoir facies. Terrestrial sandstones of Paleocene and Eocene ages are present in Takapu-1A, Tara-1, and Toroa-1. Good quality shallow marine sandstones of Early Eocene age, with porosities as high as 38%, were drilled in Rakiura-1. Shelf sands of Early Eocene age were also reported from Pakaha-1. Onshore in the Canterbury region, the Eocene-aged Homebush and Iron Creek sandstones – both shallow marine units – have porosities of 30 to 38% and permeabilities between 163 and 6,410 mD. Seismic surveys across the Great South Basin indicate that Paleocene and Eocene turbidite sandstone bodies are common in much of the deep water parts of the basin. Sandstones deposited in deep marine environments were reported from Pukaki-1.

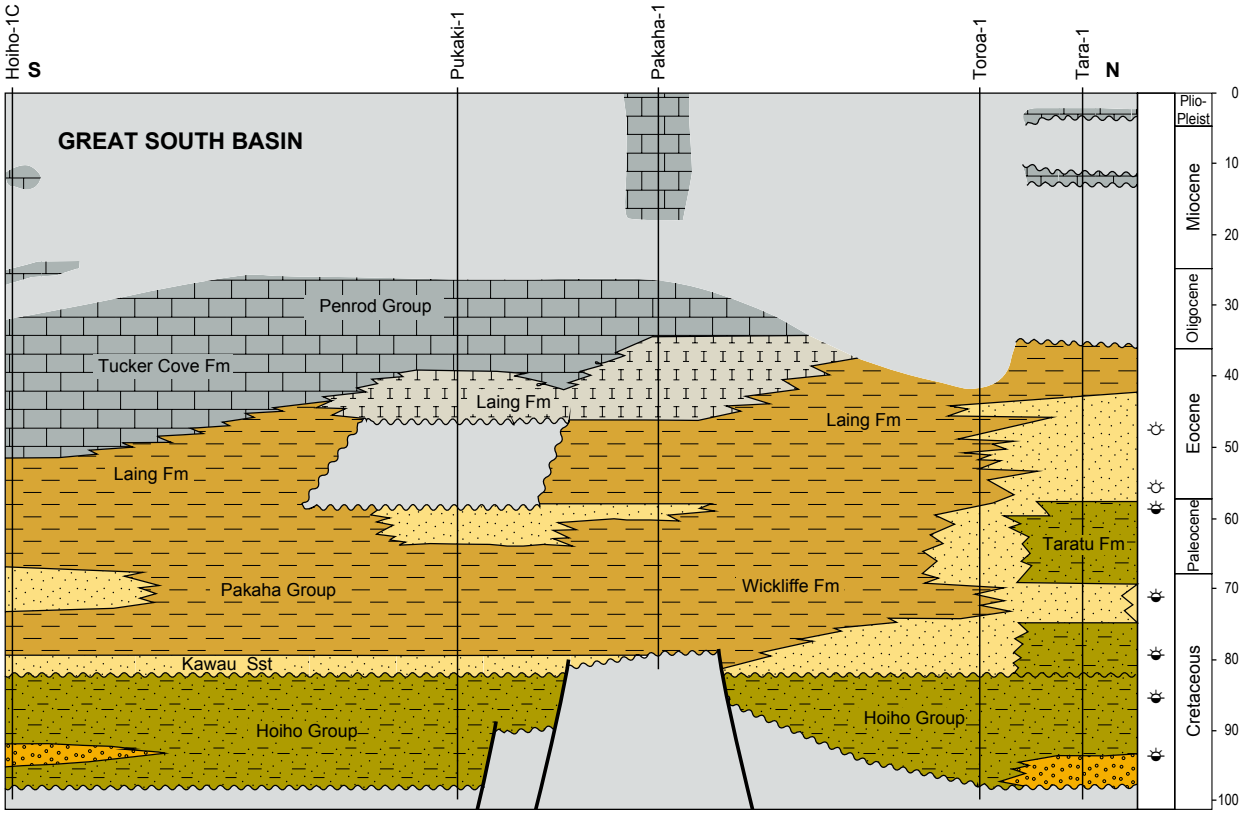
OLIGOCENE-MIOCENE RESERVOIRS

Oligocene and Miocene sandstones and limestones may also have reservoir potential in offshore Canterbury Basin, though these are perhaps best regarded as secondary reservoir targets. Miocene sandstones are generally less abundant in the Canterbury Basin, and are more lithic-rich than Cretaceous-Paleocene units. Calcareous and relatively coarse-grained reservoirs that have potential in northwest Canterbury Basin, such as the Mt Brown Formation, are unlikely to extend into offshore parts of the basin.

VOLCANIC RESERVOIRS

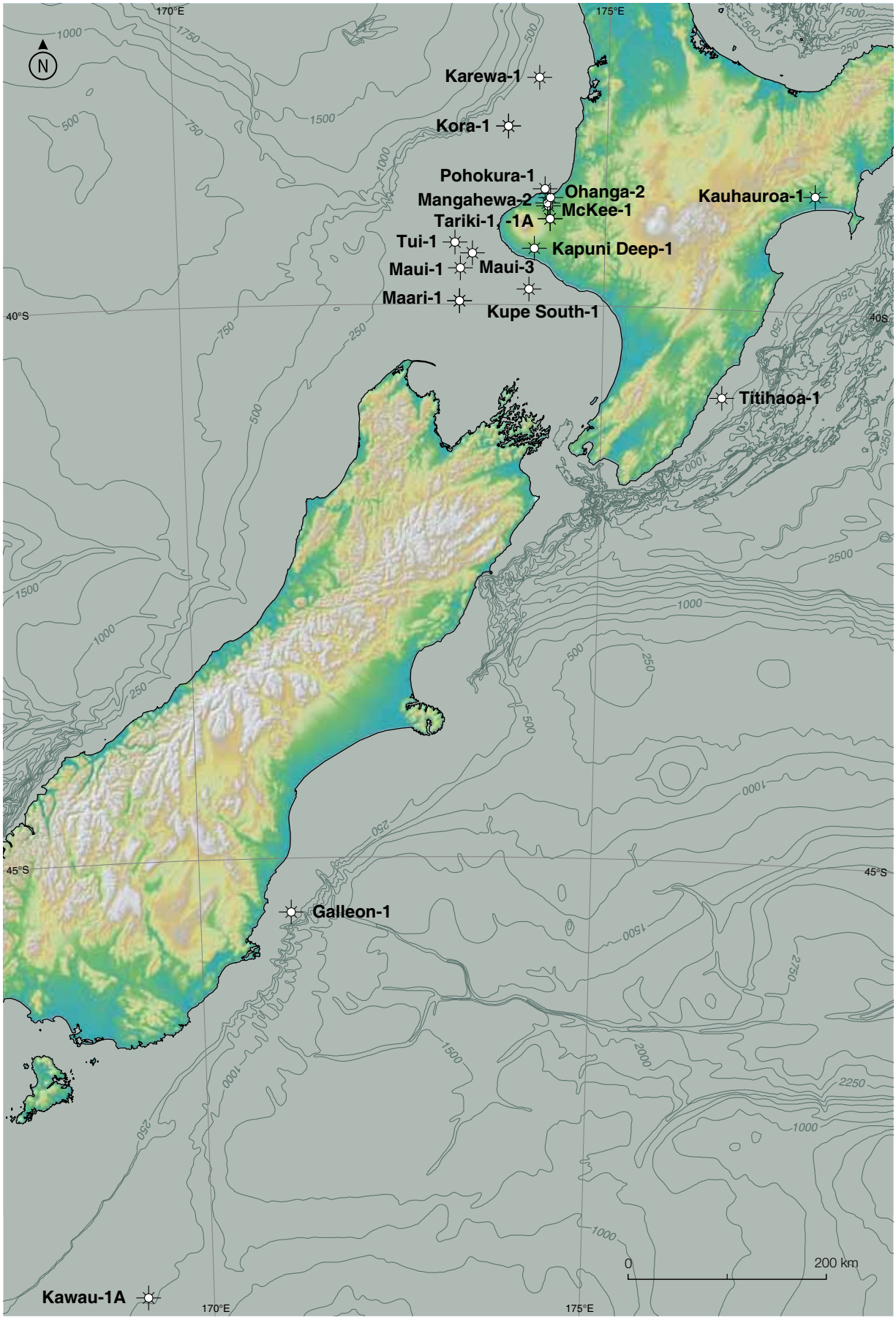
Cretaceous, Paleocene-Eocene, Oligocene, Miocene and Pliocene volcanic and volcaniclastic rocks are also present locally in the province and might offer additional reservoir potential.





Legend

Volcanic	Marl	Sandstone	Conglomeratic	Oil shows	Gas shows
Limestone	Mudstone	Coal measures	Non deposition	Oil & gas shows	


170°E
175°E
N
Karewa-1
Kora-1
Pohokura-1
Ohanga-2
Mangahewa-2
McKee-1
Tariki-1, -1A
Kauhauroa-1
Tui-1
Kapuni Deep-1
Maui-1
Maui-3
Maari-1
Kupe South-1
Titihaoa-1
40°S
45°S
Galleon-1
Kawau-1A
0
200 km

Discoveries and Developments

INTRODUCTION

The only commercial production today is from the Taranaki Basin. The first well was drilled in 1865 and petroleum has been continuously produced from the basin since about 1900. The modern era of exploration and production began in 1959 with the discovery of gas-condensate at Kapuni on the Taranaki Peninsula. Subsequently, the third offshore well drilled in New Zealand discovered the Maui gas-condensate field in 1969, which at the time was classed as one of the world's 'giant' offshore fields. Some wells have been drilled in other basins, but most remain largely unexplored.

The bigger discoveries made to date are gas-condensate. However, several oil fields have been discovered and geochemical research suggests similar oil potential of the coaly source rocks to that of prolific oil provinces in South East Asia.

At the end of 2011, total ultimately recoverable reserves (P50) in Taranaki amounted to 534.0 mmbbl of oil and condensate and 7,318.7 bcf of gas. Remaining un-produced reserves of oil and condensate were 148.9 mmbbl, and of gas were 1,803.5 bcf.

Whilst the Maui Field contained the bulk of reserves overall, this field is in decline. Two other fields, Maari and Pohokura, currently contain about 56% of the remaining oil and condensate reserves, whilst Pohokura accounts for 38% of remaining gas reserves. The rest of the reserves are distributed across 17 other producing fields.

A variety of successful structural and stratigraphic play types are present in the Taranaki Basin, and discoveries have been made at all potential reservoir levels except the Cretaceous. Of the offshore fields to have recently come on-stream, Maari and Kupe have been known for over two decades. On the other hand, the offshore Tui oilfield was brought onstream within just 4.5 years of its discovery in 2003. New discoveries have been made at a steady rate, and new play types are still being found both onshore and offshore.

Whilst several other basins have had wells drilled, these and far offshore frontier basins remain largely unexplored. Exploration began in the East Coast and Westland basins before 1900, when shallow wells were drilled near oil seeps. Prior to 1970 the majority of exploration was conducted onshore, but with the acquisition of increasing amounts of marine seismic reflection data, offshore well drilling soon followed. In offshore Canterbury, the Galleon-1 well (1985) produced a respectable 10 mmscf/d of gas and 2,300 bbl/d of condensate on test. In the Great South Basin, Kawau-1A in 1977 flowed 6.8 mmscf/d gas and has estimated reserves of 461 bcf. In 1998 an onshore gas discovery was made at Kauhauroa-1, on the North Island's East Coast. With cheap, readily accessible gas supply from Maui dominating the local marketplace, these discoveries were deemed un-commercial. Their size still precludes their economic viability, but they prove the existence of effective petroleum systems in the respective basins.

A period of concerted data acquisition by Crown Minerals since the mid-2000s, together with improved deep-water drilling and production technology, is shifting the focus of exploration further offshore, where the biggest new discoveries are anticipated. Parts of five frontier offshore basins (Great South, Canterbury, Pegasus, Deepwater Taranaki and Reinga) are currently licensed to large international companies.

This section includes information on two classes of discoveries; the first includes a selection of significant producing fields and the second is a selection of sub-commercial discoveries and shows to illustrate the range of currently known petroleum plays. Each class is ordered by date of discovery and all depths are in metres Measured Depth below Rotary Table (MD) and/or True Vertical Depth Sub Surface (TVDSS). Elevation depth datum is mean sea level. Production and reserves data are sourced from the Energy Data File 2011 and provided as production to date (1 January 2012) and remaining reserves.

Kapuni
Gas-Condensate Field

LOCATION	ONSHORE TARANAKI BASIN
CURRENT PERMIT	38839
AREA OF PERMIT	219 km²
OPERATOR	Shell Todd Oil Services
DISCOVERED	1959
YEAR ON STREAM	1970
ELEVATION	172 m
RESERVOIR DEPTH	3,300 to 3,800 m TVDSS
RESERVOIR ROCK	Mangahewa Formation
GEOLOGICAL AGE	Late Eocene
PLAY TYPE	Transgressive terrestrial to marginal marine sands in 4-way dip inverted anticline



PRODUCTION AND RESERVES

Gross cumulative production from the field as at 1 January 2013 is 1841 bcf gas and 68.3 mmbbl condensate. Estimated remaining reserves (P90) at 1 January 2013 are 116 bcf gas and 2.6 mmbbl condensate.

GEOLOGY

The Kapuni field is a structurally complex 4-way dip closure along the Manaia Anticline, a north-trending fold structure of regional scale bounded on the west side by the Manaia Fault, a major reverse-reactivated normal fault, inverted by Miocene compression events.

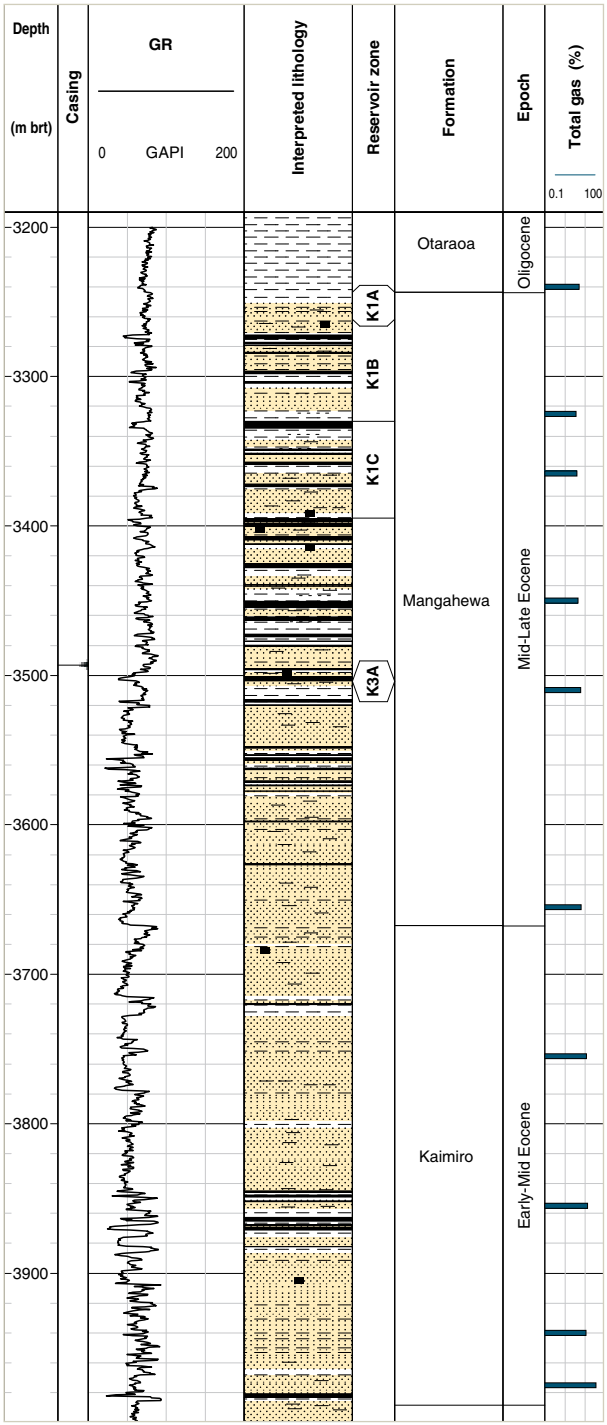
The top of the Kapuni structure shows a vertical closure of about 790 m crossed by a pronounced saddle. Some stratigraphic traps may be present where point bar sandstones terminate against flood basin siltstones. The field produces from multiple sandstone reservoirs in the 400 m-thick Eocene Mangahewa Formation. The formation consists of sandstone, coal, carbonaceous mudstone, siltstone and mudstone lithologies, which were deposited in terrestrial to marginal marine environments. It comprises a series of stacked sandstone-mudstone-coal cycles, whose origin is attributed to periods of fluctuating sea level.

RESERVOIR CHARACTERISTICS

The main reservoir zones are designated K1A, K1C, K3A, and K3E. High quality reservoir sandstone intervals are present with porosities as high as 20 to 25% and permeabilities of over 500 mD. The average porosity is 15% for depths of about 3,700 m in the field.

FACILITIES

Kapuni facilities currently consist of 16 wells located on nine well sites around the Kapuni Production Station. Low Temperature Separation Units are installed on the well sites to separate the gas from the liquids (condensate and water) and pre-condition the gas. Via separate pipeline gathering systems, gas and liquids are transported to the Kapuni processing plant, where the bulk of the gas is diverted, without significant further treatment, for sale into the NGC-owned Gas Treatment Plant. Because gas produced from the Kapuni Field contains approximately 45% CO_2, amine scrubbing is used to reduce the CO_2 content to acceptable pipeline-entry levels. NGC removes CO_2 from the gas and also recovers liquefied petroleum gas (LPG) and natural gasoline products. The in-field gas gathering system also feeds the pipeline to the co-generation facilities at the Fonterra dairy plant in Whareroa. The liquid streams separated by the well site facilities are treated in the Kapuni Production Station; treatment involves separation and stabilisation of water and condensate phases. The water is disposed of via a water injection well, while the condensate is transported by a dedicated cross-country pipeline to Port Taranaki in New Plymouth. From there it is exported by marine tankers to refining destinations.



KAPUNI DEEP-1

SPUD DATE	3 May 1983
TD	5,664 m MD

Kapuni Deep-1 was drilled to test any deeply buried reservoirs below the known producing zones of the Kapuni Group. The 1975 Kupe discovery, offshore to the south, had found coal measures in the Paleocene and seismic data indicated that these appeared to thicken to the north. Kapuni Deep-1 reached a TD of 5,664 m and drilled thick Paleocene coal measures, as expected. Kapuni Deep-1 was later re-entered as Kapuni-13 and completed as a Mangahewa Formation producer.

The well showed that the source rocks in Kapuni Deep-1 are generally richer and more oil-prone than in the equivalent sections of previously examined wells, and appear to become mature for oil generation between about 3,500 and 4,000 m. No marine source rocks were identified, but oil extracted from the drilling mud at 5,496 m had marine characteristics that did not correlate with the Kapuni Group source rocks.

Mangahewa
Gas-Condensate Field

LOCATION	ONSHORE TARANAKI BASIN
CURRENT PERMIT	38150
AREA	44 km²
OPERATOR	Todd Energy
DISCOVERED	1961
YEAR ON STREAM	2001
ELEVATION	120 m
RESERVOIR DEPTH	From 3,300 m TVDSS
RESERVOIR ROCK	Mangahewa Formation
GEOLOGICAL AGE	Late Eocene
PLAY TYPE	Transgressive terrestrial to marginal marine sands in broad, low-relief anticline



PRODUCTION AND RESERVES

The Mangahewa structure was identified as a prospect early in the exploration of Taranaki Basin and was first drilled in 1961. Mangahewa-1 found some gas in the Kapuni Group but was considered sub-economic. Further exploration in the 1990s led to the drilling of the Mangahewa-2 discovery well in 1996. This well located some 20 m of hydrocarbon reservoirs over several zones of "tight gas". Three zones were hydraulically fractured and only one zone was utilised for short and long-term production testing. Following the flow tests, acquisition of 3D seismic in 1997, and appraisal of the northern flank of the structure with the Ohanga-2 well (1998), permanent production from the field started in September 2001.

Gross cumulative production of the field at 1 January 2013 is 74.8 bcf gas and 1.79 mmbbl condensate. Estimated remaining reserves (P90) at 1 January 2013 are 97.5 bcf gas and 1.6 mmbbl condensate. Todd Energy has reassessed the reserves and appraisal/development opportunities within the Kapuni formation based on well performance and subsurface studies. This has resulted in the drilling of appraisal wells Mangahewa-03 (2007), Mangahewa-04 (2009), and Mangahewa-06 (2009). The results of those appraisal wells provided the foundation for the Mangahewa Expansion Project, which was formally announced in January 2012. This expansion project, which is expected to cost in excess of $750 million, and which targets to mature 450 PJ of contingent gas resources into 2P gas reserves, consists of: approximately 25 appraisal and development wells, gas processing facility expansion, installation of additional gas export pipeline and additional gathering lines and wellsite facilities.

GEOLOGY

Mangahewa is a gas-condensate field located west of the McKee Field and 18 km southeast of New Plymouth. The structure is a broad, low-relief anticline at Kapuni Group level, formed by regional shortening along the eastern margin of Taranaki Basin in the Early to Mid-Miocene. The field is on the same broad structural trend as Pohokura, offshore to the north.

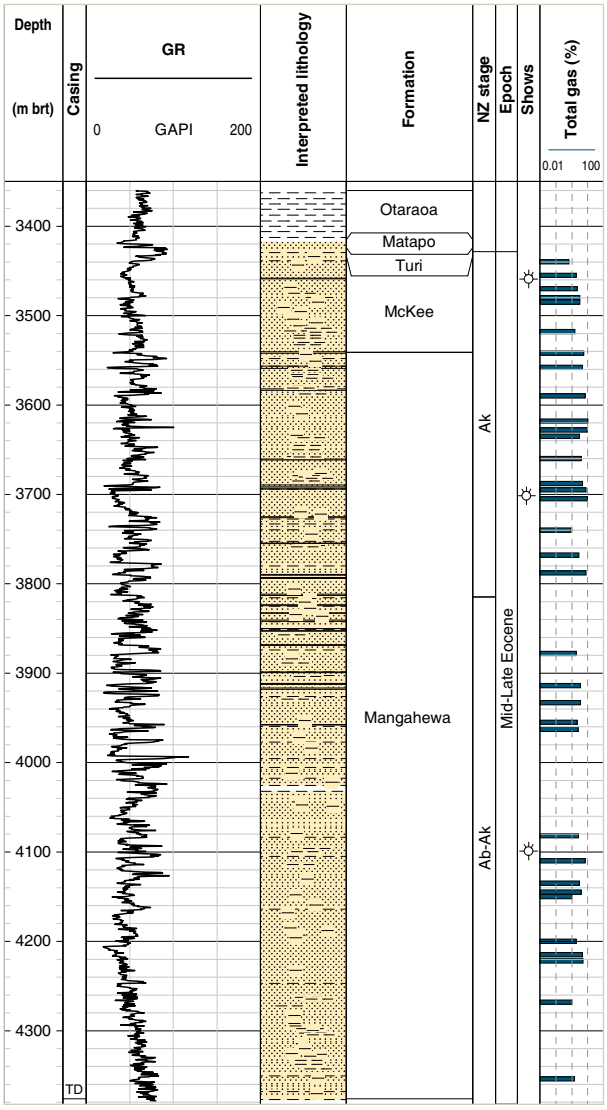
RESERVOIR CHARACTERISTICS

In the Mangahewa Field, the Eocene Mangahewa reservoir sands are sealed by intra-formational shales. Vertical isolation of the various reservoir layers is indicated by different fluid contents, gas composition and pressures.

The reservoir has been more deeply buried than it is today and reservoir porosity and permeability are relatively low as a result. Reservoir quality also varies from sand to sand, depending on grain size, diagenetic effects, and clay content. Porosities range up to 12%, and the main producing interval (MA-72) has an average porosity of 9% and average permeability of 4.5 mD. Production delivery relies on artificial fracturing of the reservoir.

FACILITIES

The Mangahewa facilities consist of eight wells with associated wellsite facilities, connected via multiple gathering lines to the McKee-Mangahewa Production Station. Fluids are treated in a single high pressure gas train. The McKee facilities are used for the provision of utilities and for treating condensate, water and LTS liquids.



MANGAHEWA-2

SPUD DATE	15 September 1996
TD	4,376 m MD

Mangahewa-2 was drilled on the crest of a large, low amplitude inversion anticline. Primary targets were the McKee and Mangahewa formations; the Kaimiro Formation was the secondary target. A gas-condensate discovery was made within Mangahewa Formation reservoirs. The lower Mangahewa Formation tends to be of lower permeability but it contains significant thicknesses of gas-bearing sandstone.

Maui
Gas-Condensate Field

LOCATION	OFFSHORE TARANAKI BASIN
CURRENT PERMIT	381012
AREA OF PERMIT	784 km²
OPERATOR	Shell Todd Oil Services
DISCOVERED	1969
YEAR ON STREAM	1979
WATER DEPTH	110 m
RESERVOIR DEPTH	From 2,700 m TVDSS
RESERVOIR ROCK	Mangahewa, Kaimiro and Farewell formations
GEOLOGICAL AGE	Paleocene and Eocene
PLAY TYPE	Transgressive terrestrial to marginal marine sands in fault-bound dual-crested anticline



PRODUCTION AND RESERVES

Gross cumulative production of the field as at 1 January 2013 is 3579 bcf gas and 185 mmbbl oil/condensate. Estimated remaining reserves (P90) at 1 January 2013 are 80 bcf gas and 3.2 mmbbl oil/condensate.

GEOLOGY

The field is a large, low relief anticline of approximately 150 km² areal extent, with the northeast referred to as Maui A, and the southwest as Maui B. The two components are separated by a saddle. The field is bounded to the west by the Whitiki Fault and to the east by the Cape Egmont Fault. Both faults have had reversals of movement, which were critical to forming the Maui structure. The field is in an area of the basin where Miocene shortening and subsequent extension overlap. The Maui structure is interpreted as an inversion structure formed by fault-bend folding during compressional reactivation of the Whitiki Fault.

The field produces from three reservoirs (C, D and F sands) within the Mangahewa, Kaimiro and Farewell formations of the Eocene to Paleocene Kapuni Group. These formations were deposited on a relatively narrow coastal plain along a fluctuating, northeast-trending shoreline. Movement of the shoreline back and forth across the Maui area, and fluctuations in the rate and locus of sediment supply, have resulted in highly cyclic, intercalated coastal plain and shallow marine strata.

RESERVOIR CHARACTERISTICS

The bulk of gas reserves in the Maui Field are contained in the C sands reservoir, with gas and thin oil columns in the D sands and oil in the F sands reservoir. Porosities in the C sands are up to 27% and permeabilities from 460 to 2,000 mD. The D sands have over 20% porosity and 200 to 500 mD permeability. In the F sands, porosities are over 20% and permeabilities up to several hundred mD.

FACILITIES

Maui Field facilities comprise four major components:

- a multiphase pipeline connects the Maui A platform to the processing plant at Oaonui
- the Maui Production Station onshore at Oaonui operates to bring gas and condensate to pipeline specifications
- the Omata Tank Farm is located at the port of New Plymouth. Processed condensate and associated products are stored here before shipping
- the Maui B platform (15 km from the Maui A) is connected to Maui A via an undersea pipeline. Oil from the F sands was formerly fed directly to Floating Production Storage and Offloading (FPSO) vessel.

MAUI-1

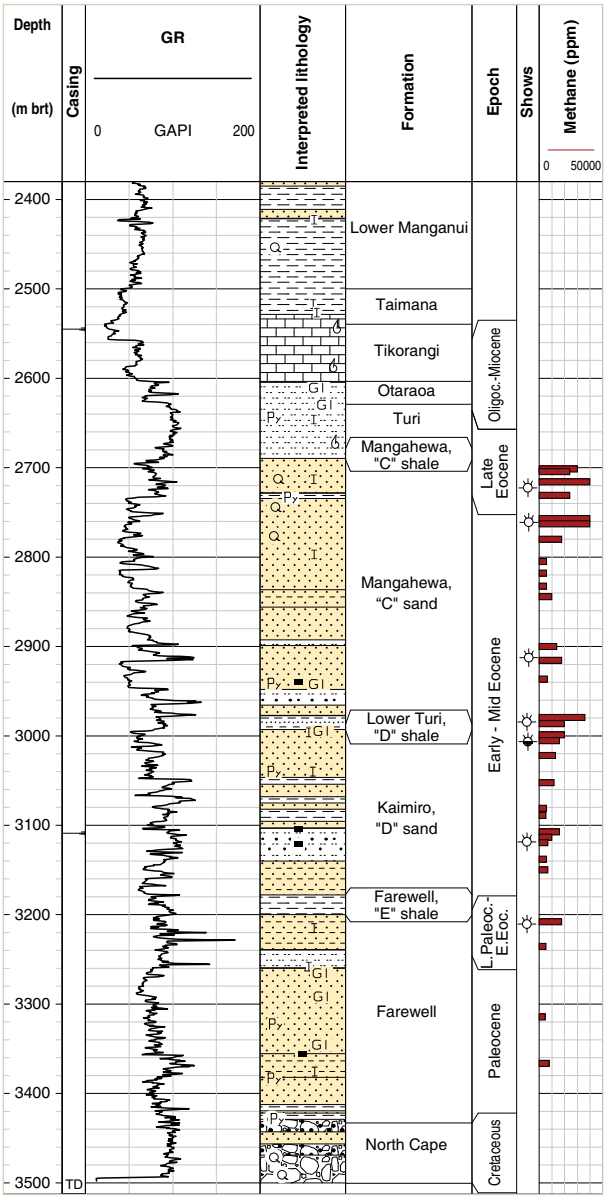
SPUD DATE 27 January 1969
TD 3,512 m MD

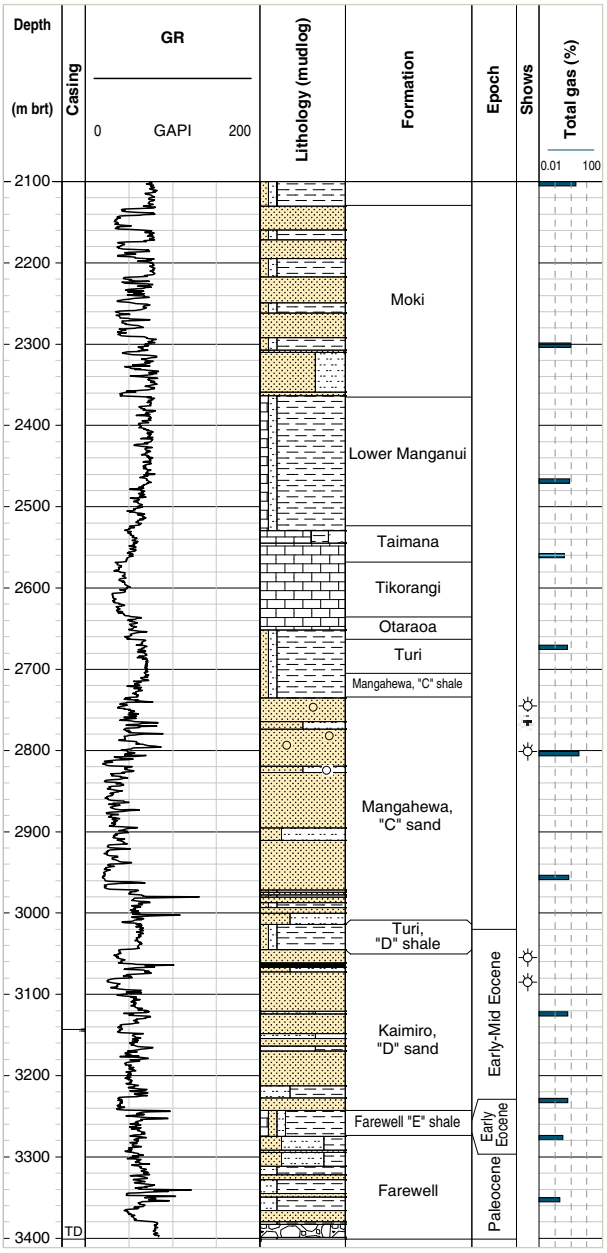
The field was discovered in 1969 by the Maui-1 well, drilled in the southern B area, and two following wells (Maui-2 and Maui-3) drilled on the A area structure. Maui-1 proved the presence of petroleum, mainly gas and condensate, but with some 10 m of net oil in Eocene sandstone reservoirs with good porosity and permeability. At the time, the Maui Field was one of the largest gas fields in the world, with initial reserves of 3,830 bcf of gas.

MAUI-3

SPUD DATE 25 December 1969
TD 3,401 m MD

The appraisal well Maui-3 was drilled 21.5 km north of Maui-1. New Zealand's premier offshore oil and gas producing facility, Maui-A, was later built nearby. It was brought on stream in 1979. The uppermost C sand was hydrocarbon-bearing, with identical GOC and OWC as in Maui-1 and -2. The D sand, with a gross hydrocarbon-bearing thickness of 41 m, had a lower OWC than in Maui-1 indicating separate reservoirs in the two culminations. The average porosity of gas-bearing sands in Maui-1 and -3 is 18.8%; permeabilities range from 460 to 2,000 mD.





McKee
Oil and Gas Field

LOCATION	ONSHORE TARANAKI BASIN
CURRENT PERMIT	38086
AREA	27.4 km²
OPERATOR	Todd Energy
DISCOVERED	1979
YEAR ON STREAM	1984
ELEVATION	90 to 150 m
RESERVOIR DEPTH	2,000 to 2,400 m TVDSS
RESERVOIR ROCK	McKee Formation
GEOLOGICAL AGE	Late Eocene
PLAY TYPE	Transgressive marginal marine sands in thrust-faulted anticline



PRODUCTION AND RESERVES

Gross cumulative production of the field at 1 January 2013 is 161 bcf gas and 47.3 mmbbl oil. Estimated remaining reserves at 1 January 2013 were 23.0 bcf gas and 0.8 mmbbl oil. Oil production from the field is primarily through water drive.

During the early 1990s oil production steadily declined, while at the same time gas and water production increased. The re-injection project was upgraded by Todd Energy in 2005, with gas from the northern part of the field being re-injected into the central part. Most down-dip oil wells require gas lift to continue oil flow, and a small water-flood operation in the centre of the field has stimulated production. Treated waste water is injected in a water disposal well.

GEOLOGY

The McKee oil and gas field, located 14 km southeast of Waitara in the northern part of Taranaki Peninsula, is a mature onshore producer.

The McKee Field is developed in a thrust-faulted anticline within the Tarata Thrust Zone. The main structure evolved through westward overthrusting on a low-angle fault, as a result of regional compression in the Early to Mid-Miocene. The field is structurally complex and is broken by faults antithetic to the main controlling west-verging fault. Reservoir strata are steeply dipping, at up to 60°. Hydrocarbons in the field are structurally trapped in north-south trending elongate culminations within individual fault-bound blocks and five distinct structural areas with varying pressure and fluid properties are identified.

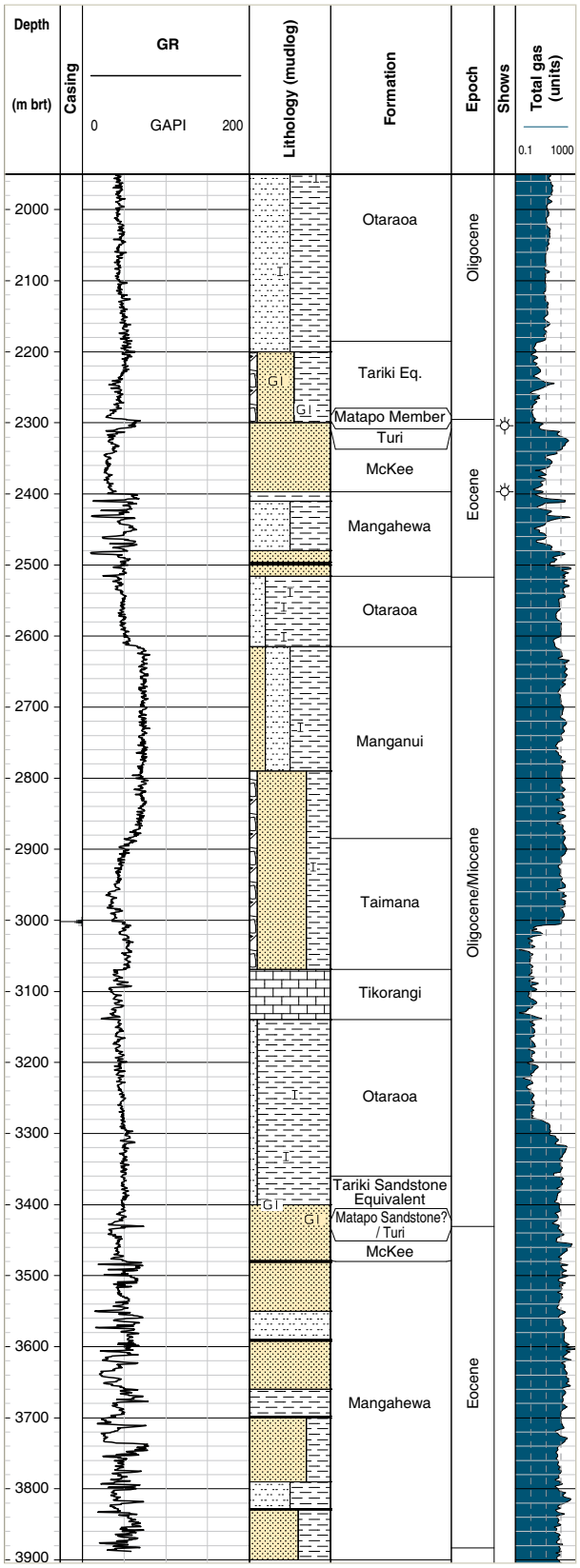
The principal reservoir is the Late Eocene McKee Formation of the Kapuni Group, a 60 to 100 m thick interval dominated by shoreline sandstones, with some interbedded mudstone and coal. Sand to shale ratio is high but variable in the McKee Formation, particularly in the southern part of the field. In central and northern parts of the field, an intra-McKee Formation shale unit is present, and is up to 15 m thick at Pouri-1A, and 18 m thick at Pukemai-1B.

RESERVOIR CHARACTERISTICS

Sandstones in the McKee Formation have porosities from 12% to 23% and low to medium permeabilities up to 500 mD, with a trend toward better rock properties in upper stratigraphic intervals. In some reservoir zones quality is reduced by widespread diagenetic modification, principally through pressure solution of quartz, the development of clay, grain fracturing due to overthrusting, and carbonate cementation.

FACILITIES

In 2010 the McKee Field includes 40 wells, linked through product-gathering lines to the McKee Production Station. At the production station, well fluids are separated into gas, oil and water steams and then stabilised. The treated gas is compressed and injected into gas export lines. The stabilised oil is pumped via a 40 km oil line to export storage facilities at Port Taranaki in New Plymouth. Treated produced water is injected in a water disposal well. In late 2011 a LPG recovery plant (27,000 tonne/annum) was commissioned, which can run on imported Pohokura gas and/or McKee-Mangahewa gas.



MCKEE-1

SPUD DATE	19 October 1979
TD	3,900 m MD

McKee-1 was drilled as an exploration well to test the McKee structure. The relatively shallow depth of the overthrust Kapuni Group made the structure particularly attractive and the well was subsequently drilled to TDwith no major problems. Waxy, light oil, paraffinic condensate and gas were tested at sub-commercial outputs.

Maari-Manaia
Oil Field

LOCATION	OFFSHORE TARANAKI BASIN
CURRENT PERMIT	38160
AREA	80 km²
OPERATOR	OMV New Zealand Limited
DISCOVERED	1983
YEAR ON STREAM	2009
WATER DEPTH	101 m
RESERVOIR DEPTH	Stacked reservoirs, between 1,200 to 1,600 m and 2,000 to 2,200 m MD.
RESERVOIR ROCK	Multiple reservoir units, including shore-face and turbidite sands
GEOLOGICAL AGE	Middle Miocene and Middle Eocene
PLAY TYPE	Southern Inversion Zone, Neogene structure with reservoirs primarily in Middle Miocene Moki Formation turbidite sands. Deeper Eocene shallow marine sands are an additional target



PRODUCTION AND RESERVES

The Maari-Manaia Field was developed on the basis of estimated recoverable reserves of about 50 mmbbl of oil in Moki Formation sands. The successful appraisal of the Eocene Mangahewa Formation in the Manaia prospect has added another 50 mmbbl oil to reserves. Initial production from the Maari-Manaia Field was about 40,000 bopd.

Gross cumulative production of the field at 1 January 2013 is 20.4 mmbbl oil. Associated gas production is not reported but is consumed on site. Estimated remaining reserves at 1 January 2013 were 20.1 mmbbl oil.

GEOLOGY

The Maari-Manaia Field is currently New Zealand's largest offshore oilfield. Discovered by the Moki-1 well in 1983, it is 80 km from shore in the Southern Inversion Zone. Additional appraisal wells were drilled in 1985, 1998 and 2003, with the final development decision made in 2005. In 2009, discovery of significant additional oil accumulations in the adjacent Manaia prospect, originally identified in the 1970s-drilled Maui-4 well, effectively doubled field reserves.

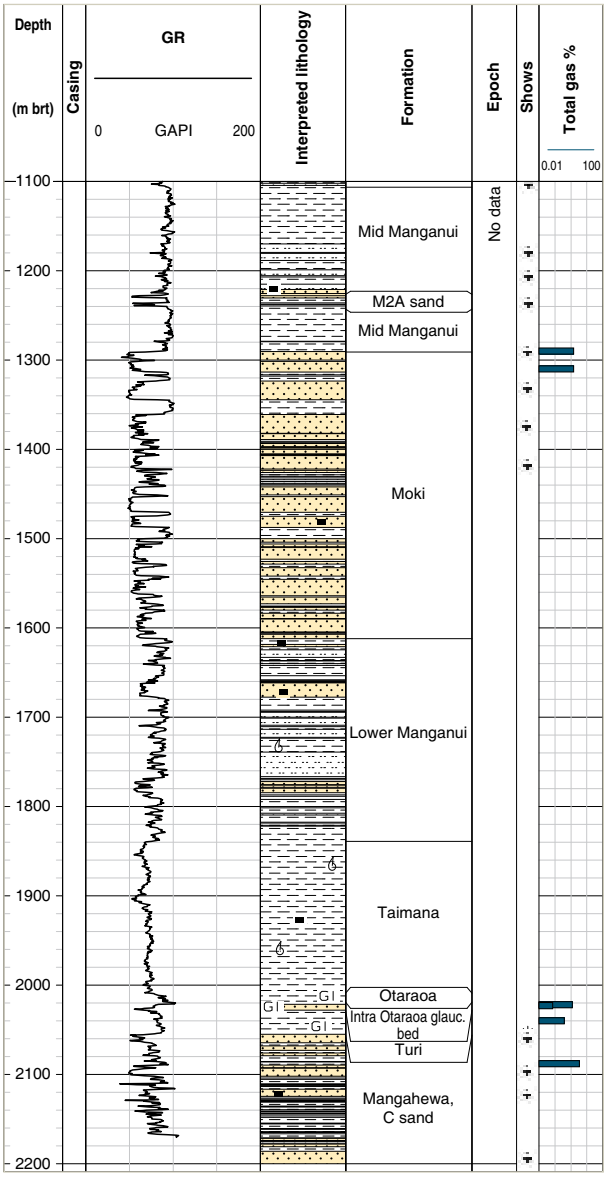
The Maari structure is a north-south trending closure. The main reservoir is Middle Miocene turbidite sands of the Moki Formation, with additional accumulations in younger M2A and older Eocene Kapuni Group sands. The closure was formed by Neogene shortening.

RESERVOIR CHARACTERISTICS

Reservoir quality at Maari is moderate, requiring some artificial lift and water injection for pressure support. Moki Formation sands tested by Maari-1 had porosities of 15 to 26% and permeabilities of 98 mD. The horizontally-drilled Maari-1A sidetrack showed that successful production could be achieved from the Moki Formation via horizontally -completed wells.

FACILITIES

The Maari-Manaia offshore production site has a wellhead platform (WHP) and an FPSO vessel, the Raroa. Five production wells are located in the Maari sector, with horizontal reservoir sections up to 2,000 m long. The Manaia-1 well in the Manaia sector of the field is the longest extended-reach well drilled in New Zealand. The 147 m-tall WHP "Tirotiro Moana" is usually un-manned. The FPSO vessel has a storage capacity of 590,000 bbl, a production capacity of 40,000 bopd, and water injection capacity of 40,000 bwpd. It is located 1.5 km from the WHP. Maari-Manaia produces a waxy oil, which requires down-hole heating, flow line insulation, and heating processes on the FPSO and WHP.



MAARI-1

SPUD DATE 30 October 1998
TD 2,200 m MD

Maari-1 was an exploration/appraisal well primarily drilled to test the Middle Miocene Moki Formation discovery of Moki-1. Secondary objectives were the younger Miocene M2A sands, and Eocene Kapuni Group sands. Moki Formation sands were found to be oil-bearing between 1,265.4 and 1,318.6 m TVDSS. The shallower M2A sand was also found to be oil-bearing, with high porosities of up to 26% and 61% oil saturation. The deeper Kapuni Group sands contained an 8 m gas column and a 33 m oil column.

Following the confirmation of moveable oil in the Moki Formation, Maari-1 was plugged back, and the Maari-1A sidetrack was horizontally drilled to test the formation. Of the 654 m drilled, 563 m was oil-bearing, with an average porosity of 24% and an oil saturation of 75%. On test, Maari-1A produced 4,370 bopd, and was subsequently suspended as an oil discovery.

TAWN
Gas-Condensate and Oil Fields

LOCATION	ONSHORE TARANAKI BASIN
CURRENT PERMIT	38138
AREA	14.4 km²
OPERATOR	NZEC (Tariki, Waihapa, Ngaere), Contact Energy (Ahunoa)
DISCOVERED	1986
YEAR ON STREAM	1996
ELEVATION	200 to 240 m
RESERVOIR DEPTH	From 2,050 m TVDSS
RESERVOIR ROCKS	Tikorangi Limestone and Tariki Sandstone
GEOLOGICAL AGE	Oligocene and Early Miocene
PLAY TYPE	Fore-deep slope turbidite sands and fractured limestones in thrust-faulted anticline



PRODUCTION AND RESERVES

Gross cumulative production of the field at 1 January 2013 is 124 bcf gas and 27.0 mmbbl oil/condensate. Remaining reserves at 1 January 2013 estimated as P90 are virtually zero.

GEOLOGY

The Tariki, Ahuroa, Waihapa and Ngaere (TAWN) fields are in the central part of Taranaki Peninsula, about 7 km east of Stratford, along the eastern boundary of Taranaki Basin.

The TAWN fields are formed in overthrust structures within the Tarata Thrust Zone, sub-parallel to the Taranaki Fault. These structures are controlled at depth by low-angle reverse faults of Early to Middle Miocene age. Younger steeply-dipping normal faults of Plio-Pleistocene age offset the shallow stratigraphic succession above the fields.

Primary reservoirs are fractured limestone of the Oligocene to Early Miocene Tikorangi Formation in the Waihapa/Ngaere oil field and the Oligocene Tariki Sandstone Member of the Otaraoa Formation in the Tariki and Ahuroa gas/condensate fields. The Waihapa Field also contains gas/condensate in the Eocene Mangahewa Formation.

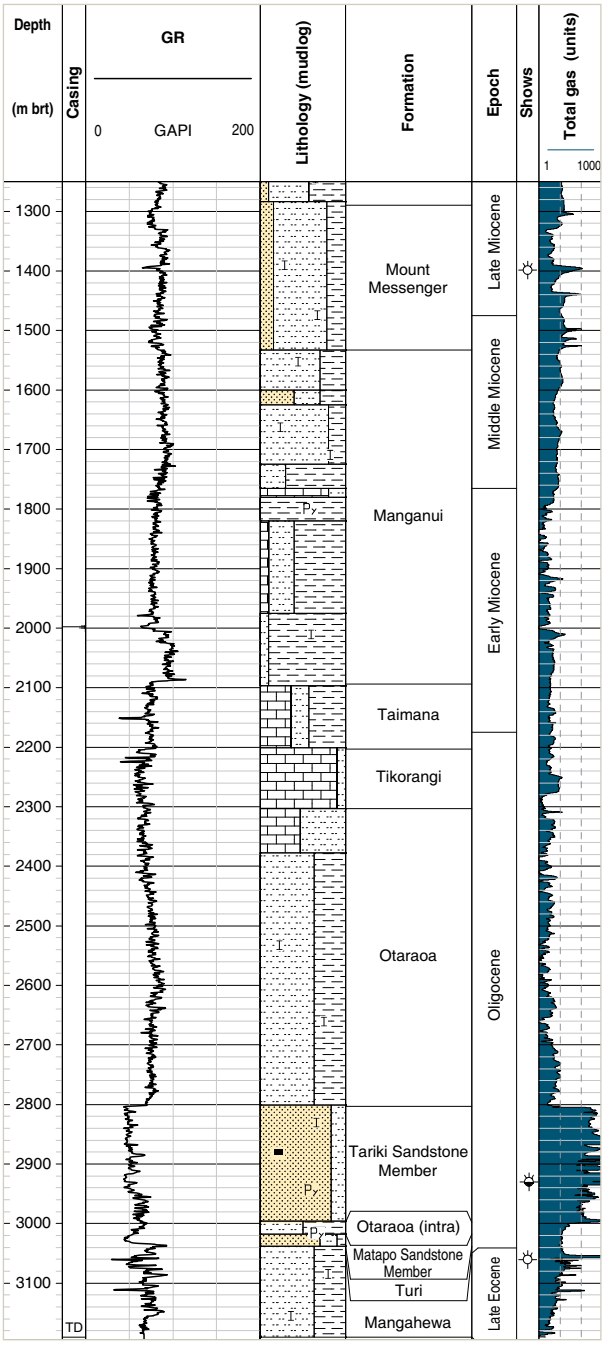
RESERVOIR CHARACTERISTICS

The Tikorangi Formation reservoir is up to 170 m thick and has a well-cemented matrix, with reservoir potential provided by fractures that resulted from the folding and faulting that formed the structural trap.

The Tariki Sandstone is up to 288 m thick in the southeastern region. Reservoir quality is best in the Ahuroa and Tariki wells. Core-derived average porosity and permeability measurements are in the ranges of 12 to 15% and 18 to 142 mD.

FACILITIES

The TAWN fields are fully developed with associated oil and gas production facilities including the Waihapa oil plant and the Tariki Ahuroa gas plant, both located at the Waihapa production station. However, the Ahuroa Field is now being used as a gas storage facility to supply the new 200 MW Stratford power station.

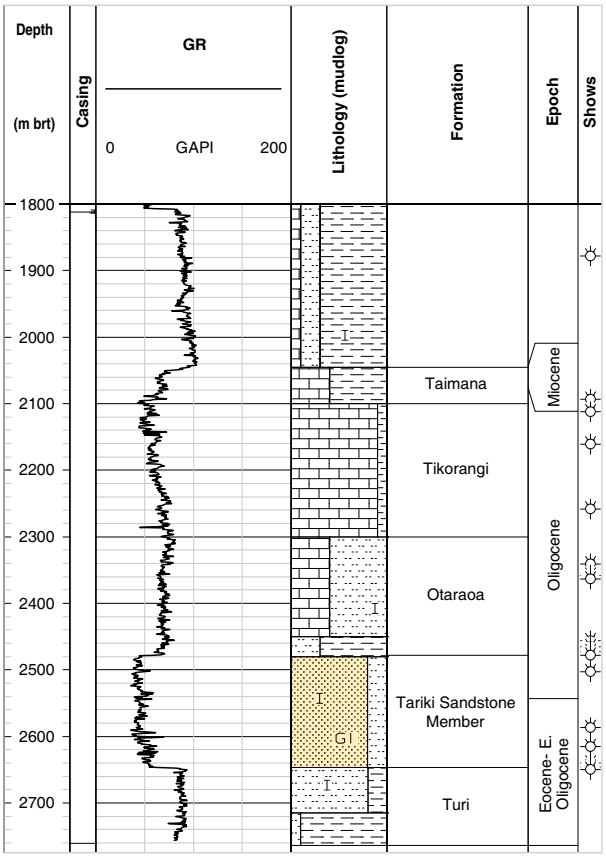


TARIKI-1 ▾ TARIKI-1A

SPUD DATES	2 March 1986	6 May 1986
TDs	3,191 m MD	2,764 m MD

Tariki-1 was drilled as a wildcat well on the west flank of the Tariki overthrust. The well discovered hydrocarbons in the 155 m MD-thick Oligocene Tariki Sandstone Member. The sandstone had good reservoir properties with 17.5% average porosity, but high water saturation averaging 51%.

The sidetrack Tariki-1A intersected the main thrust fault at the base of the Tariki Sandstone. It was drilled as an up-dip deviated appraisal well, discovering significant quantities of gas and condensate and intersecting the reservoir sand 177 m higher than in Tariki-1. Total thickness of the pay zone was 105.5 m MD (94 m TV) with an average porosity of 16% and water saturation of 13%. During deliverability testing, the well flowed 28.81 mmscf/d and 1,395 bbl/d of condensate through a 48/64" choke.



Kupe
Gas-Condensate Field

LOCATION	OFFSHORE TARANAKI BASIN
CURRENT PERMIT	38146
AREA	256 km²
OPERATOR	Origin Energy
DISCOVERED	1986
YEAR ON STREAM	2009
WATER DEPTH	35 m
RESERVOIR DEPTH	3,100 m TVDSS
RESERVOIR ROCK	Farewell Formation
GEOLOGICAL AGE	Paleocene
PLAY TYPE	Fluvial to coastal braid-plain sands in fault-bound inverted anticline



PRODUCTION AND RESERVES

Gross cumulative production of the field at 1 January 2013 is 63.6 bcf gas and 5.47 mmbbl oil/condensate. Estimated remaining reserves (P90) at 1 January 2013 are 179 bcf gas and 17.7 mmbbl oil/condensate. Production lifespan is estimated at 18 years.

GEOLOGY

This field is located 35 km off the southern Taranaki coast, on the crest of the Manaia Anticline on the same structural trend as the Kapuni Field. The Manaia Anticline is a significant inversion lineament with mainly Miocene uplift age, bound to the west and controlled by the Manaia Fault, an east-heading, steeply dipping reverse fault.

The field is compartmentalised by a series of northwest-southeast and northeast-southwest trending extensional faults. The northwest -southeast faults are the earlier of the two sets and the result of Late Oligocene to Early Miocene reactivation of a basement fault linking the Manaia and Rua faults. The northeast-southwest faults show present-day seafloor expression and have the same trend as onshore faults, some of which are seismically active today.

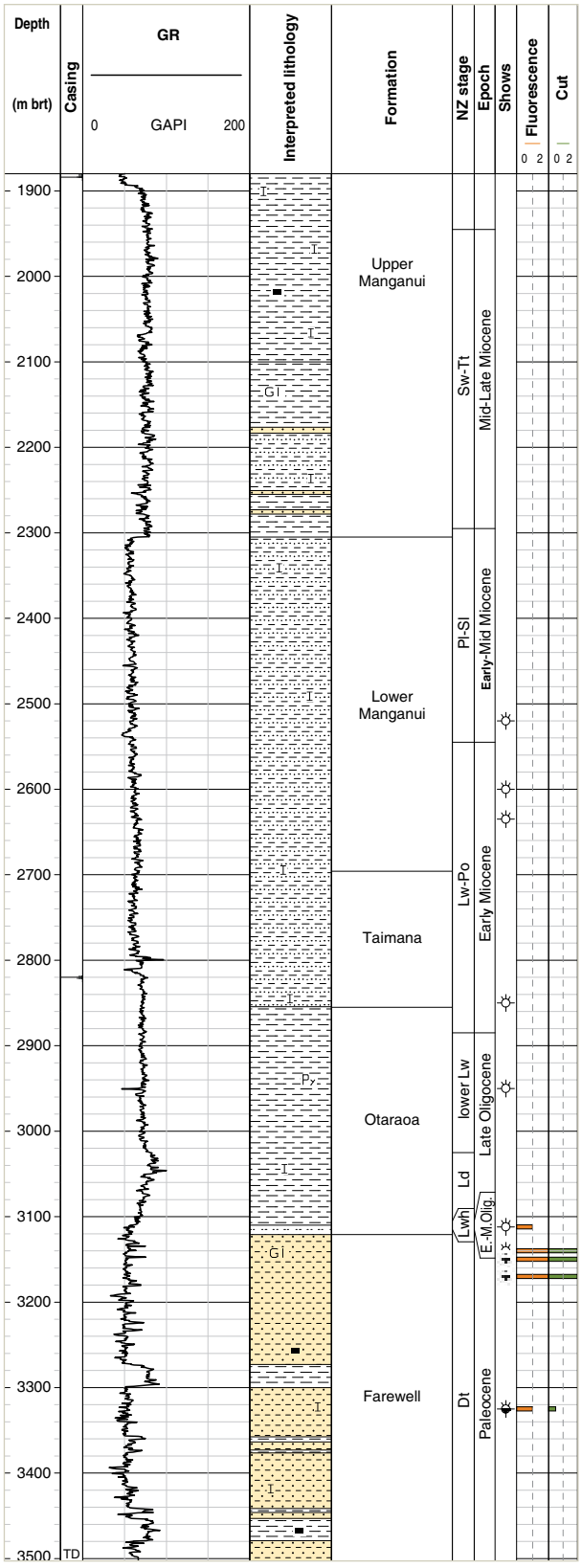
RESERVOIR CHARACTERISTICS

The Farewell Formation sandstone reservoirs are predominantly medium- to coarse-grained, with good porosities and permeabilities ranging up to one Darcy. They are compositionally immature and feldspar-rich and interpreted as having been deposited in dominantly fluvial and lacustrine settings.

FACILITIES

At present, the project includes the following:

- three production wells up to 4,000 m MD
- an unmanned offshore platform which consists of a topside deck supported by four legs above the production wells
- a 30 km subsea pipeline to bring raw gas and liquids from the platform to shore
- a subsea “umbilical”, carrying chemicals, power and fibre optic communications between the platform and the shore
- tunnels drilled through the coastal cliffs taking the pipeline and umbilical safely under the ground at the shore-face
- an onshore production station, designed to sweeten the gas and separate the liquids to produce sales quality gas, LPG and light crude
- an 11.7 km sales gas pipeline to take natural gas from the production station to Kapuni, where it is injected into the North Island transmission network and
- light crude handling, storage and export facilities near Port Taranaki.



KUPE SOUTH-1

SPUD DATE	30 October 1986
TD	3,503 m MD

The discovery well, Kupe South-1, was drilled up-dip from the Kupe-1 well, to test a flexural culmination defined by various vintages of seismic data. The reservoir sands are Paleocene Farewell Formation braided river-deltaic sandstones. The well flowed 2,000 bopd and 5.4 mmscf/d and was followed by the drilling of the Kupe South-2 well in 1987, and Kupe South-3 and 3a wells, drilled in 1988.

Kupe South-2 and Kupe South-3 both encountered a stratified hydrocarbon reservoir with a significant 20 m oil column underlying a 120 m natural gas column. These three wells formed the basis for the Kupe Central Field Area (CFA) development, proposed in 1999.

Pohokura
Gas-Condensate Field

LOCATION	OFFSHORE TARANAKI BASIN
CURRENT PERMIT	38154
AREA	156 km²
OPERATOR	Shell Exploration
DISCOVERED	2000
YEAR ON STREAM	2007
WATER DEPTH	50 m
RESERVOIR DEPTH	3,480 m TVDSS
RESERVOIR ROCK	Mangahewa Formation
GEOLOGICAL AGE	Middle and Late Eocene
PLAY TYPE	Transgressive marginal marine sands in an inverted anticline



PRODUCTION AND RESERVES

Gross cumulative production of the field at 1 January 2013 is 443 bcf gas and 31.0 mmbbl oil/condensate. Estimated remaining reserves (P90) at 1 January 2013 are 419 bcf gas and 22.5 mmbbl oil/condensate.

GEOLOGY

Pohokura, currently New Zealand's largest gas-condensate field, is located immediately offshore (about 4 km) from the northern coast of Taranaki Peninsula. It is a low-relief, north-south elongate anticline approximately 16 km long and 5 km wide. The hydrocarbon reservoir is Middle to Late Eocene shoreline sandstone of the Mangahewa Formation.

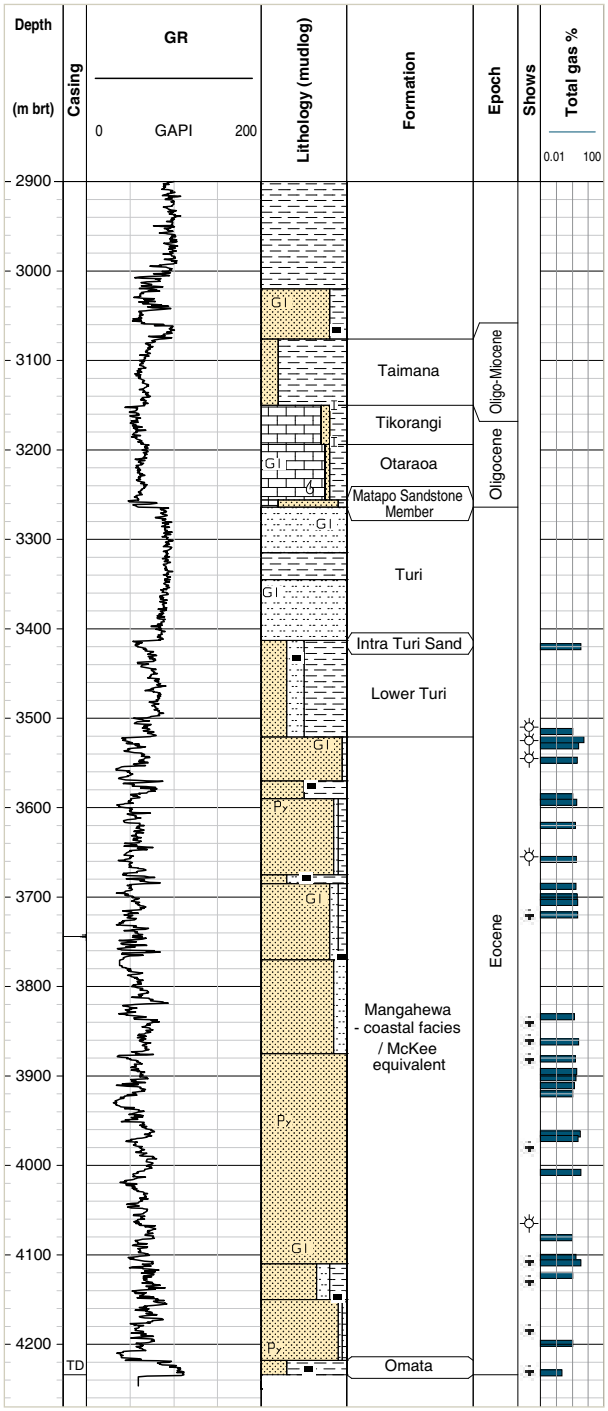
RESERVOIR CHARACTERISTICS

The field was discovered in February 2000 by the Pohokura-1 well, which intercepted a 130 m gas column within the Mangahewa Formation at a depth of 3,480 m MD and, on testing, flowed 17.6 mmscf/d gas. Pohokura-2 was drilled to test the central part of the structure 5 km north-northwest of Pohokura-1. It penetrated a 115 m gas column, which on testing flowed 35.3 mmscf/d gas. Pohokura South-1, drilled in 2001, tested the saddle area between the Pohokura and the Turangi-Mangahewa structures to the south.

FACILITIES

First commercial gas from three extended reach development wells, drilled from onshore into the southern end of the field, was fed into the North Island reticulation network in September 2006. Gas and condensate were being produced from the first offshore well by March 2007.

The offshore platform and the onshore production station are unmanned, both being remotely controlled from the operator's offices. The field lies offshore but all wells feed the onshore production station on the coast, close to the methanol plant at Motunui. Condensate, separated from the gas flow at the production station, is piped to storage tanks at Omata, near New Plymouth.



POHOKURA-1

SPUD DATE 5 February 2000
TD 4,234 m MD

The well targeted the Kapuni Group Mangahewa Formation sands in the Pohokura structure. In total, over 700 m of shallow marine sands were encountered. A gross gas column of 130 m was discovered in the upper part of the primary target. The well was deepened to access the potential of the lower Mangahewa Formation, reaching TD in the Mid-Eocene shales of the Omata Formation. However, logging confirmed no indications of commercial hydrocarbons in the lower section of the formation.

Two drill stem tests were performed in the upper section: the interval 3,625 to 3,634 m MD flowed at 3.5 mmscf/d on a 16/64" choke; the interval 3,553 to 3,570 m MD flowed 16.5 mmscf/d on a 54/64" choke. The well provided a control point for the velocity model in the field and a depth to reservoir close to the crest of the Pohokura structure.

Tui Area
Oil Fields

LOCATION	OFFSHORE TARANAKI BASIN
CURRENT PERMIT	38158
AREA	467.2 km²
OPERATOR	AWE Taranaki Limited
DISCOVERED	2003
YEAR ON STREAM	2007
WATER DEPTH	123 m
RESERVOIR DEPTH	3,660 m MD
RESERVOIR ROCK	Farewell Formation
GEOLOGICAL AGE	Late Paleocene
PLAY TYPE	Stratigraphic drapes of transgressive Late Paleocene sands over basement



PRODUCTION AND RESERVES

The Tui area oil fields were developed on the basis of 27.9 mmbbl of recoverable oil. Several resource redeterminations have been made since this time and total recoverable reserves are now estimated at 37.6 mmbbl (P90). Several prospects lie within the Tui licence area. Discoveries in these could be tied into existing facilities and prolong the life of the project.

Production rates from the Tui area oil fields have been higher than predicted. Initial production rates were as high as 50,000 bbl/d, at the limit of the FPSO. Rates are currently about 11,000 bopd. The fields also produce gas, with the FPSO Umuroa using about 80,000 m³/day to power its operations, and the remainder flared.

Gross cumulative production of the field at 1 January 2013 is 14.96 bcf gas and 35.5 mbbl oil. Estimated remaining reserves (P90) at 1 January 2013 are 3.5 mmbbl oil.

GEOLOGY

The Tui area oil fields on the Western Platform are three small oil accumulations named "Tui", "Amokura" and "Pateke". They occur in a shallow marine transgressive Farewell Formation sandstone reservoir, the Kapuni "F" sand. The traps are low relief stratigraphic drapes over pre-existing basement highs. The accumulations lie above a strong aquifer.

The Western Platform is a comparatively stable region to the west of the Cape Egmont Fault Zone and Northern Graben. The reservoirs in the field area comprise part of a west-southwest to east-northeast trending shoreline system. The area is structurally simple and the fields are buried by about 3.5 km of overlying strata.

The discovery well, Tui-1, intercepted a 10 m column of oil. Amokura-1 and Pateke-2, both drilled in 2004, encountered 12 m columns of oil. These three accumulations have been tied into a single set of production facilities. It is thought that the migration pathways that charged the Tui area fields are part of the same system that charged the Maui Field 20 to 25 km to the southeast.

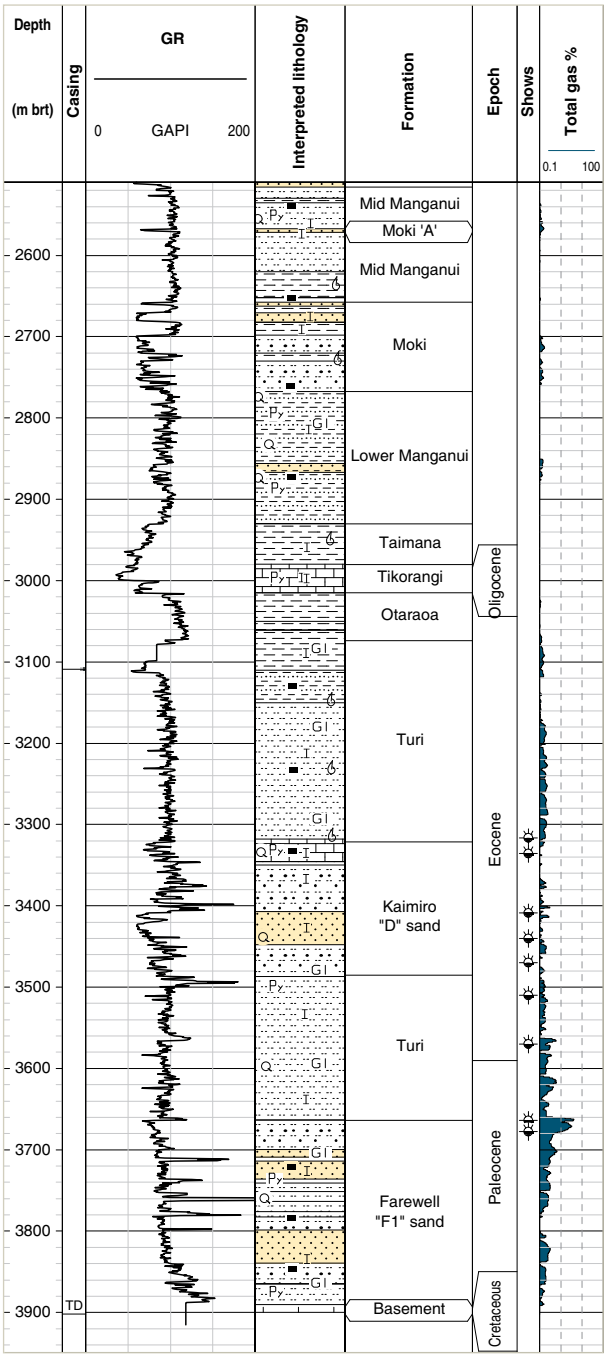
The Tui area oil fields represent a significant discovery for Taranaki Basin as they demonstrate that the Western Platform contains commercially viable petroleum accumulations. The fields were New Zealand's first stand-alone offshore oil project. They are also remarkable for the short time in which they were brought on stream, with production starting 4.5 years after discovery and 20 months after the final investment decision was made.

RESERVOIR CHARACTERISTICS

Reservoir quality is generally good, as suggested by the excellent production rates that the fields have displayed thus far. Porosities are 18 to 21%, permeabilities are 1.1 to 2.0 Darcies, and water saturations are about 21 to 32%.

FACILITIES

The Tui development is by four horizontal wells tied back to the FPSO Umuroa. The Umuroa can process up to 50,000 bbl/d oil, 25 mmscf/d gas and 120,000 bbl/d total liquids, and has a storage capacity of 770,000 bbl crude oil. Oil is offloaded to tankers, and then transported to market.



TUI-1

SPUD DATE	24 January 2003
TD	3,902 m MD

Tui-1 was drilled to test Eocene shallow marine Kaimiro Formation “D” sands with Miocene Moki “A” turbidite sands a secondary objective and Paleocene Farewell Formation “F” sands an additional target. A 10 m column of high-quality oil was drilled in the Farewell Formation, with its top at a depth of about 3,664 m MD. Tui-1 was plugged and abandoned on 15 February 2003 as an oil discovery.

Kawau
Gas-Condensate Discovery

LOCATION	OFFSHORE GREAT SOUTH BASIN
CURRENT PERMIT	Not assigned
AREA	16,390 km²
YEAR DRILLED	1977
WATER DEPTH	683 m
RESERVOIR DEPTH	3,224 m MD
RESERVOIR ROCK	Near-shore transgressive Hoiho Group sandstone and near-shore Kawau Sandstone
GEOLOGICAL AGE	Late Cretaceous
PLAY TYPE	Late Cretaceous reservoir in a fault-controlled closure; compactional drape over faulted basement



SIGNIFICANCE

This sub-commercial gas-condensate discovery proves the presence of an effective petroleum system in the Great South Basin. Hydrocarbon shows were recorded from many intervals in Kawau-1A. Significant gas shows were present in Cretaceous reservoirs between 3,565 and 3,710 m. Oil staining was reported from several intervals above the main gas-condensate sands. A drill stem test in Kawau Sandstone at 3,224 to 3,257 m MD produced up to 6.7 mmscf/d of gas with 7% CO_2 and 24 bbl/d of condensate. At that time, the overall reserve was calculated to be 461 bcf gas.

GEOLOGY

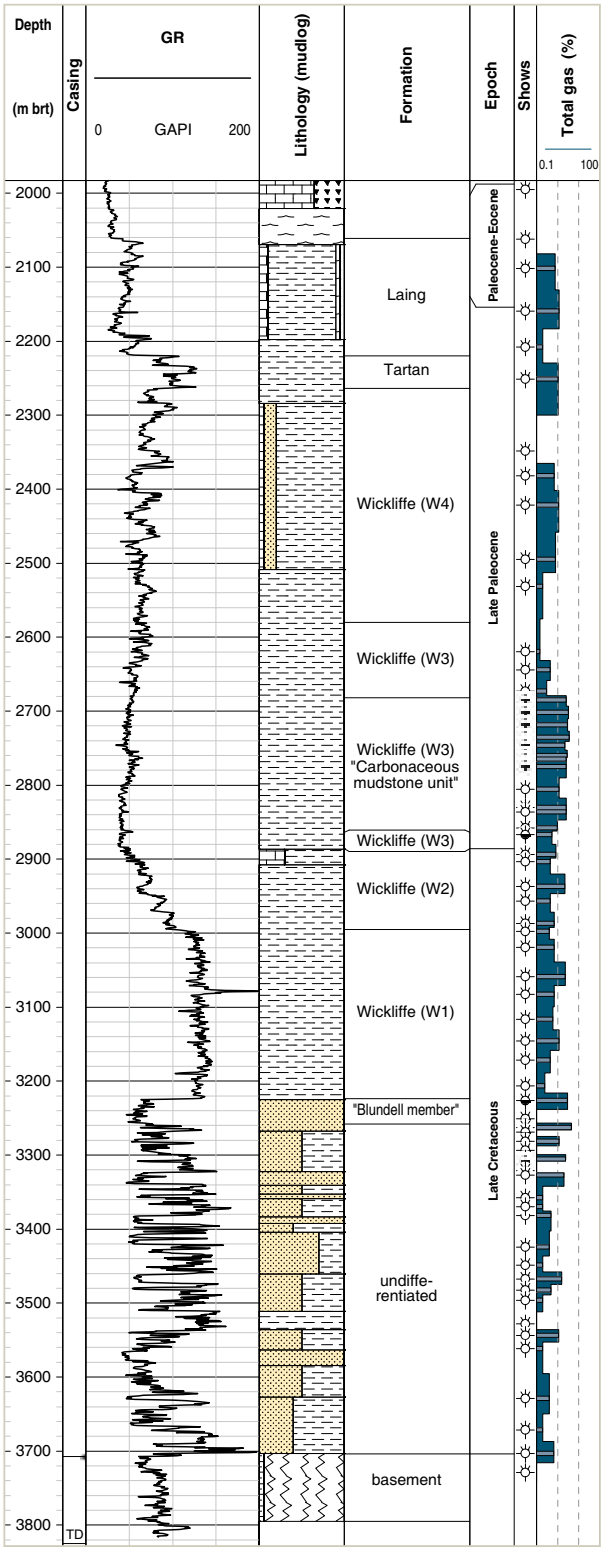
The Kawau structure is a compactional drape trap located close to the longitudinal axis of the Great South Basin. The structure is probably typical of many other potential petroleum targets in the basin. It was tested by Kawau-1A, which was spudded on 20 June 1977, and reached basement at 3,703 m MD and a TD of 3,826 m MD. The main reservoir interval represents marine transgression over block-faulted basement. Source rocks for the trap are interpreted to be mid-Cretaceous coaly facies.

RESERVOIR CHARACTERISTICS

The reservoir tested by Kawau-1A had poor to fair permeability. Uncorrected log-derived porosities were in the 17 to 20% range. Sidewall core measurements indicated porosities of 22 to 28%.

FACILITIES

No facilities or infrastructure are present in the basin.


Depth
(m brt)
Casing
GR
0
GAPI
200
Lithology (mudlog)
Formation
Epoch
Shows
Total gas (%)
0.1
100
2000
2100
2200
2300
2400
2500
2600
2700
2800
2900
3000
3100
3200
3300
3400
3500
3600
3700
3800
TD
Laing
Tartan
Wickliffe (W4)
Wickliffe (W3)
Wickliffe (W3)
"Carbonaceous
mudstone unit"
Wickliffe (W3)
Wickliffe (W2)
Wickliffe (W1)
"Blundell member"
undiffe-
rentiated
basement
Paleocene-Eocene
Late Paleocene
Late Cretaceous

Galleon
Gas-Condensate Discovery

LOCATION	OFFSHORE CANTERBURY BASIN
CURRENT PERMIT	Not assigned
AREA	1,658 km^2
YEAR DRILLED	1985
WATER DEPTH	91 m
RESERVOIR DEPTH	2,750 m MD
RESERVOIR ROCK	Herbert and Pukeiwitahi formations
GEOLOGICAL AGE	Late Cretaceous
PLAY TYPE	Late Cretaceous four-way dip-closed drape over a mid-Cretaceous basement high



SIGNIFICANCE

This sub-commercial gas-condensate discovery proves the presence of an effective petroleum system in Canterbury Basin. The structure is only partially filled and subsequent basin models suggest that the source kitchen is remote from the trap and that petroleum must have migrated some tens of kilometres from the kitchen to the trap. It might be inferred that traps closer to source may be more fully charged.

The Galleon-1 discovery was in a 21 m-thick interval of gas/condensate -bearing sandstone. Testing was carried out through a perforated liner between 2,753 and 2,763 m MD, with maximum flow rates reaching 10.1 mmscf/d gas and 2,300 bbl/d condensate. However, estimated recoverable reserves were deemed too small for commercial development at the time, and the well was plugged and abandoned as a gas-condensate discovery.

GEOLOGY

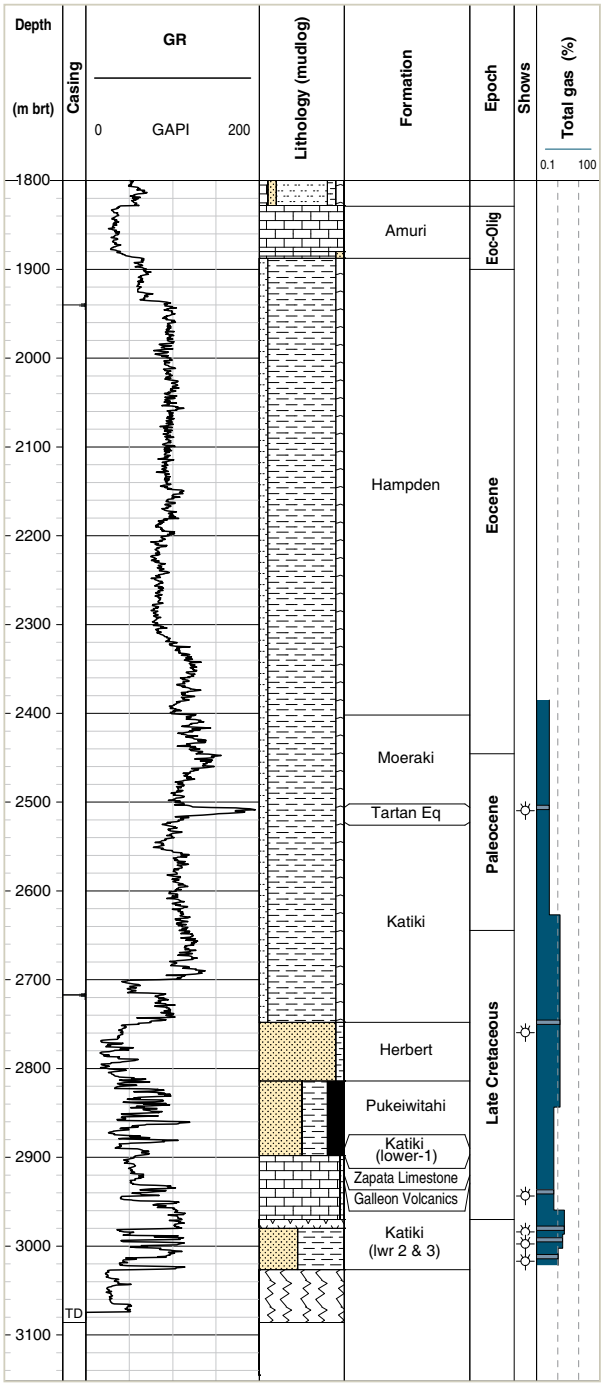
The Galleon structure is a stratigraphic drape trap of about 40 km^2 over a basement horst. It is a good example of an important play type in the offshore Canterbury Basin. The horst is thought to have formed during the mid-Cretaceous. The reservoirs comprise Late Cretaceous shallow marine sands, and post-depositional compactional folding has formed the main part of the trap. The structure was tested as Galleon-1, which was spudded on 15 September 1985, and reached basement at 3,027 m MD and a TD of 3,086 m MD.

RESERVOIR CHARACTERISTICS

Average reservoir porosity in Galleon-1 is 17%, and water saturations are 0.46%. The high flow rates during testing of Galleon-1 indicate that the reservoirs have good permeability.

FACILITIES

No facilities are present in this basin.


Depth
(m brt)
Casing
GR
0
GAPI
200
Lithology (mudlog)
Formation
Epoch
Shows
Total gas (%)
0.1
100
1800
1900
2000
2100
2200
2300
2400
2500
2600
2700
2800
2900
3000
3100
TD
Amuri
Hampden
Moeraki
Tartan Eq
Katiki
Herbert
Pukeiwitahi
Katiki
(lower-1)
Zapata Limestone
Galleon Volcanics
Katiki
(lwr 2 & 3)
Eoc-Olig
Eocene
Paleocene
Late Cretaceous

Kora
Oil Discovery

LOCATION	OFFSHORE TARANAKI BASIN
CURRENT PERMIT	Not allocated
AREA	1,400 km²
DISCOVERED	1988
WATER DEPTH	123 m
RESERVOIR DEPTH	About 1,700 to 1,800 m MD, and about 3,100 m MD
RESERVOIR ROCK	Mohakatino and Tangaroa formations
GEOLOGICAL AGE	Middle Miocene and Late Eocene to Early Oligocene
PLAY TYPE	Marine source rock-derived oils within buried volcaniclastic apron fans. Also Eocene to Oligocene mass-deposited and turbidite sands



SIGNIFICANCE

This sub-commercial oil discovery illustrates the potential of Miocene volcaniclastic reservoirs as viable plays and it shows that Paleogene marine rocks are generating oil in northern parts of the basin.

During the drilling of Kora-1, notable indications of hydrocarbons were recorded from Middle Miocene Mohakatino Formation volcanics, and from Tangaroa Formation sandstone. On test, the Miocene interval produced 1,168 bopd. A 14-day production test from the Kora-1A sidetrack achieved an average flow of 668 bopd. Biodegraded oil was noted from Kora-2 and -3 and no oil was recorded from Kora-4. The Kora Field remains undeveloped, and its small size (about 1 mmbbl) means development is unlikely.

GEOLOGY

The Kora Field is located within a trap formed by a buried Miocene volcanic edifice including complex lithofacies geometries, with doming caused by underlying intrusive igneous rocks. The discovery of oil in the field highlighted two new aspects to Taranaki Basin petroleum geology. Firstly, it illustrated that Miocene volcaniclastic deposits in the north of the basin offered reservoir potential. Secondly, it indicated that oil has been expelled from the late Paleocene Waipawa Formation marine source rock, something not previously identified within the basin. The discovery also showed that Late Eocene to Early Oligocene turbidite sandstones of the Tangaroa Formation, the original target for Kora-1, offer reservoir potential.

The field is located within a fault system that separates the Northern Graben from the Western Platform. The Northern Graben is a late Neogene extensional feature in the eastern part of the Taranaki Basin. Within parts of the graben lie a series of large volcanic edifices that developed along the eastern margin of the Taranaki Basin during the Miocene, but were especially prominent during the Middle Miocene. These mostly andesitic volcanoes formed positive relief, and generated aprons of volcaniclastic debris, the Mohakatino Formation. Rapid sedimentation and progradation of the continental shelf during the Plio-Pleistocene has buried much of the Northern Graben and the volcanic features within it. Younger rocks of the Giant Foresets Formation form top seal for the underlying Miocene volcaniclastic reservoir.

It has been suggested that oil migrated into the Kora structure along graben-bounding faults in the Late Pliocene. A lack of sufficiently impermeable seal has been postulated, with capillary entry pressure measurements suggesting the Giant Foresets Formation overlying Kora may be capable of withholding a 40 to 120 m hydrocarbon column. Other volcanic features in the basin are more deeply buried and may have more effective seals.

KORA-1

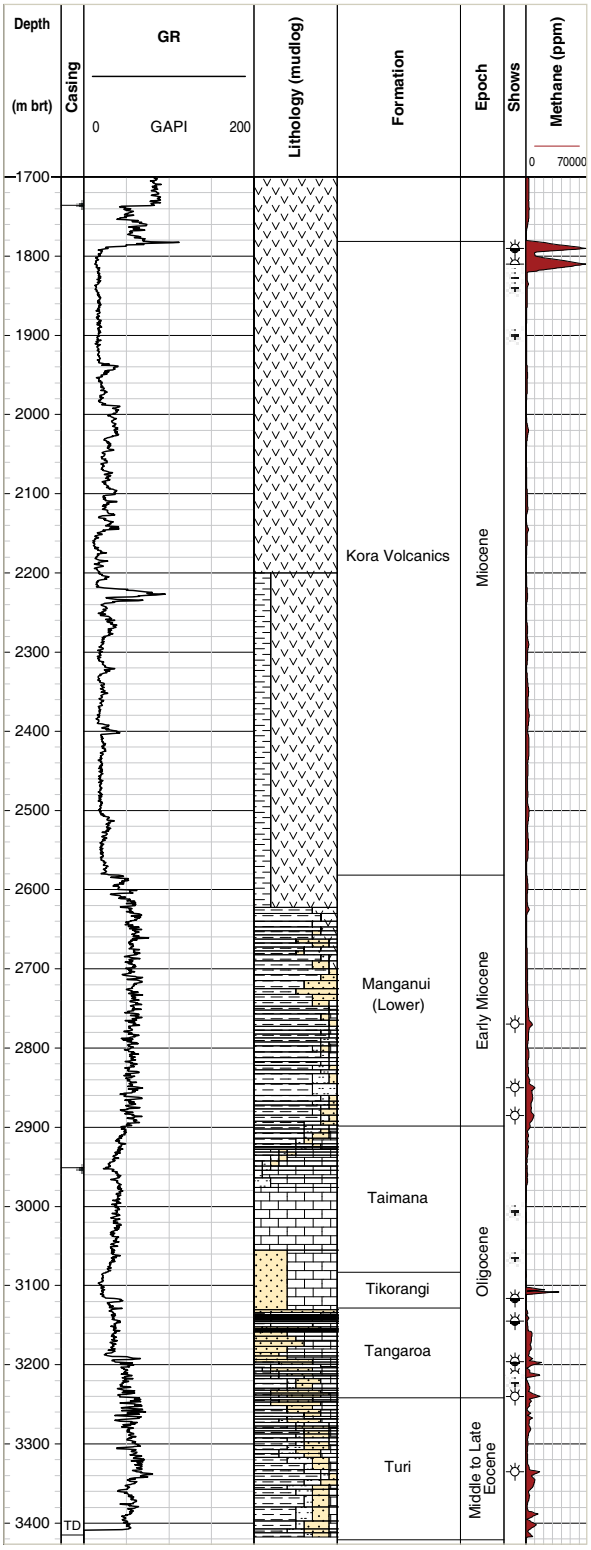
SPUD DATE	10 November 1987
TD	3,421 m MD

Kora-1 was drilled to test Eocene Tangaroa Formation sands within a large dome structure associated with a Miocene volcano. Significant hydrocarbon indications were encountered in the upper parts of a succession of Miocene volcaniclastic rocks at a depth of 1,785 m. Additional shows were also encountered in the Tangaroa Formation between 3,131 and 3,244 m. Drill stem tests of the Tangaroa Formation produced very small quantities of oil, and it was concluded that the formation was tight. The Miocene volcaniclastics flowed 1,168 bopd on test, after which Kora-1 was plugged back and the Kora-1A sidetrack drilled to evaluate the potential. A long-term production test yielded an average flow rate of 668 bopd. The well was plugged and abandoned.

RESERVOIR CHARACTERISTICS

Tangaroa Formation sandstones are typically 100 to 200 m thick and fine- to coarse-grained, commonly present in two intervals separated by a thin limestone. These sands had oil shows in Kora-1 but reservoir quality was poor, due to the presence of silica and carbonate cements, possibly associated with hydrothermal fluid migration.

The stratigraphic relationships between potential reservoir rocks within the Mohakatino Formation are complex and difficult to predict. Core measurements from the Kora-1A sidetrack indicate porosities from 9 to 28% with relatively low permeabilities of 0.1 to 200 mD. However, some high permeabilities also exist, with good flow rates obtained from clast-supported volcaniclastic conglomerates.



Titihaoa
Gas Show

LOCATION	OFFSHORE EAST COAST BASIN
CURRENT PERMIT	Not allocated
YEAR DRILLED	1994
WATER DEPTH	126 m
RESERVOIR DEPTH	About 1,500 to 2,700 m MD
RESERVOIR ROCK	Turbidite sandstones
GEOLOGICAL AGE	Middle Miocene
PLAY TYPE	Neogene mass-emplaced sandstones within a late Neogene hanging wall anticline



SIGNIFICANCE

Although deemed non-commercial, the Titihaoa dry gas accumulation represents an example of the potential for significant hydrocarbon discoveries in the offshore East Coast Basin. It illustrates the presence of thick successions of sandstone-bearing strata within large structural closures and demonstrates that generation and migration of hydrocarbons has occurred in the region. The presence of overpressured horizons demonstrates that seal capacity is good. Successful discoveries might be made in the structural lows around anticlines, where sands are inferred to be thicker and better developed, with elements of stratigraphic trapping.

Titihaoa-1 intercepted a column of gas-bearing rock over 800 m thick. Some thin sandstone beds produced significant gas kicks, and maximum total gas of 80.2% was recorded at a depth of 1,943 m MD. The gas composition was almost entirely methane, with minor propane, and the well was considered non-commercial. However, a post-drill analysis estimated that the Titihaoa structure may hold up to 400 bcf of recoverable gas.

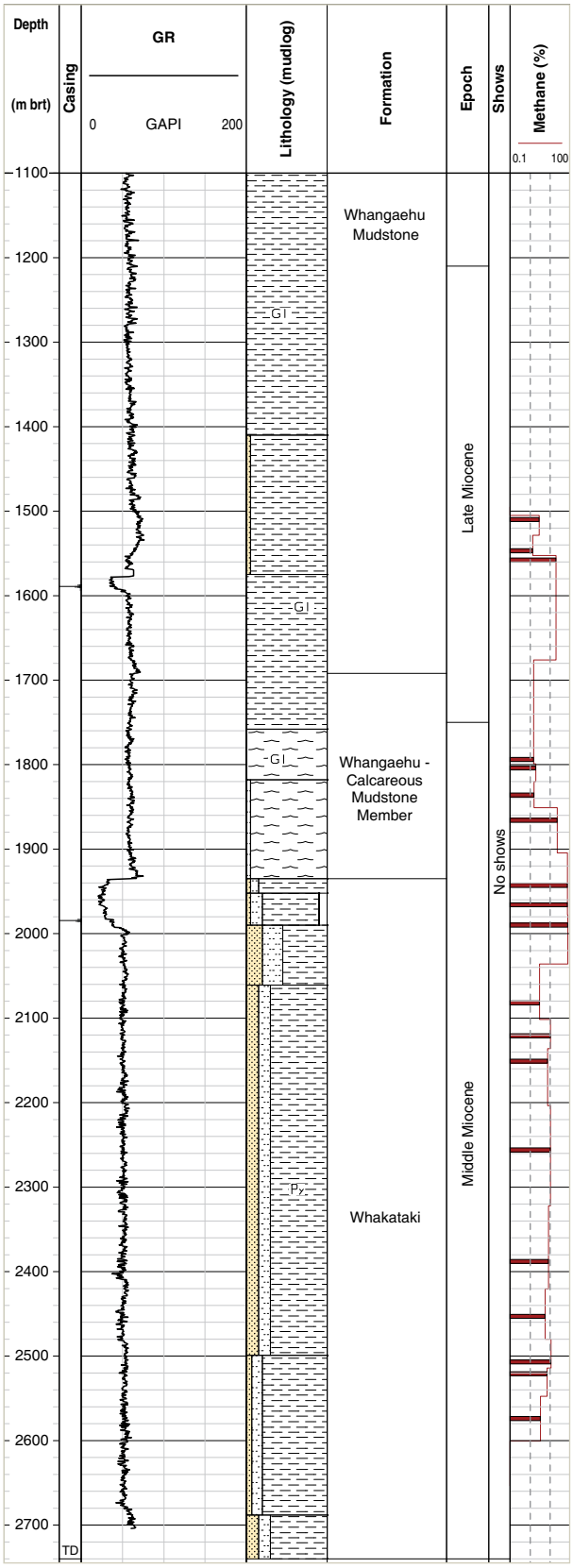
GEOLOGY

The Titihaoa structure, covering about 30 km^2, is one of many fault-bound, hanging wall, anticlinal closures that underlie offshore parts of the East Coast Basin. These structures developed during Neogene compression, a consequence of proximity to the modern subduction margin. The closure was tested by Titihaoa-1.

The Titihaoa structure lies with the accretionary prism, a structurally complex zone overlying the subducting Pacific Plate, comprising accreted sediments and slope basins imbricated during the Neogene. Anticlinal closures are common in the hanging walls of reverse faults in this area and the areas of closure vary significantly. The region is thought to include a large thickness of potential Cretaceous and Paleogene source rocks at depth. Reservoirs are primarily Neogene turbidite sandstones. Fine-grained mudstones and marls provide seal.

RESERVOIR CHARACTERISTICS

The reservoir succession in Titihaoa-1 consists of thin-bedded Middle Miocene turbidite sands. These rocks are fine-grained and moderately well sorted, with porosities of about 15 to 20%. Permeabilities, estimated from Modular Dynamic Test (MDT) tool pressure build-ups, were about 20 mD and greater.



TITIHAOA-1

SPUD DATE	11 November 1994
TD	2,740 m MD

Titihaoa-1 was a wildcat well drilled offshore from Castlepoint by Amoco NZ Exploration Company. The second well to be drilled in offshore East Coast, it was designed to test possible thick Early Miocene and Oligocene sandstone reservoirs. The well was drilled 237 m beyond the anticipated TD but the rocks penetrated were no older than Middle Miocene. Gas shows were recorded throughout the well, although the lack of well developed reservoir sands meant it was deemed non-commercial.

Kauhauroa
Gas Discovery

LOCATION	ONSHORE EAST COAST BASIN
CURRENT PERMIT	38346
AREA	5,582 km²
OPERATOR	Westech Energy NZ Ltd
DISCOVERED	1998
RESERVOIR DEPTH	About 1,100 to 1,600 m MD
RESERVOIR ROCK	Kauhauroa Limestone
GEOLOGICAL AGE	Early to Middle Miocene
PLAY TYPE	Fractured shallow marine limestones and Neogene mass-emplaced sandstones within a late Neogene fault-cored anticline



SIGNIFICANCE

The Kauhauroa sub-commercial dry gas discovery is significant in several ways. It proves the reservoir properties of sandstones and fractured limestone and demonstrates that seals are effective despite very high over-pressures encountered at shallow depths. The gas appears to have a thermogenic component, further proving maturity in the onshore East Coast Basin.

Gas was discovered mainly in fractured Kauharoa Limestone, but also in sandstone reservoirs. Tunanui Formation sandstones flowed gas in Kauhauroa-5 and the Rere Sandstone Member was tested in Kauhauroa-1. Waitahora-1, drilled in 2007 as a direct offset to Kauhauroa-1, recorded gas shows over a 70 m along-hole interval, probably from fractured limestone.

Gas is 98% methane, with negligible amounts of higher hydrocarbons. Isotopic analyses indicate that a thermogenic source is possible, although a mixed biogenic-thermogenic source is more likely. This is consistent with isotopic analyses obtained from surficial gas seeps elsewhere in the Wairoa area. When first tested, Kauhauroa-1 yielded a stabilised flow of 11.5 mmscf/d from the Kauhauroa Limestone. A following test yielded erratic flows. There are no published estimates of reserves and the field remains suspended awaiting appraisal.

GEOLOGY

The Kauhauroa Field is located in one of many northeast-southwest trending anticlines in the Wairoa area of northern Hawke's Bay, produced by late Neogene compression. The structure has been tested by eight wells. Its discovery by Kauhauroa-1 in 1998 highlighted the previously unknown potential of the Early and Middle Miocene shallow marine Kauhauroa and Kiakia limestones as fractured reservoirs in the East Coast Basin. Small flows of gas were also achieved from several sandstone units, including the Middle Miocene Tunanui Formation intercepted by Kauhauroa-5, the late-Early Miocene Rere Sandstone Member of the Waingaromia Formation in Kauhauroa-1, and the Early Miocene Whangara Sandstone in Kauhauroa-4B.

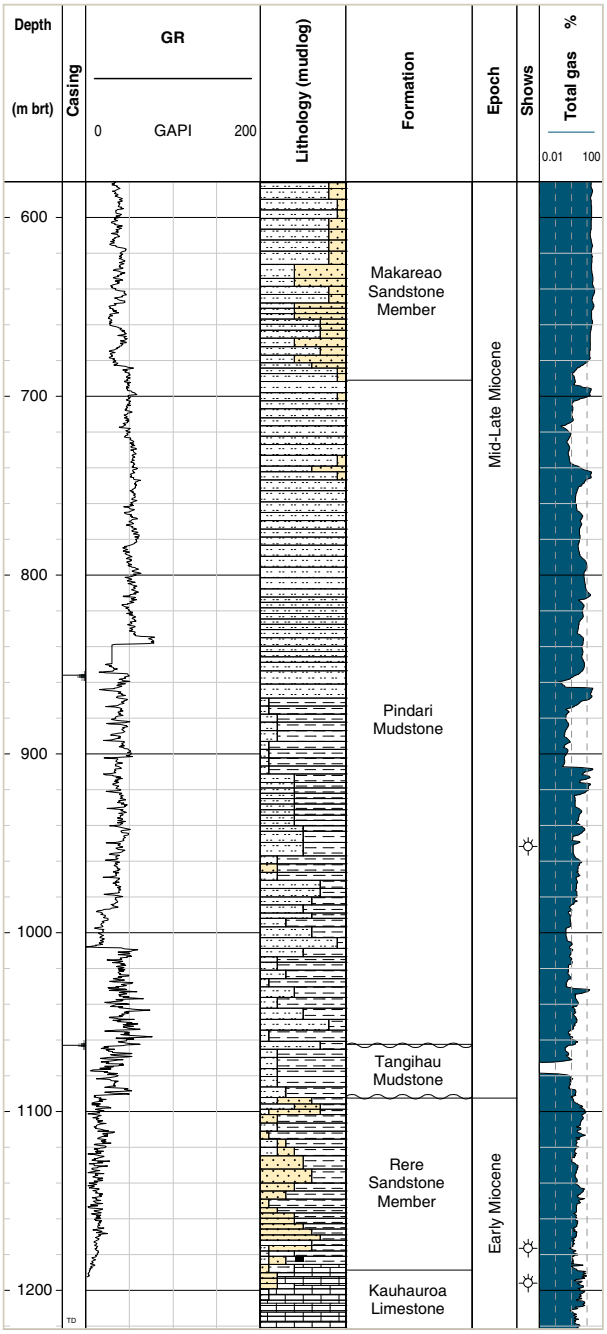
The Kauhauroa structure occurs along the crest of the asymmetric Kauhauroa Anticline, which is developed as a hanging-wall fold above a west-dipping reverse fault. Several back-thrusts verge off the primary fault in the structure, variably offsetting the main reservoir interval. The structure is 15 km long and 6 to 9 km wide. Its crest is separated by a saddle from a less tightly folded, southwestern extension. The trend of the anticline is discernable from outcrop geology.

RESERVOIR CHARACTERISTICS

Fractured Kauhauroa Limestone, the presence of which was unknown prior to the drilling of Kauhauroa-1, is the main reservoir. The Early to Late Miocene Kiakia Limestone is another fractured limestone reservoir first identified during the Wairoa drilling programme. Porosity is primarily from fractures, which are often well imaged on geophysical image logs. The Kauhauroa structure exhibited significant overpressures, indicating the effective seal capacity of the thick overlying Neogene mudstones.

FACILITIES

No infrastructure or facilities have been developed in the Kauhauroa Field area.



KAUHAUROA-1

SPUD DATE	March 4 1998
TD	1,221.6 m MD

Kauhauroa-1 was the first well to test a series of Neogene structural traps in the Wairoa area of onshore northern Hawke's Bay. Several potential Miocene to Pliocene reservoirs were targeted. The Kauhauroa-1 well tested the potential of the Kauhauroa Anticline, near the small village of Frasertown.

The well drilled through several previously unknown lithological units. Significant gas overpressures were encountered and mud weights of up to 17.2 ppg were required. Gas shows were recorded in the Makareao Sandstone Member, a then-unknown member of the Late Miocene Poha Formation. Shows were also recorded from the turbidite sands of the Early Miocene Rere Sandstone Member. The most significant gas shows were encountered in the Early Miocene Kauhauroa Limestone, a formation unknown prior to the drilling of Kauhauroa-1.
On test, up to 11.5 mmscf/d of methane flowed from this unit. Kauhauroa-1 was completed and suspended as a gas discovery.

Karewa
Gas Discovery

LOCATION	OFFSHORE TARANAKI BASIN
CURRENT PERMIT	38602
AREA	244 km²
OPERATOR	Todd Petroleum Mining Company Ltd
DISCOVERED	2003
WATER DEPTH	89 m
RESERVOIR DEPTH	About 2,000 m MD
RESERVOIR ROCK	Mangaa Formation sandstone fans
GEOLOGICAL AGE	Early to Late Pliocene
PLAY TYPE	Neogene mass-emplaced basin-floor sandstones within a late Neogene hanging wall anticline. Charged with biogenically derived gas



SIGNIFICANCE

The discovery of gas in the Karewa Field led to the identification of a new play concept in Taranaki Basin, highlighting the possibility of discovering commercial volumes of gas in shallow, relatively young rocks. Gas in the structure is inferred to be biogenic as it is more than 97% methane. However, some higher hydrocarbons were reported.

Options are being investigated to bring the field on stream. Proved and probable reserves, announced by the operator in 2007, indicated that the Mangaa Formation sands hold about 145-178 bcf of gas and that nearby prospects could yield another 130 bcf.

GEOLOGY

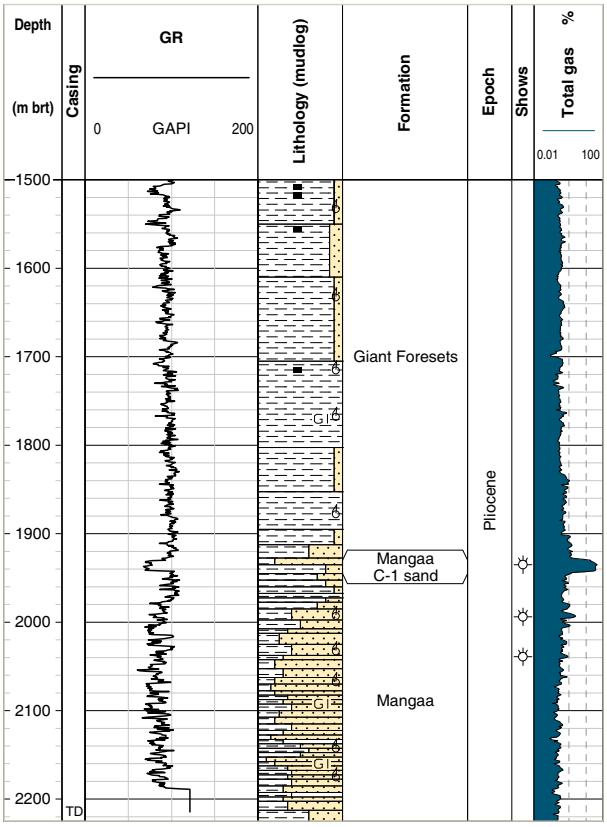
Karewa is located within the Northern Graben, about 40 km offshore from Raglan. The graben is a late Neogene extensional feature along the eastern margin of Taranaki Basin. The Karewa prospect was identified from high resolution seismic data as a possible hydrocarbon anomaly. The structure is a Pliocene feature associated with slope failure within the prograding Giant Foresets Formation above a Miocene volcanic body. Soling-out of a north-south oriented listric fault within intra-Mangaa Formation claystone helped produce a roll-over within the Mangaa Formation C-1 sand. The small four-way dip closure was mapped with about 50 m of total relief.

Pre-drill assessment was that Paleocene marine rocks were the most likely source of any hydrocarbons in the prospect. Older and more deeply buried Late Cretaceous rocks were considered unlikely to have been effective sources due to the young age of the structure. However, isotopic analysis of headspace gas from Karewa-1 indicates that the methane is biogenically derived. No liquid hydrocarbons were encountered.

The Karewa Field occurs in a similar stratigraphic position to the nearby Kahawai-1 well, where gas is also inferred to be biogenically derived. Other similar gas accumulations are likely to occur in this part of Taranaki Basin, offering opportunities for the discovery of comparatively shallow gas accumulations in basin-floor sands.

RESERVOIR CHARACTERISTICS

The reservoir is Pliocene Mangaa Formation C-1 sands. Based on wire-line log analysis, sands have an estimated average porosity of 25.5% and a water saturation of 27.7%.



KAREWA-1

SPUD DATE	26 December 2002
TD	2,225 m MD

Karewa-1 was drilled to test Mangaa Formation C-1 sands, which were subsequently encountered between 1,930 and 1,942 m. Wire-line log testing indicated the formation is gas-charged, with 11.2 m of net pay. The well was plugged and abandoned as a gas discovery.

Waka Nui-1

LOCATION	OFFSHORE NORTHLAND BASIN
CURRENT PERMIT	Not allocated
YEAR DRILLED	1999
WATER DEPTH	1454.9 m
RESERVOIR DEPTH	3464 m MD
RESERVOIR ROCK	Middle Cretaceous syn-rift sediments (amalgamated fluvial sandstones, lacustrine shoreline and deltaic facies) and basal fluvial sandstones deposited during the period of initial rift fill.
GEOLOGICAL AGE	Paleocene
PLAY TYPE	Cretaceous sediments draped across the dip slope of a large tilted fault block



SIGNIFICANCE

The Waka Nui-1 well is the only offshore petroleum exploration well north of Auckland and provides important lithologic and age control in the Northland and Reinga basins.

The well intersects a nearly complete Cenozoic sequence and Mesozoic Murihiku Supergroup rocks, all of which may be part of the petroleum system in these basins. Information from the well is critical for tying seismic data and constraining interpretations of the tectonic history and depositional environment in the entire offshore region northwest of New Zealand. Most of the recent seismic surveys in the Reinga, Northland and Deepwater Taranaki basins include lines that tie into Waka Nui-1 such is the importance of the well for stratigraphic and structural control.

GEOLOGY

The well was drilled to test Cretaceous coal measures interpreted to be draped across the dip slope of a large tilted fault block. The upper part of the drilled sequence largely matched the well prognosis however no Cretaceous-aged section was encountered in the well.

The primary target interval was found to consist of non-reservoir early Paleocene volcanigenic conglomerates and sandstones unconformably overlying indurated volcaniclastic pre-rift Murihiku strata (middle Jurassic), separated by a 95 Ma hiatus at the unconformity.

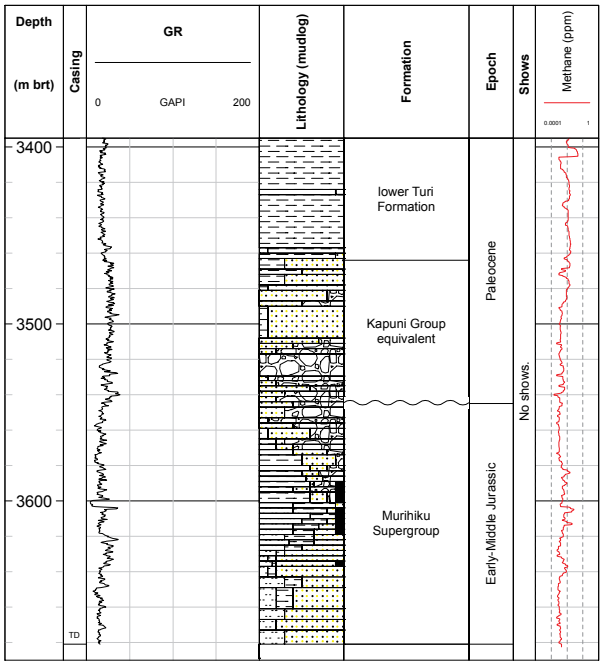
No Upper Eocene sands were encountered below the Tikorangi Formation, and the Tikorangi Formation was thicker than prognosed. The Paleocene Turi Formation, also thicker than prognosed, encompassed the entire interval prognosed to be of Late Cretaceous age, and contained a Waipawa Formation condensed interval. A transgressive sand sequence of Paleocene age (rather than late Cretaceous) was also encountered, but is highly volcanigenic and has little reservoir potential.

RESERVOIR CHARACTERISTICS

The purpose of the well was to test the reservoir potential and quality of the Middle and Upper Cretaceous sandstones, and the Late Cretaceous transgressive sandstones and Eocene turbiditic sandstone. None of these sediments were encountered in the well. Instead, the target interval consisted of non-reservoir volcanigenic sediments.

Average porosities in the well range between 10 to 20%. The interval 3455.5-3474 m contains a transgressive sand considered the best reservoir rock in the well. The wireline log response of the sand is definitely not that of clean quartz sand. This interval saw a fresh water influx while drilling, an indicator of some porosity and permeability.

Detrital quartz and feldspar grains taken from a sidewall core sample at 3482 m (a transgressive sand unit) contain oil bearing fluid inclusions. The low Grains containing Oil Inclusions (GOI) value of 0.2% is consistent with oil migration through rocks with high water saturation.



The oil inclusions occur on fractures in detrital minerals however, the timing of oil migration relative to diagenesis cannot be constrained. The low GOI value can be interpreted as migration to a trap lacking closure or an oil charge to a valid trap, but with insufficient volume to form an oil column.

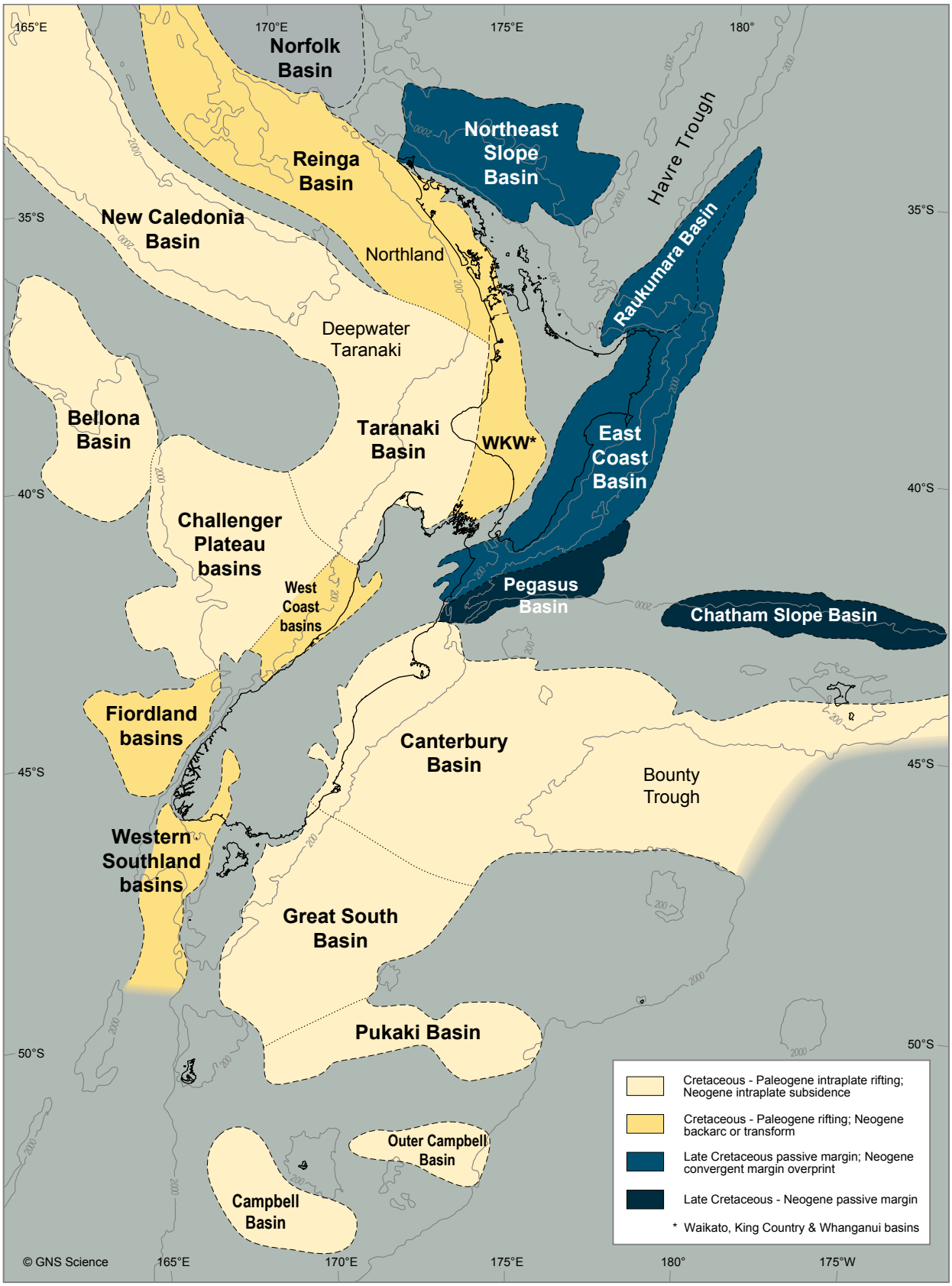
Geochemical analyses are inconclusive due to mud contamination in the samples. The one sample not largely affected by contamination was terrestrial in origin and characterised as being fully mature. The depositional environment is possibly anoxic. The source rock of the sample is believed to be a late mature coal.

WAKA NUI-1

SPUD DATE: 30 April 1999
TD: 3,681 m MD

Waka Nui-1 was a wildcat exploration well drilled approximately 100 km west of Kaipara Harbour, off the northwest coast of Northland by Conoco Northland Ltd in 1999.

The well was designed to determine the reservoir potential and quality of the Middle and Upper Cretaceous sandstones, and the Late Cretaceous transgressive sandstone and Eocene turbiditic sandstone. The upper part of the drilled sequence largely matched the well prognosis however the predicted Cretaceous coal measures were not encountered. The well reached a total depth of 3681 m MD. No oil shows and only trace amounts of gas were recorded, and the well was plugged and abandoned.


165°E
170°E
175°E
180°
175°W
35°S
40°S
45°S
50°S
Norfolk Basin
Reinga Basin
Northeast Slope Basin
Havre Trough
New Caledonia Basin
Northland
Raukumara Basin
Deepwater Taranaki
Bellona Basin
Taranaki Basin
WKW*
East Coast Basin
Challenger Plateau basins
West Coast basins
Pegasus Basin
Chatham Slope Basin
Fiordland basins
Canterbury Basin
Bounty Trough
Western Southland basins
Great South Basin
Pukaki Basin
Outer Campbell Basin
Campbell Basin
Cretaceous - Paleogene intraplate rifting; Neogene intraplate subsidence
Cretaceous - Paleogene rifting; Neogene backarc or transform
Late Cretaceous passive margin; Neogene convergent margin overprint
Late Cretaceous - Neogene passive margin
* Waikato, King Country & Whanganui basins
© GNS Science

Future Frontiers

WESTERN FRONTIER BASINS

The Bellona Basin is located along the western margin of the Challenger Plateau. The New Caledonia Basin extends northwestwards from the Taranaki shelf between the Challenger Plateau and the Northland Shelf/Norfolk Ridge.

BASIS FOR INFERRING THICK BASIN-FILL SEQUENCE

- Regional seismic (reflection and refraction) data, satellite gravity data, gravity models.

INFERRED GEOLOGY AND ANALOGUES

- Analogues – Gippsland and Bass Strait basins of SE Australia. New Caledonia Basin – Rockall Trough-Faeroes-Shetland Channel system west of the British Isles.
- Challenger Plateau and Lord Howe Rise basins – transgressive terrestrial and marginal marine sediments, deposited during post-rift regional subsidence. By the Eocene, most or all of the region was entirely marine.
- New Caledonia Basin – Cretaceous clastics and volcaniclastics deposited in fluvial and marginal marine environments. Terrigenous clays and mudstones, and authigenic limestone. Early Miocene volcanic activity.

PETROLEUM PROSPECTIVITY

- Older parts of the basins may contain terrestrial and marginal marine source rocks.
- Granitic basement along the Challenger Plateau margin – good source of high-quality reservoir rocks.
- Regional tilting may have affected migration pathways.
- Potential oil seeps along the southern flank of the Challenger Plateau identified by satellite radar studies.
- Rift sequence in New Caledonia Basin buried to a depth great enough for thermal maturation.
- Structural and stratigraphic traps may exist at a number of levels in New Caledonia Basin.

CAMPBELL PLATEAU BASINS

Pukaki, Campbell and Outer Campbell basins lie south of Great South Basin on the Campbell Plateau.

BASIS FOR INFERRING THICK BASIN-FILL SEQUENCE

- Regional seismic and satellite gravity data.

INFERRED GEOLOGY AND ANALOGUES

- Analogue – Great South Basin.
- Oldest sediments interpreted as a syn-rift and transgressive terrestrial and marginal marine sequence.
- Subsidence ended about 45 million years ago.
- Post-Paleocene, carbonate content increases.
- Post-Eocene sediments primarily carbonate oozes and chalks.

PETROLEUM PROSPECTIVITY

- Thick coal measures may be present along the outer margin of Campbell Plateau.
- Potential source and reservoir rocks – Cretaceous and Paleogene transgressive deposits and deep water turbidites.
- Larger basins could have sufficient thickness of sediments to generate hydrocarbons.

DEEP-WATER CANTERBURY BENEATH BOUNTY TROUGH

A thick sedimentary sequence believed to be present beneath the modern Bounty Trough, in water depths of 1500 to 3000 m. Considered a province of the contiguous Canterbury Basin however no structural boundary between the two basins.

BASIS FOR INFERRING THICK BASIN-FILL SEQUENCE

- Regional seismic profiles, inferred geology and analogues
- Analogue – Canterbury and Great South basins.
- Oldest sediments interpreted to be a transgressive terrestrial and marginal marine sequence.
- Subsidence ended about 45 million years ago.
- Neogene sediments are primarily carbonate oozes and chalks.

PETROLEUM PROSPECTIVITY

- Probably prospective.
- Coal measures and good reservoir sands in the rift and early transgressive sequences.
- Some areas of basin-fill are sufficiently thick to have generated hydrocarbons.

NORTHEAST SLOPE BASIN

Lies to the northeast of North Island, beneath 200 to 2,000 m of water. Covers the outer continental shelf, continental slope, and part of the Northland Plateau.

BASIS FOR INFERRING THICK BASIN-FILL SEQUENCE

- Regional seismic lines and gravity data.

INFERRED GEOLOGY AND ANALOGUES

- Analogues – Raukumara, East Coast, Pegasus, and Chatham Slope Basins.
- The Northeast Slope Basin contains relatively undisturbed sedimentary rocks overlying a more disrupted, layered to chaotic unit (Northland Allochthon).
- Upper part of Neogene succession turbidites and hemipelagic limestones.
- Early Miocene volcaniclastics and bioclastic limestone.
- Underlying units may include Cretaceous to Oligocene continental slope and basin floor clastic sediments above a Gondwana margin succession.

PETROLEUM PROSPECTIVITY

- Little is known about the prospectivity.
- Potential oil seeps are across the basin.
- Some basin-fill rocks are within the oil and gas generation windows.
- Source rocks – Cretaceous marine mudstones, Paleocene Waipawa Formation black shale.
- Reservoir rocks – Late Cretaceous quartzofeldspathic turbidite sandstones, fractured Whangai Formation siliceous mudstone, chert and marl, and Miocene turbidites, coquina limestones and fractured limestones.
- Traps – Early and Middle Miocene broad fault-bounded anticlines.

CHATHAM SLOPE BASIN

Overlies the northern slope of Chatham Rise.

BASIS FOR INFERRING THICK BASIN-FILL SEQUENCE

- Seismic lines, gravity data (large linear gravity low).
- Inferred geology and analogues.
- Analogue – none identified.
- Basal succession below Chatham Slope - Gondwana margin accretionary wedge.
- Igneous and sedimentary rocks from the Hikurangi Plateau are likely to be incorporated within, and partially subducted below, the paleo-accretionary wedge whch forms Chatham Rise.
- Following cessation of Cretaceous subduction, depositon of Late Cretaceous and Paleogene passive margin succession.
- By the Neogene, the region had subsided and only small volumes of clastic sediment reached it.

PETROLEUM PROSPECTIVITY

- Little can be inferred from present knowledge.

MURCHISON BASIN

Part of the West Coast basins group, on and offshore the West Coast of the South Island. Murchison Basin is considered to be a southern extension of the productive Taranaki Basin.

BASIS FOR INFERRING THICK BASIN-FILL SEQUENCE

- Seismic surveys , gravity surveys.

INFERRED GEOLOGY AND ANALOGUES

- Analogue – Taranaki Basin.
- Late Eocene coal measures with interbedded and overlying sandstones and mudstones unconformably overlie basement.
- Oligocene mudstones with interbedded sandstones and limestones.
- Miocene and younger mudstones, sandstones and conglomerates.
- Normal faults control Eocene source rock distribution and Oligocene seal facies and thickness.

PETROLEUM PROSPECTIVITY

- Anticlinal prospects.
- Significant number of oil and gas seeps.
- Potential reservoirs occur in all the main stratigraphic units.
- Source rocks – coal measures, carbonaceous mudstones.
- Oil source rocks lie within oil window at economically drillable depths = large oil plays with good hydrocarbon prospects.
- Seals (interbedded siltstones, mudstones and claystones) generally 2,000 m thick.
- Coal measures equivalent to the upper Kapuni Group, the major hydrocarbon bearing unit in the Taranaki Basin.

WAIKATO, KING COUNTRY AND WANGANUI BASINS

A series of mainly Neogene depocentres east of Taranaki Fault. King Country and Waikato basins are essentially entirely onshore. Wanganui Basin extends eastwards to the central North Island axial ranges and offshore from central and southern North Island.

BASIS FOR INFERRING THICK BASIN-FILL SEQUENCE

- Seismic surveys.

INFERRED GEOLOGY AND ANALOGUES

- Analogue – none identified.
- Waikato Basin - Late Eocene coal measures deposited between flanking basement highs, then overtopped by estuarine to open marine facies deposited during transgression.
- Bathyal mudstone and turbidites were deposited following subsidence in Early Miocene time.
- King Country Basin – thin sedimentary succession preserved.
- Oligocene coal measure and limestone deposition. Early Miocene erosion, then deposition of paralic rocks, including coal measures.
- Wanganui Basin – Late Miocene deposition of paralic and shallow-marine sediments. Shelf facies overlain by outer shelf to bathyal and basin-floor fan deposits, overtopped by progradational shelf systems.
- Southward depocentre migration continues.

PETROLEUM PROSPECTIVITY

- Little thermogenic oil and gas potential, perceived to be thermally immature and lacking good source rocks.
- Sub-bituminous coal seams in Waikato Basin have considerable coal seam gas potential.
- Sedimentary thicknesses are generally too thin to generate and expel petroleum.
- Eocene, Oligocene and Miocene coal measures present in King Country and Waikato regions.
- Reservoir rocks – Eocene to Quaternary fluvial sandstones, shelf sandstones, basin-floor fans, and Oligocene to earliest Miocene limestones.
- Plays – inversion structures, anticlines, stratigraphic pinch-outs, and drapes over pre-existing basement relief.

WESTERN SOUTHLAND BASINS

Onshore Te Anau and Waiau basins, and offshore from west to east, the Puysegur, Balleny, Waitutu, and Solander Basins. Southern limits of the offshore basins unknown.

BASIS FOR INFERRING THICK BASIN-FILL SEQUENCE

- Regional seismic data.

INFERRED GEOLOGY AND ANALOGUES

- Analogue – Taranaki Basin.
- Early to mid-Cretaceous terrestrial conglomerate and sandstone, with lesser lacustrine deposits. Eocene coarse-grained alluvial fan deposits, fluvial sandstones, mudstones and coals are overlain by Oligocene bathyal marine mudstones with intercalated turbidite sandstones. Miocene deep-water mudstone, sandstone and limestone.
- Marine deposition ceased in the Middle Miocene in the Te Anau Basin; younger beds are non-marine conglomerates and sandstones. In the Waiau Basin, marine mudstone deposition continued through to the Pliocene.
- Cambrian to Early Cretaceous basement rocks.
- Plio-Pleistocene arc volcanism.

PETROLEUM PROSPECTIVITY

- Gas seeps, oil shales, degraded oil in outcrop, and a show in an exploration well indicate an effective petroleum system.
- Coal seam gas found in Cretaceous and Eocene coal measures in the Waiau Basin may be commercial.
- Large number of potential traps.
- Source rocks – Cretaceous syn-rift sediments, some potential in Eocene lacustrine mudstones. More oil prone than those in Taranaki.
- Reservoir rocks – Late Cretaceous and Eocene fluvial sandstones, Oligocene turbidites, and Miocene turbidite sandstones in the offshore.
- Plays – anticlines, stratigraphic pinch-outs, and drapes over pre-existing basement relief.

Recommended Reading

REFERENCES

TARANAKI

King, P.R., Thrasher, G.P., Bland, K.J., Carthew, P., D'Cruz, D., Griffin, A.G., Jones, C.M., Strogen, D.P. 2010. Cretaceous-Cenozoic geology and petroleum systems of the Taranaki Basin, New Zealand. Digitally remastered version. GNS Science, Lower Hutt. *Institute of Geological & Nuclear Sciences monograph 13*. 1 CD.

Strogen, D.P. [compiler] 2011. Updated paleogeographic maps for the Taranaki Basin and surrounds. GNS Science, Lower Hutt. GNS Science Report 2010/53. 83 p.

Uruski, C.I., Wood, R. 1991. A new look at the New Caledonia Basin, an extension of the Taranaki Basin, offshore North Island, New Zealand. *Marine and Petroleum Geology [8]:* 379-391.

Uruski, C.I., 2008. Deepwater Taranaki, New Zealand: structural development and petroleum potential. *Exploration Geophysics 39 [2]:* 94-107.

REINGA – NORTHLAND

Bache, F., Stagpoole, V., Sutherland, R. 2013. Seismic Stratigraphy of the Reinga Basin, NW New Zealand: Tectonic and Petroleum Implications. In Rosen N.C. et al. [eds.] New understanding of the Petroleum Systems of Continental Margins of the World. GCS-SEPM Foundation. 32nd Annual Bob F. Perkins Research Conference.

Bache, F., Sutherland, R., Stagpoole, V.M., Herzer, R.H., Collot, J., Rouillard, P. 2012. Stratigraphy of the southern Norfolk Ridge and the Reinga Basin: a record of initiation of Tonga-Kermadec-Northland subduction in the southwest Pacific. *Earth and planetary science letters* 321/322: 41-53.

Bierbrauer, K., Herdy, T.J., Rek, A., Mills, K.L. 2008. Offshore Northland Basin – Frontier exploration north of the established Taranaki Hydrocarbon Province. New Zealand Petroleum Conference 2008, Auckland, New Zealand.

Stagpoole, V., Reid, E., Browne, G.H., Bland, K., Ilg, B., Griffin, A., Herzer, R.H., Uruski, C. 2009. Petroleum prospectivity of the Reinga Basin, New Zealand. GNS Science, Lower Hutt. Ministry of Economic Development New Zealand Unpublished Petroleum Report PR 4037.

Uruski, C.I., Reid, E., Stagpoole, V., Herzer, R., Griffin, A. Bland, K., Ilg, B., Browne, G. 2010. The Reinga Basin, North Island, New Zealand. *APPEA journal 50:* 287-308.

EAST COAST PROVINCE

Field, B.D., Uruski, C.I. 1997. Cretaceous-Cenozoic geology and petroleum systems of the East Coast region, New Zealand. Institute of Geological & Nuclear Sciences, Lower Hutt, New Zealand. *Institute of Geological & Nuclear Sciences monograph 19*. 301 p.

Stagpoole, V., Field, B., Sutherland, R., Uruski, C., Zink, K. 2008. Petroleum prospectivity of the Raukumara Basin, New Zealand. Ministry of Economic Development New Zealand Unpublished Petroleum Report PR 3921.

Uruski, C.I., Field, B.D., Funnell, R., Hollis, C., Nicol, A., Maslen, G. 2006. Developments in the central and northeastern East Coast Basin, North Island, New Zealand. *APPEA journal 46 [1]:* 215-235.

Uruski, C.I., Kennedy, C., Stagpoole, V., Sutherland, R., Henrys, S. 2008. The discovery of a new sedimentary basin: Offshore Raukumara, East Coast, North Island, New Zealand. *APPEA journal 48:* 53-68.

Uruski, C.I. 2010. Breaking new ground – again! The Pegasus subbasin of the East Coast Basin, North Island, New Zealand. ASEG Conference 2010, Sydney, Australia.

Uruski, C.I., Bland, K. 2011. Pegasus Basin and the prospects for oil and gas. Ministry of Economic Development New Zealand Unpublished Petroleum Report PR 4326.

GREAT SOUTH–CANTERBURY PROVINCE

Field, B.D., Browne, G.H., Davy, B.W., Herzer, R.H., Hoskins, R.H., Raine, J.I., Wilson, G.J., Sewell, R.J., Smale, D., Watters, W.A. 1989. Cretaceous and Cenozoic sedimentary basins and geological evolution of the Canterbury Region, South Island, New Zealand. New Zealand Geological Survey, Lower Hutt. *New Zealand Geological Survey Basin Studies 2*. 94 p.

Cook, R.A., Sutherland, R., Zhu, H. and others. 1999. Cretaceous – Cenozoic geology and petroleum systems of the Great South Basin, New Zealand. Institute of Geological & Nuclear Sciences, Lower Hutt, New Zealand. *Institute of Geological & Nuclear Sciences monograph 20*. 188 p. + 2 enclosures.

Dixon, B.T., Mogg, W.G., Robinson, S.J., Beckman, D.W., Pass, G.P. 2010. Offshore Canterbury Basin – Efficient use of emerging technology in the advancement of frontier exploration. *In 2010 New Zealand Petroleum Conference proceedings.* Wellington, Ministry of Economic Development. 1 CD.

Exxon Mobil Exploration Company, 2010. Great South Basin 3D/2D Seismic Interpretation Report – PEP 50117. Ministry of Economic Development New Zealand Unpublished Petroleum Report PR 4233. 42 p.

Exxon Mobil Exploration Company, 2010. Great South Basin Open Acreage 2D Seismic Interpretation Report, PEP 50117. Ministry of Economic Development New Zealand Unpublished Petroleum Report PR 4234. 18 p.

Killops, S.D., Cook, R.A., Sykes, R. 1997. Petroleum potential and oil-source correlation in the Great South and Canterbury basins. *New Zealand Journal of Geology and Geophysics (40):* 405-423.

Mogg, W.G., Aurisch, K., O'Leary, R., Pass, G.P. 2008. Offshore Canterbury Basin – beyond the shelf edge. *In 2008 New Zealand Petroleum Conference proceedings*. Wellington, Ministry of Economic Development. 24p.

Schiøler, P., Rogers, K., Sykes, R., Hollis, C.J., Ilg, B., Meadows, D., Roncaglia, L., Uruski, C. 2010. Palynofacies, organic geochemistry and depositional environment of the Tartan Formation (Late Paleocene), a potential source rock in the Great South Basin, New Zealand. *Marine and Petroleum Geology (27)*: 351-369.

Sykes, R., Funnell, R.H. 2002. Petroleum source rock potential and generation history in the offshore Canterbury Basin. *New Zealand Unpublished Openfile Petroleum Report PR 2707*. 140 p.

Uruski, C.I., Kennedy, C., Harrison, T., Maslen, G., Cook, R.A., Sutherland, R., Zhu, H. 2007. Petroleum potential of the Great South Basin, New Zealand – new seismic data improves imaging. *APPEA journal 47 (1/2)*: 145-161.

FRONTIER BASINS

Collet, T., Johnson, A., Knapp, C., Boswell, R. (eds.) 2009. Natural Gas Hydrates: Energy Resource Potential and Associated Geologic Hazards. *AAPG Memoir 89*.

Griffin, A.G. 2009. Overview of shale gas: exploration and exploitation characteristics and issues based on experience from the United States, and implications for New Zealand. GNS Science, Lower Hutt. *GNS Science Report 2009/70*. 72 p.

Kamp, P.J.J., Vonk, A.J., Bland, K.J., Hansen, R.J., Hendy, A.J.W., McIntyre, A.P., Ngatai, M., Cartwright, S.J., Hayton, S., Nelson, C.S. 2004. Neogene stratigraphic architecture and tectonic evolution of Wanganui, King Country, and eastern Taranaki basins, New Zealand. *New Zealand Journal of Geology and Geophysics 47(4)*: 625-644.

Pecher, I.A., Gorman, A.R., Henrys, S.A., Stagpoole, V., Fohrmann, M., Toulmin, S. 2010. Gas Hydrate Exploration on the Hikurangi Margin, New Zealand 2010. GNS Science, Lower Hutt. *GNS Science Report 2010/10*. 31 p.

Hoskins, R.H., Raine, J.I., Smale, D. 1986. Cretaceous and Cenozoic sedimentary basins of the West Coast Region, South Island, New Zealand. DSIR, Wellington. NZ Geological Survey Basin Studies 1. 89 p.

Turnbull. I.M., Uruski, C.I. 1993. Cretaceous and Cenozoic sedimentary basins of Western Southland, South Island, New Zealand. DSIR, Wellington. *New Zealand Geological Survey basin studies 4 (also known as Institute of Geological & Nuclear Sciences monograph 1)*. 86 p.

Uruski, C.I. 1993. Petroleum potential of Western Southland, South Island, New Zealand. *PESA journal (July 1993):* 55-66.

Uruski, C.I. 2000. Petroleum potential of New Zealand's deepwater basins. In: *2000 New Zealand Petroleum Conference proceedings.* Wellington, Ministry of Economic Development. Pp 146-152.

Uruski, C.I. 2010. New Zealand's deepwater frontier. *Marine and Petroleum Geology 27* (2010): 2005-2026.

LINKS

New Zealand Petroleum & Minerals
www.nzpam.govt.nz

Online Exploration Database
data.nzpam.govt.nz

Petroleum Basins Explorer
data.gns.cri.nz/pbe

Ministry of Business, Innovation & Employment (MBIE)
www.mbie.govt.nz

Health & Safety (MBIE)
www.osh.dol.govt.nz

Inland Revenue
www.ird.govt.nz

Maritime New Zealand
www.maritimenz.govt.nz

Environmental Protection Authority (EPA)
www.epa.govt.nz

Ministry of Primary Industries
www.mpi.govt.nz

GNS Science
www.gns.cri.nz

Petroleum Exploration & Production Association New Zealand (PEPANZ)
www.pepanz.com



NEW ZEALAND
PETROLEUM & MINERALS

New Zealand Petroleum & Minerals,
Ministry of Business, Innovation
& Employment,
PO Box 1473, Wellington 6140.

Telephone: +64 3 962 6179
Facsimile: +64 4 471 0187

Email: nzpam@mbie.govt.nz
Website: www.nzpam.govt.nz